

VECTOR SPACES (CHAPTER 1)

$\mathbb{R}^n$ AND $\mathbb{C}^n$ (SECTION 1A)

LISTS (SUB-SECTION 1A1)

DEF 1.8 (*list, length*):

A list of length n is the same as an n -tuple. We write a list of vectors of length m , $(v_1, \dots, v_m)$, as $v_1, \dots, v_m$ throughout this document without closing parentheses. Since lists are the same as tuples, two lists are only equal if they have the same length and the same elements in the same order.

DEF 1.11 ($\mathbb{F}^n$, *coordinate*):

$$\mathbb{F}^n := \left\{ (x_1, \dots, x_n) : x_k \in \mathbb{F}, k \in \{1, \dots, n\} \right\}$$

DEFINITION OF A VECTOR SPACE (SECTION 1B)

DEF 1.20 (*vector space*):

A “vector space” is a set V along with an addition on V and a scalar multiplication on V such that the following properties hold.

- **COMMUTATIVITY:** $u + v = v + u$, $\forall u, v \in V$
- **ASSOCIATIVITY:** $(u + v) + w = u + (v + w)$ and $(ab)v = a(bv)$, $\forall u, v, w \in V$ and $\forall a, b \in \mathbb{F}$
- **ADDITIVE IDENTITY:** $\exists 0 \in V : v + 0 = v$, $\forall v \in V$.
 $\exists!$ is not a requirement, but a property of vector spaces. See Theorem 1.26.
- **ADDITIVE INVERSE:** $\forall v \in V$, $\exists w \in V : v + w = 0$
- **MULTIPLICATIVE IDENTITY:** $1v = v$, $\forall v \in V$
- **DISTRIBUTIVE PROPERTIES:** $a(u + v) = au + av$, $\forall u, v \in V$ and $(a + b)v = av + bv$, $\forall a, b \in \mathbb{F}$ and $v \in V$.

DEF 1.24 (*notation $\mathbb{F}^S$*):

If S is a set, then $\mathbb{F}^S$ denotes the set of functions from S to $\mathbb{F}$. Let $f, g \in \mathbb{F}^S$ and $\lambda \in \mathbb{F}$:

- The “sum” $f + g \in \mathbb{F}^S$ is defined as follows: $(f + g)(x) := f(x) + g(x)$, $\forall x \in S$
- The “product” $\lambda f \in \mathbb{F}^S$ is defined as follows: $(\lambda f)(x) := \lambda f(x)$, $\forall x \in S$.

The vector space $\mathbb{F}^n := \mathbb{F}^{\{1, 2, \dots, n\}}$ is a special case, because each $(x_1, \dots, x_n) \in \mathbb{F}^n$ can be thought of as a function x from the set $\{1, 2, \dots, n\}$ to $\mathbb{F}$ by writing $x(k)$ instead of x_k for the k^{th} coordinate of $(x_1, \dots, x_n)$.

THM 1.26 (*unique additive identity*):

A vector space has a unique additive identity.

PROOF: Suppose 0 and $0'$ are both additive identities. Then $0' = 0' + 0 = 0 + 0' = 0$. So, $0' = 0$. $\square$

THM 1.27 (*unique additive inverse*):

Every element in a vector space has a unique additive inverse.

PROOF: Let $v \in V$. Suppose $u, w \in V$ are additive inverses of v . Using associativity and the definition of the additive inverse, we get

$$u = u + 0 = u + (v + w) = (u + v) + w = 0 + w = w.$$

Thus, $u = w$, as desired. $\square$

DEF 1.28:

For $v, w \in V$, $-v$ denotes the additive inverse of v and we define subtraction as $w - v := w + (-v)$.

DEF 1.29:

For the rest of this summary, V denotes a vector space over $\mathbb{F}$ most of the time.

THM 1.30 (*the number 0 times a vector*):

$$0v = 0 \quad \forall v \in V.$$

PROOF: Let $v \in V$. We have $0v = (0 + 0)v = 0v + 0v$ by the distributive property. Adding the additive inverse of $0v$ to both sides yields $0 = 0v$, as desired. $\square$

THM 1.31 (*a number times the vector 0*):

$$a0 = 0 \quad \forall a \in \mathbb{F}.$$

PROOF: For every $a \in \mathbb{F}$, we have $a0 = a(0 + 0) = a0 + a0$. Adding the additive inverse of $a0$ to both sides yields $0 = a0$, as desired. $\square$

THM 1.32 (*the number -1 times a vector*):

$$(-1)v = -v \quad \forall v \in V.$$

PROOF: Let $v \in V$. Hence, $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$, using the definition of the multiplicative identity, the distributive property, and Theorem 1.30. This shows that $(-1)v$ is an additive inverse of v , and therefore, $(-1)v = -v$ by the uniqueness of the additive inverse. $\square$

EXERCISES ABOUT DEFINITION OF A VECTOR SPACE 1B

- (1) Prove that $-(-v) = v \quad \forall v \in V$.

PROOF: Let $v \in V$. By definition, $v + (-v) = 0$. This means that v is an additive inverse of $-v$. Since additive inverses are unique by ??, and the unique additive inverse of $-v$ is denoted by $-(-v)$, it follows that $-(-v) = v$. $\square$

- (2) Suppose that $a \in \mathbb{F}$ and $v \in V$ satisfy $av = 0$. Prove that $a = 0$ or $v = 0$.

PROOF: There are two cases for a . If $a = 0$, we are done. If $a \neq 0$, then multiplying by $\frac{1}{a} \in \mathbb{F}$ gives:

$$av = 0 \implies \frac{1}{a}(av) = \frac{1}{a}0 \implies \left(\frac{1}{a}a\right)v = 0 \implies 1v = 0 \implies v = 0.$$

So, either $a = 0$ or $v = 0$. $\square$

- (3) Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

SOLUTION: To solve this equation for x , we add the unique additive inverse $-v$ to both sides and multiply by $\frac{1}{3} \in \mathbb{F}$, yielding $x = \frac{1}{3}(w - v)$. Since addition, additive inverses, and scalar multiplication produce unique results, this solution $x \in V$ is unique.

- (4) Show that the empty set is not a vector space.

SOLUTION: A vector space must contain at least one element, namely the additive identity 0 . The empty set contains no elements, and therefore, it cannot be a vector space.

- (5) Show that in the definition of a vector space, the additive inverse condition can be replaced with the condition that

$$0v = 0 \quad \forall v \in V.$$

PROOF: Let $v \in V$. Assuming $0v = 0$, the distributive property and multiplicative identity give:

$$\begin{aligned} 0 &= 0v = (1 + (-1))v \\ &= 1v + (-1)v \\ &= v + (-1)v. \end{aligned}$$

Thus, for every $v \in V$, the element $(-1)v$ is an additive inverse of v . □

SUBSPACES (SECTION 1C)**DEF 1.33** (*subspace*):

A subset U of V ($U \subseteq V$) is called a “*subspace*” of V if U is also a vector space with the same additive identity, addition, and scalar multiplication as on V .

THM 1.34 (*conditions for a subspace*):

A subset U of V is a subspace of V if and only if U satisfies the following three conditions:

- **ADDITIVE IDENTITY:** $0 \in U$
- **CLOSED UNDER ADDITION:** $u, w \in U \implies u + w \in U$
- **CLOSED UNDER SCALAR MULTIPLICATION:** $a \in \mathbb{F}$ and $u \in U \implies au \in U$

The additive identity condition above could be replaced by the condition that U is nonempty (because taking $u \in U$ and multiplying it by 0 implies that $0 \in U$).

PROOF: We need to show the proposition in two directions.

$\implies$ **DIRECTION:** If U is a subspace of V , then U satisfies the three conditions above by the definition of a vector space.

$\Leftarrow$ **DIRECTION:** The first condition ensures that the additive identity 0 of V is in U . The other two conditions ensure that addition and scalar multiplication are well-defined operations on U . By Theorem 1.32 and closure under scalar multiplication, every $u \in U$ has an additive inverse $-u = (-1)u \in U$. The other axioms of a vector space (associativity, commutativity, and the distributive properties) are automatically inherited by U from V . Thus, U is a vector space, and hence, a subspace of V . $\square$

SUM OF SUBSPACES (SUB-SECTION 1C1)**DEF 1.36** (*sum of subspaces*):

If $V_1, \dots, V_m$ are subspaces of V , the “*sum*” of $V_1, \dots, V_m$, denoted by $V_1 + \dots + V_m$, is the set of all possible sums of elements of $V_1, \dots, V_m$. More precisely,

$$V_1 + \dots + V_m := \{v_1 + \dots + v_m \mid v_1 \in V_1, \dots, v_m \in V_m\}.$$

THM 1.40 (*sum of subspaces is the smallest containing subspace*):

If $V_1, \dots, V_m$ are subspaces of V , then $V_1 + \dots + V_m$ is the smallest subspace of V containing $V_1, \dots, V_m$.

PROOF: We first prove that $W := \sum_{k=1}^m V_k$ is a subspace of V , and then show that it is the smallest such subspace containing each V_k . Since $0 \in V_k$ for every $k \in \{1, \dots, m\}$, we have $0 = 0 + \dots + 0 \in W$. If

$u = \sum_{k=1}^m v_k \in W$ and $w = \sum_{k=1}^m v'_k \in W$ with $v_k, v'_k \in V_k$, then

$$u + w = \sum_{k=1}^m (v_k + v'_k) \in W, \quad \text{since each } V_k \text{ is closed under addition.}$$

For any $\lambda \in \mathbb{F}$,

$$\lambda u = \lambda \sum_{k=1}^m v_k = \sum_{k=1}^m \underbrace{\lambda v_k}_{\in V_k} \in W, \quad \text{as each } V_k \text{ is closed under scalar multiplication.}$$

Thus, W contains 0, is closed under addition and scalar multiplication, and hence, is a subspace of V by Theorem 1.34. Now, we show that W is contained in every subspace of V containing $V_1, \dots, V_m$. The subspaces $V_1, \dots, V_m$ are all contained in W , i.e., $V_k \subseteq W$ for every $k \in \{1, \dots, m\}$, because every $v \in V_k$ can be written as $v = 0 + \dots + v + \dots + 0$.

Conversely, every subspace of V containing $V_1, \dots, V_m$ must contain all sums of their elements because subspaces are closed under addition. Therefore, every such subspace contains W ,

showing that W is the smallest subspace of V containing $V_1, \dots, V_m$. $\square$

DIRECT SUMS (SUB-SECTION 1C2)

DEF 1.41 (*direct sum, $\oplus$*):

$V_1 + \dots + V_m$ is called a “direct sum” if each element of $V_1 + \dots + V_m$ can be written in only one way as a sum $v_1 + \dots + v_m$, where each $v_k \in V_k$. In this case, we write

$$V_1 + \dots + V_m = V_1 \oplus \dots \oplus V_m$$

to denote that $V_1 + \dots + V_m$ is a direct sum.

EXAMPLE 1.42:

$$\mathbb{F}^3 = \left\{ (x, y, 0) \in \mathbb{F}^3 \mid x, y \in \mathbb{F} \right\} \oplus \left\{ (0, 0, z) \in \mathbb{F}^3 \mid z \in \mathbb{F} \right\}.$$

THM 1.45 (*condition for a direct sum*):

$\sum_{k=1}^m V_k$ is a direct sum (i.e., $\sum_{k=1}^m V_k = \bigoplus_{k=1}^m V_k$) if and only if the only way to express 0 as $v_1 + \dots + v_m$ with $v_k \in V_k$ is $v_1 = \dots = v_m = 0$.

PROOF: $\Rightarrow$ **DIRECTION:** First, suppose $\sum_{k=1}^m V_k$ is a direct sum. By the uniqueness of decompositions, the only way to write $0 = v_1 + \dots + v_m$ with $v_k \in V_k$ is $v_1 = \dots = v_m = 0$.

$\Leftarrow$ **DIRECTION:** Second, suppose that the only way to write 0 as a sum $w_1 + \dots + w_m$ with $w_k \in V_k$ is by taking each $w_k = 0$. Let $v \in \sum_{k=1}^m V_k$, with

$$v = v_1 + \dots + v_m = u_1 + \dots + u_m, \quad v_k, u_k \in V_k \text{ (for } k = 1, \dots, m).$$

Subtracting gives

$$0 = \underbrace{(v_1 - u_1)}_{\in V_1} + \dots + \underbrace{(v_m - u_m)}_{\in V_m}.$$

By hypothesis, each summand must be zero, so $v_1 = u_1, \dots, v_m = u_m$. Hence, every element has a unique decomposition, so $\sum_{k=1}^m V_k$ is direct. $\square$

THM 1.46 (*sum and intersection of two subspaces*):

$U + W$ is a direct sum (i.e., $U + W = U \oplus W$) if and only if $U \cap W = \{0\}$.

PROOF: Let U and W be subspaces of V .

$\Rightarrow$ **DIRECTION:** First, suppose $U + W$ is a direct sum. If $v \in U \cap W$, then $v \in U$ and $-v \in W$. We also have

$$0 = v + (-v), \text{ where } v \in U \text{ and } (-v) \in W.$$

Because $U + W$ is a direct sum and $0 = 0 + 0$ with $0 \in U$ and $0 \in W$, the uniqueness of representation implies that $v = 0$. Therefore, $U \cap W = \{0\}$.

$\Leftarrow$ **DIRECTION:** Suppose $U \cap W = \{0\}$. Let $u \in U$ and $w \in W$ be such that

$$0 = u + w. \tag{1.1}$$

By Theorem 1.45, it suffices to show that $u = w = 0$. The equation (1.1) implies that $u = -w \in W$ (since W is a subspace) and $w = -u \in U$ (since U is a subspace). Thus, $u, w \in U \cap W$. Since $U \cap W = \{0\}$, it follows that $u = w = 0$. Therefore, $U + W$ is a direct sum. $\square$

EXERCISES ABOUT SUBSPACES 1C

① For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$.

(a) $\left\{ (x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0 \right\}$

- (b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$
 (c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$
 (d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$

SOLUTION:

- (a) Let $E := \{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 + 2x_2 + 3x_3 = 0\}$. We check the subspace criteria: The additive identity $(0, 0, 0) \in E$ since $0 + 2 \cdot 0 + 3 \cdot 0 = 0$. For $u, v \in E$, we have $u_1 + 2u_2 + 3u_3 = 0$ and $v_1 + 2v_2 + 3v_3 = 0$, so $(u_1 + v_1) + 2(u_2 + v_2) + 3(u_3 + v_3) = 0$, hence $u + v \in E$. For $\lambda \in \mathbb{F}$ and $u \in E$, $\lambda u_1 + 2\lambda u_2 + 3\lambda u_3 = \lambda(u_1 + 2u_2 + 3u_3) = 0$, hence $\lambda u \in E$. Therefore, E is a subspace of $\mathbb{F}^3$ by Theorem 1.34.
- (b) It is not a subspace because it does not contain the additive identity $(0, 0, 0)$.
- (c) Let $U := \{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 x_2 x_3 = 0\}$. Then U is not closed under addition:
 $(1, 1, 0) \in U$ and $(0, 0, 1) \in U$, but $(1, 1, 0) + (0, 0, 1) = (1, 1, 1) \notin U$.
 Therefore, U is not a subspace of $\mathbb{F}^3$.
- (d) Let $U := \{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 = 5x_3\}$. Then $(0, 0, 0) \in U$. If $u, v \in U$, then $(u_1 + v_1) = 5u_3 + 5v_3 = 5(u_3 + v_3)$, so $u + v \in U$. If $\lambda \in \mathbb{F}$, then $\lambda u_1 = \lambda(5u_3) = 5(\lambda u_3)$, so $\lambda u \in U$. Hence, U is a subspace of $\mathbb{F}^3$.

② Verify all assertions about subspaces in example 1.35.

- (a) If $b \in \mathbb{F}$, then $U := \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_3 = 5x_4 + b\}$ is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

SOLUTION: If U is a subspace of $\mathbb{F}^4$, then $(0, 0, 0, 0) \in U$, which requires $0 = 5(0) + b \implies b = 0$. Conversely, if $b = 0$, then $(0, 0, 0, 0) \in U$. For $u, v \in U$, $(u_3 + v_3) = 5u_4 + 5v_4 = 5(u_4 + v_4)$, so $u + v \in U$. For $\lambda \in \mathbb{F}$, $\lambda u_3 = \lambda(5u_4) = 5(\lambda u_4)$, so $\lambda u \in U$. Thus, U is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

- (b) The set of continuous real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

SOLUTION: The constant function 0 is continuous. The sum of two continuous functions is continuous, and any scalar multiple of a continuous function is continuous. Therefore, the set of continuous real-valued functions on $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

- (c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.

SOLUTION: We check the three subspace properties:

ADDITIVE IDENTITY: The zero function $f(x) = 0$ for all $x \in \mathbb{R}$ is differentiable with derivative $f'(x) = 0$.

CLOSEDNESS UNDER ADDITION: If f and g are differentiable, then $(f + g)'(x) = f'(x) + g'(x)$ for all $x \in \mathbb{R}$, so $f + g$ is differentiable.

CLOSEDNESS UNDER SCALAR MULTIPLICATION: If f is differentiable and $c \in \mathbb{R}$, then $(cf)'(x) = cf'(x)$ for all $x \in \mathbb{R}$, so cf is differentiable.

- (d) The set S of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

SOLUTION: For the zero function $h(x) = 0$, we have $h'(x) = 0$. The condition $h'(2) = 0 = b$ holds if and only if $b = 0$. If $b = 0$, then $0 \in S$. For $f, g \in S$, $(f + g)'(2) = f'(2) + g'(2) = 0 + 0 = 0$, so $f + g \in S$. For $c \in \mathbb{R}$, $(cf)'(2) = cf'(2) = c \cdot 0 = 0$, so $cf \in S$. Thus, S is a subspace if and only if $b = 0$.

③ Show that the set S of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.

SOLUTION: We check all three subspace properties:

ADDITIVE IDENTITY: For the zero function $f(x) = 0$, we have $f'(x) = 0$, so $f'(-1) = 0 = 3(0) = 3f(2)$, which implies $0 \in S$.

CLOSED UNDER ADDITION: If $f, g \in S$ and $h := f + g$, then $h'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3h(2)$. Therefore, $h \in S$.

CLOSED UNDER SCALAR MULTIPLICATION: If $f \in S$, $c \in \mathbb{R}$, and $h := cf$, then $h'(-1) = cf'(-1) = c(3f(2)) = 3(cf(2)) = 3h(2)$. Therefore, $h \in S$.

Hence, S is a subspace of $\mathbb{R}^{(-4,4)}$.

- 8 Prove or give a counterexample: If U is a nonempty subset of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

SOLUTION: The statement is false. Consider $U := \mathbb{Z}^2 = \{(x, y) \in \mathbb{R}^2 \mid x, y \in \mathbb{Z}\} \subset \mathbb{R}^2$. Then U is nonempty, closed under addition, and closed under additive inverses. However, if $\lambda \in \mathbb{R} \setminus \mathbb{Z}$, then for any non-zero $(a, b) \in U$,

$$\lambda(a, b) = (\lambda a, \lambda b) \notin U.$$

For example, $(1, 1) \in U$ and $\frac{1}{2} \in \mathbb{R}$, but $\frac{1}{2}(1, 1) = (\frac{1}{2}, \frac{1}{2}) \notin U$. Therefore, U is not a subspace of $\mathbb{R}^2$.

- 12 Prove that the union $U \cup W$ of two subspaces U, W of V is a subspace of V if and only if one of the subspaces is contained in the other (i.e., $U \subseteq W$ or $W \subseteq U$).

PROOF: $\Leftarrow$ **DIRECTION:** Let U and W be two subspaces of V . If $U \subseteq W$, then $U \cup W = W$, which is a subspace of V . Similarly, if $W \subseteq U$, then $U \cup W = U$, which is a subspace of V .

$\Rightarrow$ **DIRECTION:** Assume that $U \cup W$ is a subspace of V , where U and W are subspaces of V .

PROOF BY CONTRADICTION: Suppose $U \not\subseteq W$ and $W \not\subseteq U$. Therefore,

$$\exists u \in U \text{ with } u \notin W, \quad \text{and} \quad \exists w \in W \text{ with } w \notin U.$$

Since $u, w \in U \cup W$ and $U \cup W$ is a subspace, we have $u + w \in U \cup W$. Thus, $u + w \in U$ or $u + w \in W$.

- **CASE** $u + w \in U$: Then $w = (u + w) - u \in U$, contradicting $w \notin U$.
- **CASE** $u + w \in W$: Then $u = (u + w) - w \in W$, contradicting $u \notin W$.

In both cases, we obtain a contradiction. Therefore, $U \subseteq W$ or $W \subseteq U$.

ALTERNATIVE DIRECT PROOF: If $U \subseteq W$, the disjunction is immediately true. So suppose $U \not\subseteq W$. Then there exists $u \in U$ such that $u \notin W$. Let $w \in W$ be arbitrary. Since $u, w \in U \cup W$ and $U \cup W$ is a subspace, $u + w \in U \cup W$. Hence, $u + w \in U$ or $u + w \in W$. If $u + w \in U$, then $w = (u + w) - u \in U$, which implies $w \in U$. If $u + w \in W$, then $u = (u + w) - w \in W$, which contradicts $u \notin W$. Therefore, it must be that $u + w \in U$, which forces $w \in U$. Since $w \in W$ was arbitrary, $W \subseteq U$. Thus, $U \subseteq W$ or $W \subseteq U$. $\square$

- 23 Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that $V = V_1 \oplus U$ and $V = V_2 \oplus U$, then $V_1 = V_2$.

SOLUTION: The statement is false. As a counterexample, consider $V := \mathbb{R}^2$ and

$$U := \{(x, 0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\},$$

which is the x -axis. Let V_1 and V_2 be defined as follows:

$$V_1 := \{(0, y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\} \quad \text{and} \quad V_2 := \{(x, x) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}.$$

Then $V_1 \cap U = \{(0, 0)\}$ and $V_2 \cap U = \{(0, 0)\}$. Furthermore, every $(x, y) \in \mathbb{R}^2$ can be uniquely decomposed as:

$$(x, y) = (x, 0) + (0, y) \in U \oplus V_1 \quad \text{and} \quad (x, y) = (x - y, 0) + (y, y) \in U \oplus V_2.$$

Thus, $V = V_1 \oplus U = V_2 \oplus U$, but $V_1 \neq V_2$.

FINITE DIMENSIONAL VECTOR SPACES (CHAPTER 2)

SPAN AND LINEAR INDEPENDENCE (SECTION 2A)

LINEAR COMBINATIONS AND SPAN (SUB-SECTION 2A1)

DEF 2.2 (*linear combination*):

A “linear combination” of a list of vectors $v_1, \dots, v_m$ in V is a vector of the form $a_1v_1 + \dots + a_mv_m$, where $a_1, \dots, a_m \in \mathbb{F}$.

DEF 2.4 (*span of a list of vectors*):

The “span” of a list of vectors $v_1, \dots, v_m$ in V is defined as follows:

$$\text{span}(v_1, \dots, v_m) := \{a_1v_1 + \dots + a_mv_m \mid a_1, \dots, a_m \in \mathbb{F}\},$$

$$\text{span}() := \{0\} \text{ (the span of the empty list } () \text{ is defined to be } \{0\}) .$$

The definition also implies that $\text{span}(0) = \{0\}$.

THM 2.6 (*the span of a list of vectors is a subspace*):

The span of a list of vectors in V is the smallest subspace of V containing all vectors in the list.

PROOF: Let $v_1, \dots, v_m \in V$. We have

$$0 = 0v_1 + \dots + 0v_m \in \text{span}(v_1, \dots, v_m),$$

and hence, the additive identity is in $\text{span}(v_1, \dots, v_m)$.

$\text{span}(v_1, \dots, v_m)$ is closed under addition, because for all $a_1, \dots, a_m, c_1, \dots, c_m \in \mathbb{F}$, we have

$$(a_1v_1 + \dots + a_mv_m) + (c_1v_1 + \dots + c_mv_m) = (a_1 + c_1)v_1 + \dots + (a_m + c_m)v_m \in \text{span}(v_1, \dots, v_m).$$

Furthermore, $\text{span}(v_1, \dots, v_m)$ is closed under scalar multiplication because for any $\lambda \in \mathbb{F}$,

$$\lambda(a_1v_1 + \dots + a_mv_m) = (\lambda a_1)v_1 + \dots + (\lambda a_m)v_m \in \text{span}(v_1, \dots, v_m).$$

Thus, $\text{span}(v_1, \dots, v_m)$ is a subspace of V by Theorem 1.34. Every subspace of V that contains each v_k is closed under scalar multiplication and addition, and therefore, contains $\text{span}(v_1, \dots, v_m)$.

Hence, $\text{span}(v_1, \dots, v_m)$ is the smallest subspace of V containing each $v_1, \dots, v_m$. $\square$

DEF 2.7 (*spanning a vector space*):

If $V = \text{span}(v_1, \dots, v_m)$, we say $v_1, \dots, v_m$ “spans” V .

DEF 2.9 (*finite-dimensional vector space*):

A vector space V is called “finite-dimensional” if some list of vectors in V spans V .

DEF 2.10 (*polynomial, $\mathcal{P}(\mathbb{F})$*):

A function $p : \mathbb{F} \rightarrow \mathbb{F}$ is called a “polynomial” with coefficients in $\mathbb{F}$ if there exist $a_0, \dots, a_m \in \mathbb{F}$ such that

$$p(z) = a_0 + a_1z + a_2z^2 + \dots + a_mz^m \quad \forall z \in \mathbb{F}.$$

The set of all polynomials with coefficients in $\mathbb{F}$ is denoted by $\mathcal{P}(\mathbb{F})$. It is a subspace of $\mathbb{F}^{\mathbb{F}}$.

DEF 2.11 (*degree of a polynomial, $\deg p$*):

A polynomial $p \in \mathcal{P}(\mathbb{F})$ is said to have “degree” m , written $\deg p = m$, if there exist $a_0, \dots, a_m \in \mathbb{F}$ with $a_m \neq 0$ such that

$$p(z) = a_0 + a_1z + \dots + a_mz^m \quad \forall z \in \mathbb{F}.$$

If p is the zero polynomial, then we define $\deg 0 := -\infty$.

DEF 2.12 (*the set of polynomials of degree at most m*):

For $m \geq 0$, $\mathcal{P}_m(\mathbb{F}) := \text{span}(1, z, \dots, z^m)$ denotes the subspace of $\mathcal{P}(\mathbb{F})$ consisting of all polynomials with degree at most m .

DEF 2.13 (*infinite-dimensional*):

A vector space is called “infinite-dimensional” if it is not finite-dimensional.

LINEAR INDEPENDENCE (SUB-SECTION 2A2)

DEF 2.15 (*linear independence*):

A list of vectors $v_1, \dots, v_m$ in V is called “linearly independent” if the only choice of scalars $a_1, \dots, a_m \in \mathbb{F}$ that satisfies

$$a_1 v_1 + \dots + a_m v_m = 0$$

is $a_1 = \dots = a_m = 0$.

The empty list $()$ is also declared to be linearly independent.

DEF 2.16 (*linear dependence*):

A list of vectors $v_1, \dots, v_m$ in V is called “linearly dependent” if it is not linearly independent. That is, there exist scalars $a_1, \dots, a_m \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + \dots + a_m v_m = 0.$$

THM 2.19 (*linear dependence lemma*):

Suppose the list $v_1, \dots, v_m \in V$ is linearly dependent. Then there is an index $k \in \{1, \dots, m\}$ such that:

- (a) $v_k \in \text{span}(v_1, \dots, v_{k-1})$.
- (b) If the k^{th} term is removed from $v_1, \dots, v_m$, then the span of the remaining list equals $\text{span}(v_1, \dots, v_m)$:

$$\text{span}(v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_m) = \text{span}(v_1, \dots, v_m).$$

PROOF: Because the list $v_1, \dots, v_m$ is linearly dependent, there exist $a_1, \dots, a_m \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + \dots + a_m v_m = 0.$$

Let k be the largest element of $\{1, \dots, m\}$ such that $a_k \neq 0$. Then

$$v_k = -\frac{a_1}{a_k} v_1 - \dots - \frac{a_{k-1}}{a_k} v_{k-1},$$

which proves that $v_k \in \text{span}(v_1, \dots, v_{k-1})$.

To prove the second assertion, suppose $u \in \text{span}(v_1, \dots, v_m)$. Then there exist $c_1, \dots, c_m \in \mathbb{F}$ such that

$$u = c_1 v_1 + \dots + c_m v_m. \tag{2.1}$$

Since $v_k \in \text{span}(v_1, \dots, v_{k-1})$, we can write $v_k = b_1 v_1 + \dots + b_{k-1} v_{k-1}$ for some $b_1, \dots, b_{k-1} \in \mathbb{F}$.

Substituting this into (2.1) yields:

$$\begin{aligned} u &= c_1 v_1 + \dots + c_k (b_1 v_1 + \dots + b_{k-1} v_{k-1}) + \dots + c_m v_m \\ &= (c_1 + c_k b_1) v_1 + \dots + (c_{k-1} + c_k b_{k-1}) v_{k-1} + c_{k+1} v_{k+1} + \dots + c_m v_m, \end{aligned}$$

which shows that $u \in \text{span}(v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_m)$. Therefore,

$$\text{span}(v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_m) = \text{span}(v_1, \dots, v_m),$$

as desired. □

THM 2.22 (*length of linearly independent list $\leq$ length of spanning list*):

In a finite-dimensional vector space, the length of every linearly independent list of vectors is less

than or equal to the length of every spanning list of vectors.

PROOF: Let $u_1, \dots, u_m \in V$ be a linearly independent list and let $w_1, \dots, w_n \in V$ be a list that spans V . We need to prove that $m \leq n$. We construct, in m steps, a spanning list of size n that incorporates $u_1, \dots, u_m$. Start with

$$B_0 := (w_1, \dots, w_n), \quad V = \text{span}(B_0).$$

STEP 1: Add u_1 to B_0 , obtaining the list $(u_1, w_1, \dots, w_n)$. Since $u_1 \in V = \text{span}(B_0)$, this enlarged list is linearly dependent. By Theorem 2.19, one of the vectors is in the span of the previous vectors; since $u_1 \neq 0$ (as the u 's are linearly independent), this vector must be one of the w 's, say w_j . Removing w_j gives a new list B_1 of length n with $u_1 \in B_1$ and $V = \text{span}(B_1)$.

STEP k (FOR $k = 2, \dots, m$): The list B_{k-1} from the previous step spans V and contains $u_1, \dots, u_{k-1}$ along with $n - (k - 1)$ of the original w 's. Adding u_k to the front of the w 's gives a list of length $n + 1$:

$$B_k^* := (u_1, \dots, u_k, \tilde{w}_1, \dots, \tilde{w}_{n-(k-1)}), \quad V = \text{span}(B_k^*).$$

Since $u_k \in V = \text{span}(B_{k-1})$, the list B_k^* is linearly dependent. By Theorem 2.19, some vector in B_k^* is in the span of the previous vectors. Because $u_1, \dots, u_k$ are linearly independent, this vector cannot be one of the u 's, so it must be one of the $\tilde{w}$'s, say $\tilde{w}_j$. Removing $\tilde{w}_j$ leaves a list B_k of length n containing $u_1, \dots, u_k$ such that $V = \text{span}(B_k)$.

CONCLUSION: After m steps, we have a list B_m of length n containing all m independent vectors $u_1, \dots, u_m$. Therefore, $m \leq n$. $\square$

THM 2.25 (finite-dimensional subspace):

Every subspace of a finite-dimensional vector space is finite-dimensional.

PROOF: Suppose V is finite-dimensional, and let U be a subspace of V . Since V is finite-dimensional, there is a spanning list of length n for V .

STEP 1: If $U = \{0\}$, then U is finite-dimensional, and we are done. Otherwise, choose a non-zero vector $u_1 \in U$.

STEP k ($k \geq 2$): If $U = \text{span}(u_1, \dots, u_{k-1})$, then U is finite-dimensional, and we are done. If $U \neq \text{span}(u_1, \dots, u_{k-1})$, choose a vector $u_k \in U$ such that

$$u_k \notin \text{span}(u_1, \dots, u_{k-1}).$$

After each step, the list $u_1, \dots, u_k$ is linearly independent by Theorem 2.19. By Theorem 2.22, the length of any linearly independent list in V cannot exceed n . Thus, the process must terminate in at most n steps, which means $U = \text{span}(u_1, \dots, u_k)$ for some $k \leq n$. Hence, U is finite-dimensional. $\square$

EXERCISES ABOUT SPAN AND LINEAR INDEPENDENCE 2A

- ① Find a list of four distinct vectors in $\mathbb{F}^3$ whose span equals

$$U := \{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}.$$

SOLUTION: The list $(1, -1, 0), (0, 1, -1), (1, 0, -1), (2, -1, -1)$ consists of four distinct vectors in U . The first two vectors span U because any $(x, y, z) \in U$ satisfies $z = -x - y$, so $(x, y, z) = x(1, -1, 0) + (x + y)(0, 1, -1)$. Since the other two vectors also belong to U , the span of all four vectors is precisely U .

- ② Prove that if v_1, v_2, v_3, v_4 spans V , then the list $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ also spans V .

PROOF: Notice that

$$\begin{aligned} v_1 &= (v_1 - v_2) + (v_2 - v_3) + (v_3 - v_4) + v_4, \\ v_2 &= (v_2 - v_3) + (v_3 - v_4) + v_4, \\ v_3 &= (v_3 - v_4) + v_4, \\ v_4 &= v_4. \end{aligned}$$

Thus, each $v_k \in \text{span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4)$ for $k = 1, 2, 3, 4$. Since v_1, v_2, v_3, v_4 spans V , it follows that

$$V = \text{span}(v_1, v_2, v_3, v_4) \subseteq \text{span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4) \subseteq V.$$

Hence, $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ spans V . □

- ③ Prove that if $v_1, \dots, v_m \in V$ and

$$w_k := v_1 + \dots + v_k \quad \text{for } k = 1, \dots, m,$$

then $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$.

PROOF: By definition, each $w_k = v_1 + \dots + v_k$ is a linear combination of $v_1, \dots, v_m$, and therefore, $w_k \in \text{span}(v_1, \dots, v_m)$. Consequently,

$$\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m). \quad (2.2)$$

Conversely, observe that

$$\begin{aligned} v_1 &= w_1, \\ v_2 &= w_2 - w_1, \\ v_3 &= w_3 - w_2, \\ &\vdots \\ v_m &= w_m - w_{m-1}. \end{aligned}$$

Thus, each $v_k \in \text{span}(w_1, \dots, w_m)$ for $k = 1, \dots, m$. It follows that

$$\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m). \quad (2.3)$$

Combining (2.2) and (2.3), we conclude that $\text{span}(w_1, \dots, w_m) = \text{span}(v_1, \dots, v_m)$. □

- ①⑥ Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbb{F})$.

SOLUTION: The list $1, z, z^2, z^3, z^4$ in $\mathcal{P}_4(\mathbb{F})$ is linearly independent and has length 5. By Theorem 2.22, the length of every linearly independent list is less than or equal to the length of every spanning list. Therefore, every spanning list of $\mathcal{P}_4(\mathbb{F})$ must contain at least 5 polynomials, so no list of four polynomials can span $\mathcal{P}_4(\mathbb{F})$.

- ②⑩ Suppose $p_0, p_1, \dots, p_m$ are polynomials in $\mathcal{P}_m(\mathbb{F})$ such that $p_k(2) = 0$ for $0 \leq k \leq m$. Prove that $p_0, p_1, \dots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbb{F})$.

PROOF: Original Approach (using linear functionals and dimension):

Define an evaluation map $\varphi : \mathcal{P}_m(\mathbb{F}) \rightarrow \mathbb{F}$ by $\varphi(p) := p(2)$. The polynomials that vanish at $z = 2$ form the null space $\text{null } \varphi$ of $\mathcal{P}_m(\mathbb{F})$. Any polynomial $p \in \text{null } \varphi$ can be factored as $p(z) = (z - 2)q(z)$ for some $q(z) \in \mathcal{P}_{m-1}(\mathbb{F})$. Therefore, $\dim \text{null } \varphi = m$. Since the list $p_0, p_1, \dots, p_m$ has length $m + 1$ in this m -dimensional subspace, it cannot be linearly independent by Theorem 2.22.

Self-Contained Version (using only Chapter 2A concepts):

Let $U := \{p \in \mathcal{P}_m(\mathbb{F}) \mid p(2) = 0\}$. Every $p \in U$ can be factored as $p(z) = (z - 2)q(z)$ for some $q \in \mathcal{P}_{m-1}(\mathbb{F})$. Since $\mathcal{P}_{m-1}(\mathbb{F}) = \text{span}(1, z, \dots, z^{m-1})$, the list

$$(z - 2), (z - 2)z, \dots, (z - 2)z^{m-1}$$

spans U and has length m . By Theorem 2.22, the length of any linearly independent list in U cannot exceed the length of a spanning list, which is m . Since the list $p_0, p_1, \dots, p_m$ consists of $m + 1$ polynomials in U , it cannot be linearly independent. $\square$

BASES (SECTION 2B)

DEF 2.26 (basis):

A “basis” of V is a list of vectors in V that is linearly independent and spans V .

THM 2.28 (criterion for a basis):

A list $v_1, \dots, v_n$ of vectors in V is a basis of V if and only if every $v \in V$ can be written uniquely in the form

$$v = a_1 v_1 + \dots + a_n v_n,$$

where $a_1, \dots, a_n \in \mathbb{F}$.

THM 2.30 (every spanning list contains a basis):

Every spanning list of vectors in a vector space V can be reduced to a basis of V .

PROOF: Suppose $V = \text{span}(v_1, \dots, v_n)$. Let $B_0 := (v_1, \dots, v_n)$.

STEP 1: If $v_1 = 0$, delete v_1 from B_0 , obtaining $B_1 := (v_2, \dots, v_n)$. If $v_1 \neq 0$, leave B_0 unchanged, so $B_1 := B_0 = (v_1, \dots, v_n)$.

STEP k ($k = 2, \dots, n$): Let B_{k-1} be the list obtained after step $k-1$. If v_k is in the span of the previous vectors in B_{k-1} , delete v_k from B_{k-1} to form B_k . Otherwise, leave B_{k-1} unchanged, setting $B_k := B_{k-1}$.

After step n , we obtain a list B_n . The list B_n spans V because our original list B_0 spanned V , and we only discarded vectors that were already in the span of the previous vectors. Furthermore, the construction ensures that no vector in B_n is in the span of the previous ones. By Theorem 2.19, B_n is linearly independent. Hence, B_n is a basis of V . $\square$

THM 2.31 (basis of a finite-dimensional vector space):

Every finite-dimensional vector space has a basis.

PROOF: By definition, every finite-dimensional vector space has a spanning list. By Theorem 2.30, this spanning list can be reduced to a basis. $\square$

THM 2.32 (linearly independent list extends to a basis):

Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.

PROOF: Let $u_1, \dots, u_m \in V$ be linearly independent. Since V is finite-dimensional, there is a spanning list $w_1, \dots, w_n$ of V . Then

$$V = \text{span}(u_1, \dots, u_m, w_1, \dots, w_n).$$

Applying the reduction procedure of Theorem 2.30 to this list produces a basis of V . Because $u_1, \dots, u_m$ is linearly independent, none of the u_j are deleted during the first m steps. Thus, the resulting basis contains all the vectors $u_1, \dots, u_m$. $\square$

THM 2.33 (every subspace of V is part of a direct sum equal to V):

Suppose V is finite-dimensional and U is a subspace of V . Then there is a subspace W of V such that $V = U \oplus W$.

PROOF: Suppose U is a subspace of a finite-dimensional vector space V . By Theorem 2.25, U is finite-dimensional, and therefore, U has a basis $u_1, \dots, u_m$ by Theorem 2.31. By Theorem 2.32, we can extend this to a basis $u_1, \dots, u_m, w_1, \dots, w_n$ of V . Set

$$W := \text{span}(w_1, \dots, w_n).$$

By Theorem 1.46, to show that $V = U \oplus W$, it suffices to prove that $V = U + W$ and $U \cap W = \{0\}$.

PROOF OF $V = U + W$: Let $v \in V$. Since $u_1, \dots, u_m, w_1, \dots, w_n$ is a basis of V , there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbb{F}$ such that

$$v = \underbrace{(a_1 u_1 + \dots + a_m u_m)}_{\in U} + \underbrace{(b_1 w_1 + \dots + b_n w_n)}_{\in W}.$$

Hence, $v \in U + W$, so $V \subseteq U + W$. Conversely, since $U, W \subseteq V$, we have $U + W \subseteq V$. Therefore, $V = U + W$.

PROOF OF $U \cap W = \{0\}$: Let $v \in U \cap W$. Then there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbb{F}$ such that

$$v = a_1 u_1 + \dots + a_m u_m \quad \text{and} \quad v = b_1 w_1 + \dots + b_n w_n.$$

Thus,

$$a_1 u_1 + \dots + a_m u_m - b_1 w_1 - \dots - b_n w_n = 0.$$

Because $u_1, \dots, u_m, w_1, \dots, w_n$ is linearly independent, all scalars a_i and b_j must be zero. Therefore, $v = 0$, which proves that $U \cap W = \{0\}$.

Hence, $V = U \oplus W$, as desired. $\square$

EXERCISES ABOUT BASES (SUB-SECTION 2B0)

- ⑥ Prove that if v_1, v_2, v_3, v_4 is a basis of V , then

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of V .

PROOF: STEP 1: We first show linear independence. Let $a_1, a_2, a_3, a_4 \in \mathbb{F}$ such that

$$a_1(v_1 + v_2) + a_2(v_2 + v_3) + a_3(v_3 + v_4) + a_4 v_4 = 0.$$

Rearranging terms gives:

$$a_1 v_1 + (a_1 + a_2)v_2 + (a_2 + a_3)v_3 + (a_3 + a_4)v_4 = 0.$$

Since v_1, v_2, v_3, v_4 is linearly independent, all coefficients must be zero:

$$a_1 = 0, \quad a_1 + a_2 = 0, \quad a_2 + a_3 = 0, \quad a_3 + a_4 = 0.$$

Substituting $a_1 = 0$ successively yields $a_2 = 0$, $a_3 = 0$, and $a_4 = 0$. Thus, the list $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ is linearly independent.

STEP 2: Next, we show that $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ spans V . Notice that:

$$v_4 = v_4,$$

$$v_3 = (v_3 + v_4) - v_4,$$

$$v_2 = (v_2 + v_3) - (v_3 + v_4) + v_4,$$

$$v_1 = (v_1 + v_2) - (v_2 + v_3) + (v_3 + v_4) - v_4.$$

Thus, each $v_k \in \text{span}(v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4)$ for $k = 1, 2, 3, 4$. Since v_1, v_2, v_3, v_4 spans V , it follows that $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ spans V .

Therefore, $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ is a basis of V . $\square$

- ⑦ Prove or give a counterexample: If v_1, v_2, v_3, v_4 is a basis of V and U is a subspace of V such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then v_1, v_2 is a basis of U .

SOLUTION:

The statement is **false**. Here is a counterexample:

Let $V = \mathbb{F}^4$ with standard basis e_1, e_2, e_3, e_4 , and let

$$U := \text{span}(e_1, e_2, e_3 + e_4).$$

Then $e_1, e_2 \in U$, but $e_3 \notin U$ and $e_4 \notin U$. However, e_1, e_2 does not span U because $e_3 + e_4 \in U \setminus \text{span}(e_1, e_2)$. Therefore, e_1, e_2 is not a basis of U .

8 Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k := v_1 + \dots + v_k.$$

Show that $v_1, \dots, v_m$ is a basis of V if and only if $w_1, \dots, w_m$ is a basis of V .

PROOF: Self-Contained Version (using only Chapter 2B concepts):

From Exercise 2A.3, we already know that $\text{span}(w_1, \dots, w_m) = \text{span}(v_1, \dots, v_m)$. Thus, $w_1, \dots, w_m$ spans V if and only if $v_1, \dots, v_m$ spans V .

It remains to show that $w_1, \dots, w_m$ is linearly independent if and only if $v_1, \dots, v_m$ is linearly independent.

$\Rightarrow$ **DIRECTION:** Suppose $v_1, \dots, v_m$ is linearly independent. Let $c_1, \dots, c_m \in \mathbb{F}$ satisfy

$$c_1 w_1 + c_2 w_2 + \dots + c_m w_m = 0.$$

Substituting $w_k = v_1 + \dots + v_k$ and collecting coefficients yields:

$$(c_1 + c_2 + \dots + c_m)v_1 + (c_2 + \dots + c_m)v_2 + \dots + c_m v_m = 0.$$

Since $v_1, \dots, v_m$ is linearly independent, all coefficients are zero:

$$c_m = 0, \quad c_{m-1} + c_m = 0 \implies c_{m-1} = 0, \quad \dots, \quad c_1 = 0.$$

Thus, $w_1, \dots, w_m$ is linearly independent.

$\Leftarrow$ **DIRECTION:** Suppose $w_1, \dots, w_m$ is linearly independent. Let $d_1, \dots, d_m \in \mathbb{F}$ satisfy

$$d_1 v_1 + d_2 v_2 + \dots + d_m v_m = 0.$$

Substituting $v_1 = w_1$ and $v_k = w_k - w_{k-1}$ for $k \geq 2$ gives:

$$(d_1 - d_2)w_1 + (d_2 - d_3)w_2 + \dots + (d_{m-1} - d_m)w_{m-1} + d_m w_m = 0.$$

Since $w_1, \dots, w_m$ is linearly independent, $d_m = 0$, $d_{m-1} - d_m = 0 \implies d_{m-1} = 0, \dots, d_1 = 0$.

Thus, $v_1, \dots, v_m$ is linearly independent.

Therefore, $v_1, \dots, v_m$ is a basis of V if and only if $w_1, \dots, w_m$ is a basis of V .

Original Approach (using dimension theorems from Chapter 2C):

$\Rightarrow$ Assume $v_1, \dots, v_m$ is a basis of V . Then it spans V and is linearly independent. Since each v_k can be written as a linear combination of $w_1, \dots, w_k$, any vector in $\text{span}((v_1, \dots, v_m))$ is also in $\text{span}((w_1, \dots, w_m))$. Therefore, $w_1, \dots, w_m$ spans V . Furthermore, since $v_1, \dots, v_m$ is a basis, $\dim V = m$. Since $w_1, \dots, w_m$ spans V and has length $m = \dim V$, it must also be linearly independent (by Theorem 2.42), and therefore, is a basis of V .

$\Leftarrow$ Assume $w_1, \dots, w_m$ is a basis of V . Then it spans V and is linearly independent. Since each w_k can be written as a linear combination of $v_1, \dots, v_k$, any vector in $\text{span}((w_1, \dots, w_m))$ is also in $\text{span}((v_1, \dots, v_m))$. Therefore, $v_1, \dots, v_m$ spans V . Since $w_1, \dots, w_m$ is a basis, $\dim V = m$. Since $v_1, \dots, v_m$ spans V and has length $m = \dim V$, it must also be linearly independent, and therefore, is a basis of V . $\square$

DIMENSION (SECTION 2C)

THM 2.34 (*basis length*):

Any two bases of a finite-dimensional vector space have the same length.

PROOF: Let B_1 and B_2 be two bases of a finite-dimensional vector space V . Since B_1 is linearly independent and B_2 spans V , we have $|B_1| \leq |B_2|$ by Theorem 2.22. Reversing roles, since B_2 is linearly independent and B_1 spans V , we obtain $|B_2| \leq |B_1|$. Therefore, $|B_1| = |B_2|$. $\square$

DEF 2.35 (*dimension*):

The “*dimension*” of a finite-dimensional vector space V , denoted by $\dim V$, is the length of any basis of V . We also set $\dim\{0\} := 0$.

THM 2.37 (*dimension of a subspace*):

If U is a subspace of V , then

$$\dim U \leq \dim V.$$

PROOF: By Theorem 2.25, U is finite-dimensional, so it has a basis. Any basis of U is a linearly independent list in V , and any basis of V is a spanning list of V . By Theorem 2.22, the length of a basis of U is less than or equal to the length of a basis of V . Thus, $\dim U \leq \dim V$. $\square$

THM 2.38 (*linearly independent list of the right length is a basis*):

Suppose V is finite-dimensional. Every linearly independent list of vectors in V of length $\dim V$ is a basis of V .

PROOF: Let $\dim V = n$, and let $v_1, \dots, v_n$ be a linearly independent list in V . This list can be extended to a basis of V by Theorem 2.32. However, every basis of V has length n by Theorem 2.34, so no additional vectors can be added. Hence, $v_1, \dots, v_n$ is already a basis of V . $\square$

THM 2.39 (*subspace of full dimension*):

If U is a subspace of a finite-dimensional vector space V with $\dim U = \dim V$, then $U = V$.

PROOF: Let $u_1, \dots, u_n$ be a basis of U . Thus, $n = \dim U$, and by hypothesis, $n = \dim V$. Hence, $u_1, \dots, u_n$ is a linearly independent list in V of length $\dim V$. By Theorem 2.38, $u_1, \dots, u_n$ is a basis of V . In particular, every vector in V is a linear combination of $u_1, \dots, u_n$. Therefore, $U = V$. $\square$

THM 2.42 (*spanning list of the right length is a basis*):

Suppose V is finite-dimensional. Every spanning list of vectors in V of length $\dim V$ is a basis of V .

PROOF: Suppose $\dim V = n$, and let $v_1, \dots, v_n$ be a list that spans V . By Theorem 2.30, this list can be reduced to a basis of V . Since every basis of V has length n by Theorem 2.34, no vector can be removed during the reduction process. Thus, $v_1, \dots, v_n$ is a basis of V . $\square$

THM 2.43 (*dimension of a sum of subspaces*):

If V_1 and V_2 are subspaces of a finite-dimensional vector space V , then

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

PROOF: Let $r_1, \dots, r_m$ be a basis of $V_1 \cap V_2$, so $\dim(V_1 \cap V_2) = m$. (If $V_1 \cap V_2 = \{0\}$, the list is empty and $m = 0$).

Since $r_1, \dots, r_m$ is linearly independent in V_1 and in V_2 , it can be extended to a basis

$$r_1, \dots, r_m, u_1, \dots, u_j \quad \text{of } V_1,$$

and to a basis

$$r_1, \dots, r_m, w_1, \dots, w_k \quad \text{of } V_2.$$

Thus, $\dim V_1 = m + j$ and $\dim V_2 = m + k$. We will show that the list

$$r_1, \dots, r_m, u_1, \dots, u_j, w_1, \dots, w_k. \quad (2.4)$$

The span of the list (2.4) contains V_1 and V_2 , and therefore, equals $V_1 + V_2$. To show that it is a basis, it remains only to prove linear independence. Suppose scalars $a_i, b_i, c_i \in \mathbb{F}$ satisfy:

$$\sum_{i=1}^m a_i r_i + \sum_{i=1}^j b_i u_i + \sum_{i=1}^k c_i w_i = 0. \quad (2.5)$$

Then $\sum_{i=1}^k c_i w_i = -\sum_{i=1}^m a_i r_i - \sum_{i=1}^j b_i u_i \in V_1$. Since $\sum_{i=1}^k c_i w_i \in V_2$ as well, this sum lies in $V_1 \cap V_2$. Because $r_1, \dots, r_m$ is a basis of $V_1 \cap V_2$, there exist scalars $d_1, \dots, d_m \in \mathbb{F}$ such that:

$$\sum_{i=1}^k c_i w_i = \sum_{i=1}^m d_i r_i \iff \sum_{i=1}^k c_i w_i - \sum_{i=1}^m d_i r_i = 0.$$

Because $r_1, \dots, r_m, w_1, \dots, w_k$ is a basis of V_2 , it is linearly independent, which forces all $c_i = 0$ (and $d_i = 0$). Substituting $c_1 = \dots = c_k = 0$ into (2.5) leaves:

$$\sum_{i=1}^m a_i r_i + \sum_{i=1}^j b_i u_i = 0.$$

Since $r_1, \dots, r_m, u_1, \dots, u_j$ is a basis of V_1 , it is linearly independent, so all $a_i = 0$ and all $b_i = 0$. Thus, all coefficients in (2.5) are zero, proving linear independence and completing the proof. $\square$

EXERCISES ABOUT DIMENSION 2C (SUB-SECTION 2C0)

- 20 Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if V_1, V_2, V_3 are subspaces of a finite-dimensional vector space V , then

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim(V_1) + \dim(V_2) + \dim(V_3) \\ &\quad - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ &\quad + \dim(V_1 \cap V_2 \cap V_3). \end{aligned}$$

Then either prove the formula above or give a counterexample.

SOLUTION: The formula is **wrong**. Consider the following subspaces of $\mathbb{R}^2$:

- $V_1 := \{(x, 0) \mid x \in \mathbb{R}\}$ (the x -axis)
- $V_2 := \{(x, x) \mid x \in \mathbb{R}\}$ (the line $y = x$)
- $V_3 := \{(0, y) \mid y \in \mathbb{R}\}$ (the y -axis)

The right-hand side of the formula gives: $1 + 1 + 1 - 0 - 0 - 0 + 0 = 3$, whereas the left-hand side is $\dim(V_1 + V_2 + V_3) = 2$. So the formula **DOES NOT HOLD** for this example.

However, **IT DOES HOLD** in every case where one of the subspaces is contained in another; hence, without loss of generality, assume that $V_1 \subseteq V_3$. For this situation, we have the following lemma.

LEMMA: If V_1, V_2, V_3 are finite-dimensional subspaces of a vector space, and $V_1 \subseteq V_3$, then

$$(V_1 + V_2) \cap V_3 = (V_1 \cap V_3) + (V_2 \cap V_3)$$

PROOF: If $v \in (V_1 \cap V_3) + (V_2 \cap V_3)$, then we can write $v = u + w$, where $u \in V_1 \cap V_3$ and $w \in V_2 \cap V_3$. Because V_3 is closed under addition and both u and w are in V_3 , so is v . Hence, $v \in V_3$ and $v = u + w$, such that $u \in V_1$ and $w \in V_2$. Therefore, $v \in (V_1 + V_2) \cap V_3$. We conclude:

$$(V_1 \cap V_3) + (V_2 \cap V_3) \subseteq (V_1 + V_2) \cap V_3 \quad (2.6)$$

Conversely, if $v \in (V_1 + V_2) \cap V_3$, then we can write $v = u + w$, where $u \in V_1$, $w \in V_2$, and $v \in V_3$.

Since $V_1 \subseteq V_3$ (by our assumption), we can rewrite

$$w = \underbrace{v}_{\in V_3} - \underbrace{u}_{\in V_3}.$$

Since both u and v belong to V_3 , so does w . Therefore, $v = u + w$ such that $u \in V_1 \cap V_3$ and $w \in V_2 \cap V_3$. Hence, $v \in (V_1 \cap V_3) + (V_2 \cap V_3)$. We conclude:

$$(V_1 + V_2) \cap V_3 \subseteq (V_1 \cap V_3) + (V_2 \cap V_3) \quad (2.7)$$

Inclusion (2.6) and (2.7) together yield

$$(V_1 + V_2) \cap V_3 = (V_1 \cap V_3) + (V_2 \cap V_3)$$

for $V_1 \subseteq V_3$. □

Using the standard dimension formula for the sum of two subspaces (Theorem 2.43), we obtain

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim((V_1 + V_2) + V_3) \\ &= \dim(V_1 + V_2) + \dim(V_3) - \dim((V_1 + V_2) \cap V_3) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) \\ &\quad - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3). \end{aligned} \quad (2.8)$$

Using the assumption that $V_1 \subseteq V_3$, we calculate:

$$\begin{aligned} \dim((V_1 + V_2) \cap V_3) &= \dim((V_1 \cap V_3) + (V_2 \cap V_3)) \\ &= \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3) - \underbrace{\dim((V_1 \cap V_3) \cap (V_2 \cap V_3))}_{= \dim(V_1 \cap V_2 \cap V_3)}. \end{aligned}$$

Substituting this result into (2.8) yields:

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim(V_1) + \dim(V_2) + \dim(V_3) - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) - \dim(V_1 \cap V_2) \\ &\quad - (\dim(V_1 \cap V_3) + \dim(V_2 \cap V_3) - \dim(V_1 \cap V_2 \cap V_3)) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) \\ &\quad - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ &\quad + \dim(V_1 \cap V_2 \cap V_3). \end{aligned}$$

So, under certain conditions (e.g., when $V_1 \subseteq V_3$), the formula is **true**.

- 21** Show that for a finite-dimensional vector space V with subspaces V_1, V_2, V_3 , the following formula holds:

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim V_1 + \dim V_2 + \dim V_3 \\ &\quad - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ &\quad - \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}. \end{aligned}$$

SOLUTION: Recall that for any two subspaces U and W of a finite-dimensional vector space,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Applying this formula to the three groupings of $V_1 + V_2 + V_3$ yields:

$$\begin{aligned} \dim((V_1 + V_2) + V_3) &= \dim(V_1 + V_2) + \dim V_3 - \dim((V_1 + V_2) \cap V_3) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3), \\ \dim(V_1 + (V_2 + V_3)) &= \dim V_1 + \dim(V_2 + V_3) - \dim(V_1 \cap (V_2 + V_3)) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3) - \dim((V_2 + V_3) \cap V_1), \\ \dim((V_1 + V_3) + V_2) &= \dim(V_1 + V_3) + \dim V_2 - \dim((V_1 + V_3) \cap V_2) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_3) - \dim((V_1 + V_3) \cap V_2). \end{aligned}$$

Adding these three equations together yields:

$$\begin{aligned} 3 \dim(V_1 + V_2 + V_3) &= 3 \dim V_1 + 3 \dim V_2 + 3 \dim V_3 \\ &\quad - (\dim(V_1 \cap V_2) + \dim(V_2 \cap V_3) + \dim(V_1 \cap V_3)) \\ &\quad - (\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)). \end{aligned}$$

Dividing both sides by 3 gives the desired formula.

LINEAR MAPS (CHAPTER 3)

VECTOR SPACE OF LINEAR MAPS (SECTION 3A)

DEFINITION AND EXAMPLES OF LINEAR MAPS (SUB-SECTION 3A1)

DEF 3.1 (*linear map*):

A “linear map” from V to W is a function $T : V \rightarrow W$ with the following properties:

- **ADDITIVITY:** $T(u + v) = Tu + Tv \quad \forall u, v \in V$
- **HOMOGENEITY:** $T(\lambda v) = \lambda(Tv) \quad \forall \lambda \in \mathbb{F}, \forall v \in V$

DEF 3.2 (*notation $\mathcal{L}(V, W)$, $\mathcal{L}(V)$*):

The set of all linear maps from V to W is denoted by $\mathcal{L}(V, W)$. For the set of all linear maps from V to itself, we write $\mathcal{L}(V) := \mathcal{L}(V, V)$.

THM 3.4 (*linear map lemma*):

Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_n \in W$. Then there exists a unique linear map $T \in \mathcal{L}(V, W)$ such that

$$Tv_k = w_k \quad \forall k \in \{1, \dots, n\}.$$

PROOF: Since the list $v_1, \dots, v_n$ is a basis of V , every $v \in V$ has a unique expansion

$$v = c_1 v_1 + \dots + c_n v_n,$$

where $c_1, \dots, c_n \in \mathbb{F}$. Define $T : V \rightarrow W$ by

$$T(c_1 v_1 + \dots + c_n v_n) := c_1 w_1 + \dots + c_n w_n.$$

By uniqueness of the basis expansion, this rule defines a well-defined function $T : V \rightarrow W$. In particular, choosing $v = v_k$ (so $c_k = 1$ and $c_i = 0$ for $i \neq k$) gives

$$T(v_k) = w_k.$$

Next, we verify that T is linear. Let $u = a_1 v_1 + \dots + a_n v_n \in V$, $v = b_1 v_1 + \dots + b_n v_n \in V$, and $\lambda \in \mathbb{F}$. Then

$$\begin{aligned} T(u + v) &= T((a_1 + b_1)v_1 + \dots + (a_n + b_n)v_n) \\ &= (a_1 + b_1)w_1 + \dots + (a_n + b_n)w_n \\ &= (a_1 w_1 + \dots + a_n w_n) + (b_1 w_1 + \dots + b_n w_n) \\ &= T(u) + T(v), \end{aligned}$$

and similarly,

$$\begin{aligned} T(\lambda v) &= T(\lambda b_1 v_1 + \dots + \lambda b_n v_n) \\ &= \lambda b_1 w_1 + \dots + \lambda b_n w_n \\ &= \lambda(b_1 w_1 + \dots + b_n w_n) \\ &= \lambda T(v). \end{aligned}$$

Hence, T is linear, so $T \in \mathcal{L}(V, W)$.

Finally, to prove uniqueness, suppose $S \in \mathcal{L}(V, W)$ is any linear map such that $S(v_k) = w_k$ for each $k \in \{1, \dots, n\}$. Then for any $v = c_1 v_1 + \dots + c_n v_n \in V$, we have

$$\begin{aligned} S(v) &= c_1 S(v_1) + \dots + c_n S(v_n) \\ &= c_1 w_1 + \dots + c_n w_n \\ &= T(v), \end{aligned}$$

so $S = T$. Therefore, T is unique. □

DEF 3.5 (addition and scalar multiplication on $\mathcal{L}(V, W)$):

Suppose $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. The sum $S + T \in \mathcal{L}(V, W)$ and the product $\lambda T \in \mathcal{L}(V, W)$ are defined by:

- $(S + T)(v) := Sv + Tv \quad \forall v \in V$
- $(\lambda T)(v) := \lambda(Tv) \quad \forall v \in V$

ALGEBRAIC OPERATIONS ON $\mathcal{L}(V, W)$ (SUB-SECTION 3A2)

THM 3.6 ($\mathcal{L}(V, W)$ is a vector space):

With the operations of addition and scalar multiplication defined above, $\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$.

DEF 3.7 (product of linear maps):

If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then the “product” $ST \in \mathcal{L}(U, W)$ is defined by

$$(ST)(u) := S(Tu) \quad \forall u \in U.$$

THM 3.8 (algebraic properties of products of linear maps):

Let U, V, W, X be vector spaces over $\mathbb{F}$. Then:

- **ASSOCIATIVITY:** $(T_1 T_2) T_3 = T_1 (T_2 T_3)$, whenever $T_3 \in \mathcal{L}(X, U)$, $T_2 \in \mathcal{L}(U, V)$, and $T_1 \in \mathcal{L}(V, W)$.
- **IDENTITY:** $T I_V = I_W T = T$ for all $T \in \mathcal{L}(V, W)$.
- **DISTRIBUTIVE PROPERTIES:** $(S_1 + S_2)T = S_1 T + S_2 T$ and $S(T_1 + T_2) = ST_1 + ST_2$ for all maps with compatible domains and codomains.

THM 3.10 (linear maps take 0 to 0):

If $T \in \mathcal{L}(V, W)$, then $T(0) = 0$.

PROOF: By additivity, $T(0) = T(0 + 0) = T(0) + T(0)$. Adding the additive inverse of $T(0)$ to both sides yields $T(0) = 0$, as desired. $\square$

EXERCISES ABOUT VECTOR SPACE OF LINEAR MAPS 3A

- (7) Show that every linear map from a one-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V)$, then there exists $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$ for all $v \in V$.

PROOF: Since $\dim V = 1$, choose a basis v_1 of V . Then every $v \in V$ can be written uniquely as $v = cv_1$ for some $c \in \mathbb{F}$. Because $T(v_1) \in V = \text{span}(v_1)$, there exists a scalar $\lambda \in \mathbb{F}$ such that

$$T(v_1) = \lambda v_1.$$

Then for any $v = cv_1 \in V$, linearity of T implies:

$$T(v) = T(cv_1) = cT(v_1) = c(\lambda v_1) = \lambda(cv_1) = \lambda v.$$

Hence, $Tv = \lambda v$ for all $v \in V$, as desired. $\square$

- (12) Suppose U is a subspace of V with $U \neq V$. Let $S \in \mathcal{L}(U, W)$ with $S \neq 0$ (so $Su \neq 0$ for some $u \in U$). Define $T : V \rightarrow W$ by

$$Tv := \begin{cases} Sv & \text{if } v \in U, \\ 0 & \text{if } v \in V \setminus U. \end{cases}$$

Prove that T is not a linear map on V .

PROOF: Since $S \neq 0$, choose $u \in U$ such that $Su \neq 0$. Because $U \neq V$, there exists $v \in V \setminus U$. We

claim that $u + v \notin U$. Indeed, if $u + v \in U$, then $v = (u + v) - u \in U$, which contradicts $v \notin U$.

Therefore, by definition of T ,

$$T(u + v) = 0.$$

On the other hand,

$$Tu + Tv = Su + 0 = Su \neq 0.$$

Thus, $T(u + v) \neq Tu + Tv$, showing that T is not additive, and therefore, T is not a linear map on V . $\square$

- 13** Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V . In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that

$$Tu = Su \quad \forall u \in U.$$

PROOF: Let U be a subspace of a finite-dimensional vector space V , and let $S \in \mathcal{L}(U, W)$. By Theorem 2.33, there is a subspace X of V such that $V = U \oplus X$. Let $u_1, \dots, u_m$ be a basis of U , and let $x_{m+1}, \dots, x_n$ be a basis of X . Then

$$u_1, \dots, u_m, x_{m+1}, \dots, x_n$$

is a basis of V . By Theorem 3.4, there exists a unique linear map $T \in \mathcal{L}(V, W)$ satisfying:

$$T(u_i) = S(u_i) \quad \text{for } 1 \leq i \leq m, \quad \text{and} \quad T(x_j) = 0 \quad \text{for } m+1 \leq j \leq n.$$

For any $u \in U$, writing $u = c_1 u_1 + \dots + c_m u_m$ gives:

$$T(u) = \sum_{i=1}^m c_i T(u_i) = \sum_{i=1}^m c_i S(u_i) = S\left(\sum_{i=1}^m c_i u_i\right) = S(u).$$

Thus, $T \in \mathcal{L}(V, W)$ extends S , as required. $\square$

- 15** Suppose $v_1, \dots, v_m$ is a linearly dependent list of vectors in V . Suppose also that $W \neq \{0\}$. Prove that there exist $w_1, \dots, w_m \in W$ such that no $T \in \mathcal{L}(V, W)$ satisfies

$$Tv_k = w_k \quad \text{for all } k \in \{1, \dots, m\}.$$

SOLUTION: Since $v_1, \dots, v_m$ is linearly dependent, there exist scalars $a_1, \dots, a_m \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + \dots + a_m v_m = 0. \tag{3.1}$$

Choose an index $k \in \{1, \dots, m\}$ such that $a_k \neq 0$. Since $W \neq \{0\}$, choose a non-zero vector $w \in W$. Define the list $w_1, \dots, w_m$ in W by:

$$w_k := w \quad \text{and} \quad w_j := 0 \quad \text{for all } j \neq k.$$

Suppose, for the sake of contradiction, that there exists a linear map $T \in \mathcal{L}(V, W)$ such that $Tv_j = w_j$ for all $j \in \{1, \dots, m\}$. Applying T to both sides of (3.1) gives:

$$0 = T(0) = T(a_1 v_1 + \dots + a_m v_m) = a_1 T(v_1) + \dots + a_m T(v_m) = a_1 w_1 + \dots + a_m w_m = a_k w.$$

Since $a_k \neq 0$, multiplying by $\frac{1}{a_k}$ yields $w = 0$, contradicting the choice of $w \neq 0$. Therefore, no such linear map $T \in \mathcal{L}(V, W)$ can exist.

NULL SPACES AND RANGES (SECTION 3B)

NULL SPACE AND INJECTIVITY (SUB-SECTION 3B1)

DEF 3.11 (*null space, null T*):

For $T \in \mathcal{L}(V, W)$, the “null space” of T , denoted by $\text{null } T$, is the subset of V defined by

$$\text{null } T := \{v \in V \mid Tv = 0\}.$$

THM 3.13 (*null space is a subspace*):

If $T \in \mathcal{L}(V, W)$, then $\text{null } T$ is a subspace of V .

PROOF: We have $0 \in \text{null } T$ because $T(0) = 0$ by Theorem 3.10. Suppose $u, v \in \text{null } T$ and $\lambda \in \mathbb{F}$. Then $T(u + v) = Tu + Tv = 0 + 0 = 0$, so $u + v \in \text{null } T$. Similarly, $T(\lambda u) = \lambda Tu = \lambda 0 = 0$, so $\lambda u \in \text{null } T$. Therefore, $\text{null } T$ is a subspace of V by Theorem 1.34. $\square$

DEF 3.14 (*injectivity*):

A function $T : V \rightarrow W$ is called “injective” if $Tu = Tv \implies u = v$ for all $u, v \in V$.

THM 3.15 (*injectivity criterion via null space*):

Let $T \in \mathcal{L}(V, W)$. Then T is injective if and only if $\text{null } T = \{0\}$.

PROOF: $\implies$ **DIRECTION:** First, assume that T is injective. Since $\text{null } T$ is a subspace of V (by Theorem 3.13), we have $\{0\} \subseteq \text{null } T$. To prove that $\text{null } T \subseteq \{0\}$, suppose $v \in \text{null } T$. Then $T(v) = 0$. Since $T(0) = 0$ by Theorem 3.10 and T is injective, it follows that $v = 0$. Thus, $\text{null } T = \{0\}$.

$\Leftarrow$ **DIRECTION:** Second, assume $\text{null } T = \{0\}$. Let $u, v \in V$ satisfy $Tu = Tv$. Then

$$0 = Tu - Tv = T(u - v).$$

Thus, $u - v \in \text{null } T = \{0\}$, so $u - v = 0$, which gives $u = v$. Therefore, T is injective. $\square$

RANGE AND SURJECTIVITY (SUB-SECTION 3B2)

DEF 3.16 (*range*):

For $T \in \mathcal{L}(V, W)$, the “range” of T , denoted by $\text{range } T$, is the subset of W defined by

$$\text{range } T := \{Tv \mid v \in V\}.$$

THM 3.18 (*the range is a subspace*):

If $T \in \mathcal{L}(V, W)$, then $\text{range } T$ is a subspace of W .

PROOF: Let $T \in \mathcal{L}(V, W)$. By Theorem 3.10, $T(0) = 0$, so $0 \in \text{range } T$. If $w_1, w_2 \in \text{range } T$, then there exist $v_1, v_2 \in V$ such that $Tv_1 = w_1$ and $Tv_2 = w_2$. Then $w_1 + w_2 = Tv_1 + Tv_2 = T(v_1 + v_2) \in \text{range } T$, so $\text{range } T$ is closed under addition. If $w \in \text{range } T$ and $\lambda \in \mathbb{F}$, then there exists $v \in V$ such that $w = Tv$. Then $\lambda w = \lambda Tv = T(\lambda v) \in \text{range } T$, so $\text{range } T$ is closed under scalar multiplication. By Theorem 1.34, $\text{range } T$ is a subspace of W . $\square$

DEF 3.19 (*surjectivity*):

A function $T : V \rightarrow W$ is called “surjective” (or “onto”) if $\text{range } T = W$.

FUNDAMENTAL THEOREM OF LINEAR MAPS (SUB-SECTION 3B3)

THM 3.21 (*Fundamental Theorem of Linear Maps*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $\text{range } T$ is finite-dimensional and $\dim V = \dim \text{null } T + \dim \text{range } T$.

PROOF: Let $u_1, \dots, u_m$ be a basis of $\text{null } T$, so $\dim \text{null } T = m$. By Theorem 2.32, we can extend this to a basis

$$u_1, \dots, u_m, v_1, \dots, v_n \quad \text{of } V.$$

Thus, $\dim V = m + n$. To complete the proof, it suffices to show that $Tv_1, \dots, Tv_n$ is a basis of $\text{range } T$.

Let $v \in V$. Then there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbb{F}$ such that

$$v = a_1 u_1 + \dots + a_m u_m + b_1 v_1 + \dots + b_n v_n.$$

Applying T and using $T(u_i) = 0$ yields:

$$Tv = b_1 Tv_1 + \dots + b_n Tv_n.$$

Since this holds for every $v \in V$, the list $Tv_1, \dots, Tv_n$ spans $\text{range } T$. In particular, $\text{range } T$ is finite-dimensional.

To show that $Tv_1, \dots, Tv_n$ is linearly independent, suppose scalars $c_1, \dots, c_n \in \mathbb{F}$ satisfy:

$$c_1 Tv_1 + \dots + c_n Tv_n = 0.$$

Then $T(c_1 v_1 + \dots + c_n v_n) = 0$, so $c_1 v_1 + \dots + c_n v_n \in \text{null } T$. Because $u_1, \dots, u_m$ is a basis of $\text{null } T$, there exist scalars $d_1, \dots, d_m \in \mathbb{F}$ such that

$$c_1 v_1 + \dots + c_n v_n = d_1 u_1 + \dots + d_m u_m,$$

or equivalently,

$$-d_1 u_1 - \dots - d_m u_m + c_1 v_1 + \dots + c_n v_n = 0.$$

Because $u_1, \dots, u_m, v_1, \dots, v_n$ is a basis of V , it is linearly independent, which forces all $c_i = 0$ (and $d_i = 0$). Thus, $Tv_1, \dots, Tv_n$ is linearly independent and forms a basis of $\text{range } T$. Hence, $\dim \text{range } T = n$, and $\dim V = m + n = \dim \text{null } T + \dim \text{range } T$. $\square$

THM 3.22 (*linear map to a lower-dimensional space is not injective*):

Suppose V and W are finite-dimensional vector spaces such that $\dim V > \dim W$. Then no linear map from V to W is injective.

PROOF: Let $T \in \mathcal{L}(V, W)$ and assume $\dim V > \dim W$. By Theorem 3.21,

$$\begin{aligned} \dim \text{null } T &= \dim V - \dim \text{range } T \\ &\geq \dim V - \dim W \\ &> 0, \end{aligned}$$

where the first inequality holds because $\text{range } T$ is a subspace of W (so $\dim \text{range } T \leq \dim W$ by Theorem 2.37). Since $\dim \text{null } T > 0$, $\text{null } T \neq \{0\}$. Thus, T is not injective by Theorem 3.15. $\square$

THM 3.24 (*linear map to a higher-dimensional space is not surjective*):

Suppose V and W are finite-dimensional vector spaces such that $\dim V < \dim W$. Then no linear map from V to W is surjective.

PROOF: Let $T \in \mathcal{L}(V, W)$ and assume $\dim V < \dim W$. By Theorem 3.21,

$$\begin{aligned} \dim \text{range } T &= \dim V - \dim \text{null } T \\ &\leq \dim V \\ &< \dim W. \end{aligned}$$

Thus, $\dim \text{range } T < \dim W$, which implies $\text{range } T \neq W$. Hence, T is not surjective by Theorem 3.19. $\square$

THM 3.26 (*homogeneous system of linear equations*):

A homogeneous system of linear equations with more variables than equations has non-zero solu-

tions.

PROOF: Let positive integers m and n satisfy $n > m$, and let $A_{j,k} \in \mathbb{F}$ for $j \in \{1, \dots, m\}$ and $k \in \{1, \dots, n\}$.

Consider the system:

$$\sum_{k=1}^n A_{j,k} x_k = 0 \quad \text{for } j = 1, \dots, m.$$

Define $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by

$$T(x_1, \dots, x_n) := \left(\sum_{k=1}^n A_{1,k} x_k, \dots, \sum_{k=1}^n A_{m,k} x_k \right).$$

Then T is a linear map. Since $\dim \mathbb{F}^n = n > m = \dim \mathbb{F}^m$, Theorem 3.22 implies that T is not injective. By Theorem 3.15, $\text{null } T \neq \{0\}$, so there exists a non-zero solution $(x_1, \dots, x_n) \in \mathbb{F}^n$. $\square$

THM 3.28 (*system of linear equations with more equations than variables*):

An inhomogeneous system of linear equations with more equations than variables has no solutions for some choice of constant terms.

PROOF: Let positive integers m and n satisfy $m > n$, and let $A_{j,k} \in \mathbb{F}$ for $j \in \{1, \dots, m\}$ and $k \in \{1, \dots, n\}$.

Define $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by

$$T(x_1, \dots, x_n) := \left(\sum_{k=1}^n A_{1,k} x_k, \dots, \sum_{k=1}^n A_{m,k} x_k \right).$$

Since $\dim \mathbb{F}^n = n < m = \dim \mathbb{F}^m$, Theorem 3.24 implies that T is not surjective. Therefore, $\text{range } T \neq \mathbb{F}^m$, so there exists $(c_1, \dots, c_m) \in \mathbb{F}^m \setminus \text{range } T$ for which the system has no solution. $\square$

EXERCISES ABOUT NULL SPACES AND RANGES 3B

- ① Give an example of a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $\text{null } T = \text{range } T$.

SOLUTION: Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T(x, y) := (y, 0).$$

Then T is linear. We determine its null space and range:

- $T(x, y) = (0, 0) \iff (y, 0) = (0, 0) \iff y = 0$. Thus, $\text{null } T = \{(x, 0) \mid x \in \mathbb{R}\}$.
- $\text{range } T = \{T(x, y) \mid x, y \in \mathbb{R}\} = \{(y, 0) \mid y \in \mathbb{R}\}$.

Hence, $\text{null } T = \text{range } T = \{(x, 0) \mid x \in \mathbb{R}\}$.

- ② Suppose $S, T \in \mathcal{L}(V)$ are such that $\text{range } S \subseteq \text{null } T$. Prove that $(ST)^2 = 0$.

PROOF: For any $v \in V$,

$$(ST)^2(v) = (ST)(ST)(v) = S(T(S(Tv))).$$

Since $Tv \in V$, we have $S(Tv) \in \text{range } S \subseteq \text{null } T$, which implies $T(S(Tv)) = 0$. Applying S gives:

$$(ST)^2(v) = S(0) = 0.$$

Therefore, $(ST)^2 = 0$. $\square$

- ⑧ Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \dots, v_n$ is linearly independent in V . Prove that $Tv_1, \dots, Tv_n$ is linearly independent in W .

PROOF: Suppose scalars $a_1, \dots, a_n \in \mathbb{F}$ satisfy:

$$a_1 Tv_1 + \dots + a_n Tv_n = 0.$$

By linearity of T , this equation can be rewritten as:

$$T(a_1 v_1 + \dots + a_n v_n) = 0.$$

Since T is injective, $\text{null } T = \{0\}$ by Theorem 3.15, so

$$a_1 v_1 + \cdots + a_n v_n = 0.$$

Because $v_1, \dots, v_n$ is linearly independent in V , this forces $a_1 = \cdots = a_n = 0$. Hence, $Tv_1, \dots, Tv_n$ is linearly independent in W . $\square$

- ⑨ Suppose $v_1, \dots, v_n$ spans V and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, \dots, Tv_n$ spans $\text{range } T$.

PROOF: Let $w \in \text{range } T$. Then $w = Tv$ for some $v \in V$. Since $v_1, \dots, v_n$ spans V , there exist scalars $a_1, \dots, a_n \in \mathbb{F}$ such that

$$v = a_1 v_1 + \cdots + a_n v_n.$$

Applying T and using linearity gives:

$$w = Tv = T(a_1 v_1 + \cdots + a_n v_n) = a_1 Tv_1 + \cdots + a_n Tv_n \in \text{span}(Tv_1, \dots, Tv_n).$$

Thus, $\text{range } T \subseteq \text{span}(Tv_1, \dots, Tv_n)$.

Conversely, each $Tv_k \in \text{range } T$, and since $\text{range } T$ is a subspace of W , we have $\text{span}(Tv_1, \dots, Tv_n) \subseteq \text{range } T$. Therefore, $\text{range } T = \text{span}(Tv_1, \dots, Tv_n)$. $\square$

- ⑫ Suppose T is a linear map from $\mathbb{F}^4$ to $\mathbb{F}^2$ such that

$$\text{null } T = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_1 = 5x_2 \text{ and } x_3 = 7x_4\}.$$

Prove that T is surjective.

PROOF: Every element of $\text{null } T$ can be written as:

$$(5x_2, x_2, 7x_4, x_4) = x_2(5, 1, 0, 0) + x_4(0, 0, 7, 1) \quad \text{for } x_2, x_4 \in \mathbb{F}.$$

Since $(5, 1, 0, 0)$ and $(0, 0, 7, 1)$ are linearly independent, they form a basis of $\text{null } T$, so $\dim \text{null } T = 2$. By Theorem 3.21,

$$\dim \text{range } T = \dim \mathbb{F}^4 - \dim \text{null } T = 4 - 2 = 2.$$

Since $\text{range } T$ is a subspace of $\mathbb{F}^2$ and $\dim \text{range } T = \dim \mathbb{F}^2 = 2$, Theorem 2.39 implies that $\text{range } T = \mathbb{F}^2$. Therefore, T is surjective. $\square$

- ⑬ Suppose U is a three-dimensional subspace of $\mathbb{R}^8$ and $T \in \mathcal{L}(\mathbb{R}^8, \mathbb{R}^5)$ satisfies $\text{null } T = U$. Prove that T is surjective.

PROOF: By Theorem 3.21,

$$\dim \text{range } T = \dim \mathbb{R}^8 - \dim \text{null } T = 8 - 3 = 5.$$

Since $\text{range } T \subseteq \mathbb{R}^5$ and $\dim \text{range } T = \dim \mathbb{R}^5 = 5$, Theorem 2.39 implies $\text{range } T = \mathbb{R}^5$. Hence, T is surjective. $\square$

- ⑭ Prove that there does not exist a linear map from $\mathbb{F}^5$ to $\mathbb{F}^2$ whose null space equals

$$S := \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{F}^5 \mid x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}.$$

PROOF: Every element of S has the form:

$$(3x_2, x_2, x_3, x_3, x_3) = x_2(3, 1, 0, 0, 0) + x_3(0, 0, 1, 1, 1) \quad \text{for } x_2, x_3 \in \mathbb{F}.$$

The two vectors $(3, 1, 0, 0, 0)$ and $(0, 0, 1, 1, 1)$ are linearly independent, so $\dim S = 2$. For any linear map $T \in \mathcal{L}(\mathbb{F}^5, \mathbb{F}^2)$, Theorem 3.21 gives:

$$\dim \text{null } T = \dim \mathbb{F}^5 - \dim \text{range } T = 5 - \dim \text{range } T \geq 5 - \dim \mathbb{F}^2 = 5 - 2 = 3.$$

Since $\dim \text{null } T \geq 3 \neq 2 = \dim S$, no such linear map T can have $\text{null } T = S$. $\square$

- ⑮ Suppose there exists a linear map $T \in \mathcal{L}(V, W)$ such that its null space $\text{null } T$ and range $\text{range } T$ are both finite-dimensional. Prove that V is finite-dimensional.

PROOF: Original Approach (using Rank-Nullity):

According to the rank-nullity theorem, $\dim V = \dim \text{range } T + \dim \text{null } T$. Since both $\dim \text{range } T$ and $\dim \text{null } T$ are finite by assumption, their sum is also finite. Hence, $\dim V < \infty$, so V is finite-dimensional.

NOTE ON CIRCULARITY: The Fundamental Theorem of Linear Maps (Theorem 3.21) includes *"Suppose V is finite-dimensional"* in its hypothesis. Therefore, a complete proof must show that V is spanned by a finite list without invoking the Rank-Nullity Theorem.

Rigorous Version (constructing a finite spanning list directly):

Let $u_1, \dots, u_m$ be a basis of $\text{null } T$, and let $w_1, \dots, w_n$ be a basis of $\text{range } T$. Since $w_1, \dots, w_n \in \text{range } T$, there exist vectors $v_1, \dots, v_n \in V$ such that $Tv_j = w_j$ for each $j \in \{1, \dots, n\}$.

We claim that the list

$$u_1, \dots, u_m, v_1, \dots, v_n$$

spans V . To see this, let $v \in V$. Then $Tv \in \text{range } T$, so there exist scalars $c_1, \dots, c_n \in \mathbb{F}$ such that

$$Tv = c_1 w_1 + \dots + c_n w_n = c_1 Tv_1 + \dots + c_n Tv_n = T(c_1 v_1 + \dots + c_n v_n).$$

By linearity, this implies $T(v - (c_1 v_1 + \dots + c_n v_n)) = 0$, so

$$v - (c_1 v_1 + \dots + c_n v_n) \in \text{null } T.$$

Since $u_1, \dots, u_m$ is a basis of $\text{null } T$, there exist scalars $a_1, \dots, a_m \in \mathbb{F}$ such that

$$v - (c_1 v_1 + \dots + c_n v_n) = a_1 u_1 + \dots + a_m u_m,$$

which gives:

$$v = a_1 u_1 + \dots + a_m u_m + c_1 v_1 + \dots + c_n v_n.$$

Thus, $V = \text{span}(u_1, \dots, u_m, v_1, \dots, v_n)$. Since V is spanned by a finite list, V is finite-dimensional. $\square$

- 16** Suppose V and W are both finite-dimensional. Prove that there exists an injective linear map $T \in \mathcal{L}(V, W)$ if and only if $\dim V \leq \dim W$.

PROOF: $\Rightarrow$ DIRECTION: If there exists an injective linear map $T \in \mathcal{L}(V, W)$, then $\text{null } T = \{0\}$ by Theorem 3.15. By Theorem 3.21 and Theorem 2.37,

$$\dim V = \dim \text{null } T + \dim \text{range } T = \dim \text{range } T \leq \dim W.$$

$\Leftarrow$ DIRECTION: Suppose $\dim V \leq \dim W$. Let $v_1, \dots, v_n$ be a basis of V , where $n = \dim V$. Since $\dim W \geq n$, there exists a linearly independent list $w_1, \dots, w_n$ in W . By Theorem 3.4, there exists a unique linear map $T \in \mathcal{L}(V, W)$ such that $Tv_k = w_k$ for each $k \in \{1, \dots, n\}$. If $v = \sum_{k=1}^n c_k v_k \in \text{null } T$, then $0 = Tv = \sum_{k=1}^n c_k w_k$, which forces all $c_k = 0$ by linear independence of the w_k . Thus, $\text{null } T = \{0\}$, so T is injective. $\square$

- 17** Suppose V and W are both finite-dimensional. Prove that there exists a surjective linear map from V onto W if and only if $\dim V \geq \dim W$.

PROOF: $\Rightarrow$ DIRECTION: Suppose there exists a surjective linear map $T \in \mathcal{L}(V, W)$. Then $\text{range } T = W$, so by Theorem 3.21,

$$\dim V = \dim \text{null } T + \dim \text{range } T = \dim \text{null } T + \dim W \geq \dim W.$$

$\Leftarrow$ DIRECTION: Suppose $\dim V \geq \dim W$. Let $m = \dim W$ and $n = \dim V$, so $n \geq m$. Let $v_1, \dots, v_n$ be

a basis of V and $w_1, \dots, w_m$ be a basis of W . By Theorem 3.4, define $T \in \mathcal{L}(V, W)$ by:

$$Tv_k := \begin{cases} w_k & \text{if } 1 \leq k \leq m, \\ 0 & \text{if } m < k \leq n. \end{cases}$$

Then for every $w = \sum_{k=1}^m a_k w_k \in W$, setting $v = \sum_{k=1}^m a_k v_k \in V$ yields $Tv = w$. Hence, T is surjective. $\square$

- 18** Suppose V and W are finite-dimensional, and U is a subspace of V . Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\text{null } T = U$ if and only if $\dim U \geq \dim V - \dim W$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose $T \in \mathcal{L}(V, W)$ satisfies $\text{null } T = U$. By Theorem 3.21 and Theorem 2.37,

$$\dim V = \dim U + \dim \text{range } T \leq \dim U + \dim W \iff \dim U \geq \dim V - \dim W.$$

$\Leftarrow$ **DIRECTION:** Suppose $\dim U \geq \dim V - \dim W$. Then $\dim V - \dim U \leq \dim W$. Let $u_1, \dots, u_k$ be a basis of U , where $k = \dim U$. Extend this to a basis $u_1, \dots, u_k, v_1, \dots, v_m$ of V , where $m = \dim V - \dim U$. Since $m \leq \dim W$, we can choose a linearly independent list $w_1, \dots, w_m$ in W . By Theorem 3.4, define $T \in \mathcal{L}(V, W)$ by:

$$T(u_i) := 0 \quad \text{for } 1 \leq i \leq k, \quad \text{and} \quad T(v_j) := w_j \quad \text{for } 1 \leq j \leq m.$$

Clearly, $U \subseteq \text{null } T$. Conversely, if $v = \sum_{i=1}^k a_i u_i + \sum_{j=1}^m b_j v_j \in \text{null } T$, then $0 = Tv = \sum_{j=1}^m b_j w_j$, so all $b_j = 0$ by linear independence of the w_j . Thus, $v = \sum_{i=1}^k a_i u_i \in U$. Therefore, $\text{null } T = U$. $\square$

- 19** Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $ST = I_V$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose T is injective. Then $\dim V \leq \dim W < \infty$, so V is finite-dimensional. Let $v_1, \dots, v_m$ be a basis of V . By Theorem 3.15 and exercise 7, $Tv_1, \dots, Tv_m$ is linearly independent in W . Extend $Tv_1, \dots, Tv_m$ to a basis $Tv_1, \dots, Tv_m, w_{m+1}, \dots, w_n$ of W . By Theorem 3.4, define $S \in \mathcal{L}(W, V)$ by:

$$S(Tv_k) := v_k \quad \text{for } 1 \leq k \leq m, \quad \text{and} \quad S(w_j) := 0 \quad \text{for } m+1 \leq j \leq n.$$

Then for any $v = \sum_{k=1}^m c_k v_k \in V$,

$$ST(v) = S\left(\sum_{k=1}^m c_k Tv_k\right) = \sum_{k=1}^m c_k S(Tv_k) = \sum_{k=1}^m c_k v_k = v.$$

Thus, $ST = I_V$.

$\Leftarrow$ **DIRECTION:** Suppose $S \in \mathcal{L}(W, V)$ satisfies $ST = I_V$. If $v \in \text{null } T$, then $Tv = 0$, so $v = I_V(v) = (ST)(v) = S(Tv) = S(0) = 0$. Hence, $\text{null } T = \{0\}$, so T is injective. $\square$

- 20** Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is surjective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $TS = I_W$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose T is surjective. Let $w_1, \dots, w_m$ be a basis of W . Since T is surjective, for each $k \in \{1, \dots, m\}$ there exists $v_k \in V$ such that $Tv_k = w_k$. By Theorem 3.4, define $S \in \mathcal{L}(W, V)$ by $S(w_k) := v_k$ for $1 \leq k \leq m$. Then for any $w = \sum_{k=1}^m a_k w_k \in W$,

$$TS(w) = T\left(\sum_{k=1}^m a_k v_k\right) = \sum_{k=1}^m a_k Tv_k = \sum_{k=1}^m a_k w_k = w.$$

Thus, $TS = I_W$.

$\Leftarrow$ **DIRECTION:** Suppose $S \in \mathcal{L}(W, V)$ satisfies $TS = I_W$. For any $w \in W$, setting $v = S(w) \in V$ gives $Tv = T(S(w)) = (TS)(w) = w$. Therefore, T is surjective. $\square$

21 Suppose V is finite-dimensional, U is a subspace of V , and $T \in \mathcal{L}(V, W)$. Prove that

$$X := \{v \in V \mid Tv \in U\}$$

is a subspace of V and

$$\dim X = \dim \text{null } T + \dim(U \cap \text{range } T).$$

PROOF: First, we show that X is a subspace of V :

- Since $T(0) = 0 \in U$, we have $0 \in X$.
- If $v_1, v_2 \in X$, then $T(v_1), T(v_2) \in U$, so $T(v_1 + v_2) = T(v_1) + T(v_2) \in U$, hence $v_1 + v_2 \in X$.
- If $v \in X$ and $\lambda \in \mathbb{F}$, then $T(\lambda v) = \lambda T(v) \in U$, hence $\lambda v \in X$.

Thus, X is a subspace of V .

Now consider the restriction $T|_X: X \rightarrow W$. By Theorem 3.21,

$$\dim X = \dim \text{null}(T|_X) + \dim \text{range}(T|_X).$$

Since $0 \in U$, $\text{null } T \subseteq X$, so $\text{null}(T|_X) = \text{null } T$. Furthermore, $\text{range}(T|_X) = \{Tv \mid v \in X\} = \{Tv \mid v \in V \text{ and } Tv \in U\} = U \cap \text{range } T$. Substituting these identities yields:

$$\dim X = \dim \text{null } T + \dim(U \cap \text{range } T).$$

□

22 (a) Suppose U, V, W are finite-dimensional vector spaces, $S \in \mathcal{L}(V, W)$, and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T.$$

PROOF: Applying Theorem 3.21 to T and to ST gives:

$$\dim U = \dim \text{null } T + \dim \text{range } T,$$

$$\dim U = \dim \text{null } ST + \dim \text{range } ST.$$

Equating these expressions yields:

$$\dim \text{null } ST = \dim \text{null } T + \dim \text{range } T - \dim \text{range } ST. \quad (3.2)$$

Now consider the restriction $S|_{\text{range } T}: \text{range } T \rightarrow W$. Its range is $\text{range}(S|_{\text{range } T}) = \text{range } ST$.

Applying Theorem 3.21 to $S|_{\text{range } T}$ gives:

$$\dim \text{range } T = \dim \text{null}(S|_{\text{range } T}) + \dim \text{range } ST.$$

Since $\text{null}(S|_{\text{range } T}) = \text{null } S \cap \text{range } T \subseteq \text{null } S$, we have $\dim \text{null}(S|_{\text{range } T}) \leq \dim \text{null } S$.

Therefore,

$$\dim \text{range } T - \dim \text{range } ST = \dim \text{null}(S|_{\text{range } T}) \leq \dim \text{null } S.$$

Substituting this into (3.2) yields:

$$\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T.$$

□

(b) Give an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \text{range } TS = 2$.

SOLUTION: Define $S, T \in \mathcal{L}(\mathbb{F}^5)$ by:

$$S(x_1, x_2, x_3, x_4, x_5) := (0, 0, 0, x_4, x_5),$$

$$T(x_1, x_2, x_3, x_4, x_5) := (x_4, x_5, 0, 0, 0).$$

Then for any $x \in \mathbb{F}^5$,

$$ST(x) = S(x_4, x_5, 0, 0, 0) = (0, 0, 0, 0, 0) = 0,$$

so $ST = 0$. On the other hand,

$$TS(x) = T(0, 0, 0, x_4, x_5) = (x_4, x_5, 0, 0, 0).$$

Thus, $\text{range } TS = \{(x_4, x_5, 0, 0, 0) \mid x_4, x_5 \in \mathbb{F}\}$, which has dimension 2.

- (24) Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$. Prove that $\dim \text{range } TS \leq 2$.

PROOF:

Since $ST = 0$, we have $S(Tv) = 0$ for all $v \in V$, so $\text{range } T \subseteq \text{null } S$. Therefore,

$$\dim \text{range } T \leq \dim \text{null } S.$$

By Theorem 3.21, $\dim \text{null } S = 5 - \dim \text{range } S$, so

$$\dim \text{range } T \leq 5 - \dim \text{range } S \iff \dim \text{range } S + \dim \text{range } T \leq 5.$$

Since $\text{range } TS = T(\text{range } S) \subseteq \text{range } T$, we have $\dim \text{range } TS \leq \dim \text{range } T$. Also, $\dim \text{range } TS \leq \dim \text{range } S$. Summing these two inequalities gives:

$$2 \dim \text{range } TS \leq \dim \text{range } S + \dim \text{range } T \leq 5 \implies \dim \text{range } TS \leq \frac{5}{2}.$$

Since $\dim \text{range } TS$ must be an integer, $\dim \text{range } TS \leq 2$. □

- (25) Suppose W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\text{null } S \subseteq \text{null } T$ if and only if there exists $E \in \mathcal{L}(W)$ such that $T = ES$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose $\text{null } S \subseteq \text{null } T$. Let $w_1, \dots, w_m$ be a basis of $\text{range } S \subseteq W$. Since each $w_j \in \text{range } S$, there exist $v_1, \dots, v_m \in V$ such that $Sv_j = w_j$ for $j \in \{1, \dots, m\}$. Extend $w_1, \dots, w_m$ to a basis $w_1, \dots, w_m, w_{m+1}, \dots, w_n$ of W . By Theorem 3.4, define $E \in \mathcal{L}(W)$ by:

$$E(w_j) := T(v_j) \quad \text{for } 1 \leq j \leq m, \quad \text{and} \quad E(w_k) := 0 \quad \text{for } m+1 \leq k \leq n.$$

Now let $v \in V$. Then $Sv \in \text{range } S$, so $Sv = \sum_{j=1}^m c_j w_j = \sum_{j=1}^m c_j Sv_j = S\left(\sum_{j=1}^m c_j v_j\right)$. This implies $S\left(v - \sum_{j=1}^m c_j v_j\right) = 0$, so $v - \sum_{j=1}^m c_j v_j \in \text{null } S \subseteq \text{null } T$. Therefore,

$$T\left(v - \sum_{j=1}^m c_j v_j\right) = 0 \implies Tv = \sum_{j=1}^m c_j T(v_j).$$

On the other hand,

$$ES(v) = E\left(\sum_{j=1}^m c_j w_j\right) = \sum_{j=1}^m c_j E(w_j) = \sum_{j=1}^m c_j T(v_j) = Tv.$$

Thus, $T = ES$.

$\Leftarrow$ **DIRECTION:** Suppose there exists $E \in \mathcal{L}(W)$ such that $T = ES$. If $v \in \text{null } S$, then $Sv = 0$, so $Tv = E(Sv) = E(0) = 0$ by Theorem 3.10. Thus, $v \in \text{null } T$, proving $\text{null } S \subseteq \text{null } T$. □

- (26) Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\text{range } S \subseteq \text{range } T$ if and only if there exists $E \in \mathcal{L}(V)$ such that $S = TE$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose $\text{range } S \subseteq \text{range } T$. Let $v_1, \dots, v_n$ be a basis of V . For each $k \in \{1, \dots, n\}$, $Sv_k \in \text{range } S \subseteq \text{range } T$, so there exists $u_k \in V$ such that $T(u_k) = Sv_k$. By Theorem 3.4, there exists a unique linear map $E \in \mathcal{L}(V)$ such that $E(v_k) = u_k$ for all $k \in \{1, \dots, n\}$. Then for any $v = \sum_{k=1}^n c_k v_k \in V$,

$$TE(v) = T\left(\sum_{k=1}^n c_k E(v_k)\right) = \sum_{k=1}^n c_k T(u_k) = \sum_{k=1}^n c_k Sv_k = S(v).$$

Thus, $S = TE$.

$\Leftarrow$ **DIRECTION:** Suppose there exists $E \in \mathcal{L}(V)$ such that $S = TE$. Then for any $w \in \text{range } S$,

$w = Sv = T(Ev) \in \text{range } T$. Hence, $\text{range } S \subseteq \text{range } T$. □

- 27 Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \text{null } P \oplus \text{range } P$.

PROOF: For any $v \in V$, we can write:

$$v = (v - Pv) + Pv.$$

Applying P to the first term gives:

$$P(v - Pv) = Pv - P^2v = Pv - Pv = 0,$$

which shows that $v - Pv \in \text{null } P$. Since $Pv \in \text{range } P$, we have $V = \text{null } P + \text{range } P$.

To show that the sum is direct, suppose $u \in \text{null } P \cap \text{range } P$. Since $u \in \text{range } P$, there exists $w \in V$ such that $u = Pw$. Since $u \in \text{null } P$, we also have $Pu = 0$. Therefore,

$$u = Pw = P^2w = P(Pw) = Pu = 0.$$

Thus, $\text{null } P \cap \text{range } P = \{0\}$. By ??, $V = \text{null } P \oplus \text{range } P$. □

MATRICES (SECTION 3C)

REPRESENTING A LINEAR MAP BY A MATRIX (SUB-SECTION 3C1)

DEF 3.31 (*matrix of a linear map, $\mathcal{M}(T)$*):

Let $T \in \mathcal{L}(V, W)$, let $v_1, \dots, v_n$ be a basis of V , and let $w_1, \dots, w_m$ be a basis of W . The “matrix of T ” with respect to these bases is the m -by- n matrix $\mathcal{M}(T)$ whose entries $A_{j,k}$ are defined by

$$Tv_k = A_{1,k}w_1 + \dots + A_{m,k}w_m = \sum_{j=1}^m A_{j,k}w_j.$$

If the bases are not clear from the context, we write $\mathcal{M}(T, (v_1, \dots, v_n), (w_1, \dots, w_m))$. One can remember how to construct $\mathcal{M}(T)$ by writing the basis of W on the left and the basis of V on top:

$$\mathcal{M}(T) := \begin{array}{c} w_1 \\ \vdots \\ w_m \end{array} \begin{pmatrix} v_1 & \cdots & v_k & \cdots & v_n \\ & & A_{1,k} & & \\ & & \vdots & & \\ & & A_{m,k} & & \end{pmatrix}$$

DEF (*matrix notation for operators*):

For $T \in \mathcal{L}(V)$, we use the notation

$$\mathcal{M}(T, (v_1, \dots, v_n)) := \mathcal{M}(T, (v_1, \dots, v_n), (v_1, \dots, v_n)).$$

ADDITION AND SCALAR MULTIPLICATION OF MATRICES (SUB-SECTION 3C2)

THM 3.35 (*matrix of the sum of linear maps*):

$$\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T) \text{ for all } S, T \in \mathcal{L}(V, W).$$

THM 3.38 (*matrix of a scalar times a linear map*):

$$\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T) \text{ for all } \lambda \in \mathbb{F} \text{ and } T \in \mathcal{L}(V, W).$$

DEF 3.39 (*notation $\mathbb{F}^{m,n}$*):

$\mathbb{F}^{m,n}$ denotes the set of all m -by- n matrices with entries in $\mathbb{F}$, where $m, n \in \mathbb{N}$.

THM 3.40 (*dimension of $\mathbb{F}^{m,n}$*):

$\mathbb{F}^{m,n}$ is a vector space with $\dim \mathbb{F}^{m,n} = mn$.

PROOF: It is straightforward to verify that $\mathbb{F}^{m,n}$ is a vector space. The list of length mn that consists of distinct m -by- n matrices that have 0 in all entries except a 1 in one entry is a basis of $\mathbb{F}^{m,n}$. Hence, $\dim \mathbb{F}^{m,n} = mn$. $\square$

MATRIX MULTIPLICATION (SUB-SECTION 3C3)

Let $v_1, \dots, v_n$ be a basis of V , let $w_1, \dots, w_m$ be a basis of W , and let $u_1, \dots, u_p$ be a basis of U . Consider linear maps $T : U \rightarrow V$ and $S : V \rightarrow W$. The composition ST is a linear map from U to W . Is it the case that

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)?$$

Matrix multiplication is defined specifically so that this question has an affirmative answer. Let

$A := \mathcal{M}(S) \in \mathbb{F}^{m,n}$ and $B := \mathcal{M}(T) \in \mathbb{F}^{n,p}$. The matrices of S and T are given by:

$$A := \mathcal{M}(S) = \begin{matrix} & v_1 & \cdots & v_r & \cdots & v_n \\ \begin{matrix} w_1 \\ \vdots \\ w_j \\ \vdots \\ w_m \end{matrix} & \begin{pmatrix} & & & A_{1,r} & & \\ & & & \vdots & & \\ & & & A_{j,r} & & \\ & & & \vdots & & \\ & & & A_{m,r} & & \end{pmatrix} \end{matrix}, \quad B := \mathcal{M}(T) = \begin{matrix} & u_1 & \cdots & u_k & \cdots & u_p \\ \begin{matrix} v_1 \\ \vdots \\ v_r \\ \vdots \\ v_n \end{matrix} & \begin{pmatrix} & & & B_{1,k} & & \\ & & & \vdots & & \\ & & & B_{r,k} & & \\ & & & \vdots & & \\ & & & B_{n,k} & & \end{pmatrix} \end{matrix}$$

For $1 \leq k \leq p$, we have:

$$\begin{aligned} (ST)u_k &= S(Tu_k) = S\left(\sum_{r=1}^n B_{r,k}v_r\right) = \sum_{r=1}^n B_{r,k}Sv_r \\ &= \sum_{r=1}^n B_{r,k} \sum_{j=1}^m A_{j,r}w_j = \sum_{j=1}^m \left(\sum_{r=1}^n A_{j,r}B_{r,k}\right)w_j. \end{aligned}$$

Thus, $\mathcal{M}(ST)$ is the m -by- p matrix whose entry in row j , column k equals $\sum_{r=1}^n A_{j,r}B_{r,k}$:

$$\mathcal{M}(ST) = \begin{matrix} & u_1 & \cdots & u_k & \cdots & u_p \\ \begin{matrix} w_1 \\ \vdots \\ w_j \\ \vdots \\ w_m \end{matrix} & \begin{pmatrix} & & & \sum_{r=1}^n A_{1,r}B_{r,k} & & \\ & & & \vdots & & \\ & & & \sum_{r=1}^n A_{j,r}B_{r,k} & & \\ & & & \vdots & & \\ & & & \sum_{r=1}^n A_{m,r}B_{r,k} & & \end{pmatrix} \end{matrix}$$

DEF 3.41 (*matrix multiplication*):

Let $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$. Then $AB \in \mathbb{F}^{m,p}$ is the matrix whose entry in row j , column k is given by

$$(AB)_{j,k} := \sum_{r=1}^n A_{j,r}B_{r,k}.$$

The Octave code for calculating the entry (j,k) of the matrix product AB using the standard summation scheme is as follows:

```

1  function retval = standard_matrix_mult (A,B)
2      m = rows(A);
3      n = columns(A);
4      p = columns(B);
5      retval=zeros(m,p)
6      if (columns(A) != rows(B))
7          error('columns of A and rows of B dont match');
8      end
9      for (j=1:m)
10         for (k=1:p)
11             for (r=1:n)
12                 retval(j,k)=retval(j,k)+A(j,r)*B(r,k);
13             end
14         end
15     end
16 end
    
```

THM 3.43 (*matrix of product of linear maps*):

If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T).$$

PROOF: See the derivation at the beginning of this subsection. □

DEF 3.44 (notation: $A_{j,\bullet}, A_{\bullet,k}$):

Let $A \in \mathbb{F}^{m,n}$. Then:

- If $1 \leq j \leq m$, then $A_{j,\bullet}$ denotes the 1-by- n matrix consisting of row j of A .
- If $1 \leq k \leq n$, then $A_{\bullet,k}$ denotes the m -by-1 matrix consisting of column k of A .

THM 3.46 (entry of matrix product equals row times column):

If $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$, then for $1 \leq j \leq m$ and $1 \leq k \leq p$,

$$(AB)_{j,k} = A_{j,\bullet} B_{\bullet,k}.$$

The Octave code for multiplying rows and columns directly at entry (j,k) is as follows:

```

1  function retval = row_wise_matrix_mult (A,B)
2      m = rows(A);
3      n = columns(A);
4      p = columns(B);
5      retval=zeros(m,p)
6      if (n != rows(B))
7          error('columns of A and rows of B dont match');
8      end
9      for (j=1:m)
10         for (k=1:p)
11             retval(j,k)=A(j,:)*B(:,k);
12         end
13     end
14 end
    
```

THM 3.48 (column of matrix product equals matrix times column):

If $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$, then for $1 \leq k \leq p$,

$$(AB)_{\bullet,k} = AB_{\bullet,k}.$$

In other words, column k of AB equals A times column k of B . The Octave code is as follows:

```

1  function retval = matrix_times_columns_matrix_mult (A,B)
2      m = rows(A);
3      p = columns(B);
4      retval=zeros(m,p)
5      if (columns(A) != rows(B))
6          error('columns of A and rows of B dont match');
7      end
8      for (k=1:p)
9          retval(:,k)=A*B(:,k);
10     end
11 end
    
```

THM 3.50 (linear combination of columns):

Suppose $A \in \mathbb{F}^{m,n}$ and $b = (b_1, \dots, b_n)^t \in \mathbb{F}^{n,1}$. Then

$$Ab = b_1 A_{\bullet,1} + \dots + b_n A_{\bullet,n}.$$

See Theorem 3.54 for the definition of the transpose A^t .

THM 3.51 (matrix multiplication as linear combinations of columns or rows):

Let $C \in \mathbb{F}^{m,c}$ and $R \in \mathbb{F}^{c,n}$.

COLUMN-WISE: If $k \in \{1, \dots, n\}$, then column k of $CR \in \mathbb{F}^{m,n}$ is a linear combination of the columns of

C , with coefficients coming from column k of R ($R_{\bullet,k}$):

$CR =$

$$\left(\underbrace{\begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,1} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,1}}_{\text{1st column}} \cdots \underbrace{\begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,k} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,k}}_{\text{kth column}} \cdots \underbrace{\begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,n} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,n}}_{\text{last column}} \right)$$

The Octave code for column-wise multiplication is as follows:

```

1  function retval = column_wise_matrix_mult (C,R)
2      m = rows(C);
3      c = columns(C);
4      n = columns (R);
5      retval=zeros(m,n);
6      if (c != rows(R))
7          error('columns of C and rows of R dont match');
8      end
9      for (k=1:n)
10         for (i=1:c)
11             retval(:,k)=retval(:,k)+C(:,i)*R(i,k);
12         end
13     end
14 end
    
```

Row-wise: If $j \in \{1, \dots, m\}$, then row j of CR is a linear combination of the rows of R , with coefficients coming from row j of C ($C_{j,\bullet}$):

$$CR = \begin{pmatrix} C_{1,1}(R_{1,1}, \dots, R_{1,n}) + C_{1,2}(R_{2,1}, \dots, R_{2,n}) + \cdots + C_{1,c}(R_{c,1}, \dots, R_{c,n}) \\ C_{2,1}(R_{1,1}, \dots, R_{1,n}) + C_{2,2}(R_{2,1}, \dots, R_{2,n}) + \cdots + C_{2,c}(R_{c,1}, \dots, R_{c,n}) \\ \vdots \\ C_{j,1}(R_{1,1}, \dots, R_{1,n}) + C_{j,2}(R_{2,1}, \dots, R_{2,n}) + \cdots + C_{j,c}(R_{c,1}, \dots, R_{c,n}) \\ \vdots \\ C_{m,1}(R_{1,1}, \dots, R_{1,n}) + C_{m,2}(R_{2,1}, \dots, R_{2,n}) + \cdots + C_{m,c}(R_{c,1}, \dots, R_{c,n}) \end{pmatrix}$$

The Octave code for row-wise multiplication is as follows:

```

1  function retval = row_wise_matrix_mult (C,R)
2      m = rows(C);
3      c = columns(C);
4      n = columns (R);
5      retval=zeros(m,n);
6      if (c != rows(R))
7          error('columns of C and rows of R dont match');
8      end
9      for (j=1:m)
10         for (i=1:c)
11             retval(j,:) = retval(j,:) + C(j,i) * R(i,:);
12         end
13     end
14 end
    
```

COLUMN-ROW FACTORIZATION AND RANK OF A MATRIX (SUB-SECTION 3C4)

DEF 3.52 (*column rank, row rank*):

For $A \in \mathbb{F}^{m,n}$, the “column rank” of A is the dimension of the span of the columns of A in $\mathbb{F}^{m,1}$. The

“row rank” of A is the dimension of the span of the rows of A in $\mathbb{F}^{1,n}$.

DEF 3.54 (*transpose, A^t*):

The “transpose” of a matrix $A \in \mathbb{F}^{m,n}$, denoted by $A^t \in \mathbb{F}^{n,m}$, is obtained by interchanging rows and columns. Its entries are given by

$$(A^t)_{k,j} := A_{j,k} \quad \text{for } 1 \leq k \leq n \text{ and } 1 \leq j \leq m.$$

THM 3.56 (*column-row factorization*):

Suppose $A \in \mathbb{F}^{m,n}$ has column rank $c \geq 1$. Then there exist $C \in \mathbb{F}^{m,c}$ and $R \in \mathbb{F}^{c,n}$ such that

$$A = CR.$$

THM 3.57 (*column rank and row rank*):

If $A \in \mathbb{F}^{m,n}$, then the column rank of A equals the row rank of A .

DEF 3.58 (*rank*):

The “rank” of a matrix $A \in \mathbb{F}^{m,n}$ is the column rank of A .

EXERCISES ABOUT MATRICES 3C

INVERTIBILITY AND ISOMORPHISMS (SECTION 3D)

INVERTIBLE LINEAR MAPS (SUB-SECTION 3D1)

DEF 3.59 (*invertible, inverse*):

A linear map $T \in \mathcal{L}(V, W)$ is called “invertible” if there exists a linear map $S \in \mathcal{L}(W, V)$ such that $ST = I_V$ and $TS = I_W$. Such a linear map S is called an “inverse” of T .

THM 3.60 (*inverse is unique*):

An invertible linear map has a unique inverse.

PROOF: Suppose $T \in \mathcal{L}(V, W)$ is invertible and $S_1, S_2 \in \mathcal{L}(W, V)$ are inverses of T . Then

$$S_1 = S_1 I_W = S_1 (TS_2) = (S_1 T) S_2 = I_V S_2 = S_2.$$

Therefore, $S_1 = S_2$. □

DEF 3.61 (*notation: T^{-1}*):

If $T \in \mathcal{L}(V, W)$ is invertible, its unique inverse is denoted by $T^{-1} \in \mathcal{L}(W, V)$. We have:

$$T^{-1}T = I_V \quad \text{and} \quad TT^{-1} = I_W.$$

THM 3.63 (*invertibility is equivalent to injectivity and surjectivity*):

A linear map is invertible if and only if it is injective and surjective.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose $T \in \mathcal{L}(V, W)$ is invertible, with inverse T^{-1} . If $u, v \in V$ satisfy $Tu = Tv$, applying T^{-1} to both sides gives $u = v$, so T is injective. For each $w \in W$, we have $w = T(T^{-1}w)$, so $w \in \text{range } T$, which proves T is surjective.

$\Leftarrow$ **DIRECTION:** Suppose T is injective and surjective. For each $w \in W$, surjectivity implies there is at least one $v \in V$ with $Tv = w$, and injectivity implies this v is unique. Define $S : W \rightarrow V$ by setting $Sw := v$. Then $S(Tw) = v$ and $T(Sw) = w$, and it is straightforward to verify that S is linear. Thus, T is invertible. □

THM 3.65 (*injectivity is equivalent to surjectivity for spaces of equal dimension*):

Suppose V and W are finite-dimensional vector spaces with $\dim V = \dim W$, and let $T \in \mathcal{L}(V, W)$. Then the following are equivalent:

- (a) T is invertible;
- (b) T is injective;
- (c) T is surjective.

PROOF: By the Fundamental Theorem of Linear Maps (Theorem 3.21), we have:

$$\dim V = \dim \text{null } T + \dim \text{range } T. \tag{3.3}$$

If T is injective, then $\text{null } T = \{0\}$, so $\dim \text{null } T = 0$. Thus,

$$\dim \text{range } T = \dim V = \dim W.$$

Since $\text{range } T$ is a subspace of W with $\dim \text{range } T = \dim W$, Theorem 2.39 implies $\text{range } T = W$, so T is surjective.

Conversely, if T is surjective, then $\text{range } T = W$, so $\dim \text{range } T = \dim W = \dim V$. Equation (3.3) then gives:

$$\dim \text{null } T = \dim V - \dim \text{range } T = \dim V - \dim V = 0,$$

which implies $\text{null } T = \{0\}$, and so T is injective by Theorem 3.15.

Finally, since (b) and (c) are equivalent, either condition implies that T is both injective and surjective, which is equivalent to T being invertible. □

THM 3.68 ($ST = I \iff TS = I$ on vector spaces of the same dimension):

Suppose V and W are finite-dimensional vector spaces of the same dimension. Let $S \in \mathcal{L}(W, V)$ and $T \in \mathcal{L}(V, W)$. Then $ST = I_V$ if and only if $TS = I_W$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose $ST = I_V$. If $v \in V$ satisfies $Tv = 0$, then

$$v = I_V v = (ST)v = S(Tv) = S(0) = 0.$$

Thus, $\text{null } T = \{0\}$, and therefore, T is injective by Theorem 3.15. Because $\dim V = \dim W$, T is invertible by Theorem 3.65. Multiplying both sides of $ST = I_V$ on the right by T^{-1} gives $S = T^{-1}$. Hence, $TS = TT^{-1} = I_W$, as desired.

$\Leftarrow$ **DIRECTION:** Reversing the roles of S and T (and of V and W) in the forward implication shows that $TS = I_W$ implies $ST = I_V$. $\square$

ISOMORPHIC VECTOR SPACES (SUB-SECTION 3D2)

DEF 3.69 (isomorphism, isomorphic):

An “isomorphism” is an invertible linear map. Two vector spaces are called “isomorphic” if there exists an isomorphism from one vector space onto the other.

THM 3.70 (dimension determines whether vector spaces are isomorphic):

Two finite-dimensional vector spaces V and W over $\mathbb{F}$ are isomorphic if and only if $\dim V = \dim W$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose V and W are isomorphic, and let $T \in \mathcal{L}(V, W)$ be an isomorphism.

Since T is injective, $\dim \text{null } T = 0$, and since T is surjective, $\text{range } T = W$. By Theorem 3.21,

$$\dim V = \dim \text{null } T + \dim \text{range } T = 0 + \dim W = \dim W.$$

$\Leftarrow$ **DIRECTION:** Suppose $\dim V = \dim W = n$. Let $v_1, \dots, v_n$ be a basis of V , and let $w_1, \dots, w_n$ be a basis of W . By Theorem 3.4, there exists a unique linear map $T \in \mathcal{L}(V, W)$ such that $Tv_k = w_k$ for each $k \in \{1, \dots, n\}$. Since T maps a basis to a basis, T is injective and surjective, and thus an isomorphism. $\square$

THM 3.71 ($\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$ are isomorphic):

Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . Then the matrix mapping $\mathcal{M} : \mathcal{L}(V, W) \rightarrow \mathbb{F}^{m,n}$ is an isomorphism.

PROOF: Linearity of $\mathcal{M}$ follows from $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$ and $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$. If $\mathcal{M}(T) = 0$, then $Tv_k = 0$ for all k , which implies $T = 0$, so $\mathcal{M}$ is injective. Given any $A \in \mathbb{F}^{m,n}$, Theorem 3.4 guarantees the existence of a linear map $T \in \mathcal{L}(V, W)$ with $Tv_k = \sum_{j=1}^m A_{j,k} w_j$, so $\mathcal{M}(T) = A$, proving surjectivity. $\square$

THM 3.72 ($\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$):

Suppose V and W are finite-dimensional. Then $\mathcal{L}(V, W)$ is finite-dimensional and

$$\dim \mathcal{L}(V, W) = (\dim V)(\dim W).$$

PROOF: Let $n = \dim V$ and $m = \dim W$. By Theorem 3.71, $\mathcal{L}(V, W)$ is isomorphic to $\mathbb{F}^{m,n}$. By Theorem 3.70 and Theorem 3.40,

$$\dim \mathcal{L}(V, W) = \dim \mathbb{F}^{m,n} = mn = (\dim V)(\dim W).$$

$\square$

LINEAR MAPS THOUGHT OF AS MATRIX MULTIPLICATION (SUB-SECTION 3D3)

DEF 3.73 (*matrix of a vector, $\mathcal{M}(u)$*):

Let $u \in V$ and let $v_1, \dots, v_n$ be a basis of V such that

$$u = b_1 v_1 + \dots + b_n v_n.$$

We define the “matrix of u ” with respect to this basis by

$$\mathcal{M}(u) := \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

THM 3.75 ($\mathcal{M}(T)_{\bullet,k} = \mathcal{M}(Tv_k)$):

Suppose $T \in \mathcal{L}(V, W)$, let $v_1, \dots, v_n$ be a basis of V , and let $w_1, \dots, w_m$ be a basis of W . For $1 \leq k \leq n$, the k^{th} column of $\mathcal{M}(T)$ equals $\mathcal{M}(Tv_k)$:

$$\mathcal{M}(T)_{\bullet,k} = \mathcal{M}(Tv_k).$$

THM 3.76 (*linear maps act like matrix multiplication*):

Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. If $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W , then

$$\mathcal{M}(Tv) = \mathcal{M}(T)\mathcal{M}(v).$$

THM 3.78 (*dimension of range T and rank of $\mathcal{M}(T)$*):

Suppose V and W are finite-dimensional, and let $T \in \mathcal{L}(V, W)$. Then

$$\dim \text{range } T = \text{rank } \mathcal{M}(T).$$

CHANGE OF BASIS (SUB-SECTION 3D4)

DEF 3.80 (*invertible matrix, inverse, A^{-1}*):

A square matrix $A \in \mathbb{F}^{n,n}$ is called “invertible” if there exists a matrix $B \in \mathbb{F}^{n,n}$ such that

$$AB = BA = I_n.$$

In this case, B is the unique inverse of A , denoted by $A^{-1} := B$.

THM 3.82 (*matrix of identity operator with respect to two bases*):

Suppose $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Then the matrices

$$\mathcal{M}(I_V, (u_1, \dots, u_n), (v_1, \dots, v_n)) \quad \text{and} \quad \mathcal{M}(I_V, (v_1, \dots, v_n), (u_1, \dots, u_n))$$

are invertible, and each is the inverse of the other.

PROOF: By Theorem 3.43, we have:

$$\mathcal{M}(I_V, (v_1, \dots, v_n), (u_1, \dots, u_n)) \cdot \mathcal{M}(I_V, (u_1, \dots, u_n), (v_1, \dots, v_n)) = \mathcal{M}(I_V, (u_1, \dots, u_n), (u_1, \dots, u_n)) = I_n.$$

Interchanging the roles of the u ’s and v ’s gives the product in reverse order, which also equals I_n . □

THM 3.84 (*change-of-basis formula*):

Let $T \in \mathcal{L}(V)$. Suppose $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Let

$$A := \mathcal{M}(T, (u_1, \dots, u_n)),$$

$$B := \mathcal{M}(T, (v_1, \dots, v_n)), \quad \text{and}$$

$$C := \mathcal{M}(I_V, (u_1, \dots, u_n), (v_1, \dots, v_n)).$$

Then

$$A = C^{-1}BC.$$

THM 3.86:

If $v_1, \dots, v_n$ is a basis of V and $T \in \mathcal{L}(V)$ is invertible, then

$$\mathcal{M}(T^{-1}) = (\mathcal{M}(T))^{-1},$$

where both matrices are with respect to the basis $v_1, \dots, v_n$.

EXERCISES ABOUT INVERTIBILITY AND ISOMORPHISMS 3D

- ① Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that T^{-1} is invertible and $(T^{-1})^{-1} = T$.

PROOF: T^{-1} is the unique element in $\mathcal{L}(W, V)$ such that

$$T^{-1}T = I_V \text{ and } TT^{-1} = I_W.$$

But this makes T the unique element in $\mathcal{L}(V, W)$ such that

$$T(T^{-1}) = I_W \text{ and } T^{-1}T = I_V.$$

This makes T the inverse of T^{-1} . Since the inverse is unique, we conclude that $(T^{-1})^{-1} = T$. $\square$

- ② Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

PROOF: Let $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ be both invertible linear maps. We compute:

$$(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SI_VS^{-1} = SS^{-1} = I_W$$

$$(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}I_VT = T^{-1}T = I_U$$

Therefore, (ST) is invertible and its inverse is $(ST)^{-1} = T^{-1}S^{-1}$. $\square$

- ③ Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent:

- (a) T is invertible.
- (b) $Tv_1, \dots, Tv_n$ is a basis of V for every basis $v_1, \dots, v_n$ of V .
- (c) $Tv_1, \dots, Tv_n$ is a basis of V for some basis $v_1, \dots, v_n$ of V .

PROOF: For all proof-steps, let $T \in \mathcal{L}(V)$ and V be a finite-dimensional vector space. In each step of the proof, we show one equivalence listed above.

STEP 1: Clearly, (b) implies (c).

STEP 2: We start by showing that (c) implies (a). Actually, the proof that (b) implies (a) would look the same. Suppose $Tv_1, \dots, Tv_n$ is a basis of V for some basis $v_1, \dots, v_n$. Since $\dim V = n$, we have $\text{range } T = V$. It follows that $Tv_1, \dots, Tv_n$ is a basis of $\text{range } T$. So T is a surjective map on V by the definition 3.19 of surjectivity. Using theorem 3.65, we conclude that T is invertible. It states that for a linear map from a vector space to another one of the same dimension, invertibility, injectivity and surjectivity are equivalent.

STEP 3: Now we want to show that (c) implies (b). So let $Tv_1, \dots, Tv_n$ be a basis of V for some basis $v_1, \dots, v_n$ of V . Let $u_1, \dots, u_n$ be another basis of V . We already know that (c) $\implies$ (a), so T is injective. This means, that $Tu_1, \dots, Tu_n$ is a basis of V as well. Since $u_1, \dots, u_n$ was arbitrary, we have shown that (c) $\implies$ (b).

STEP 4: Lastly, we want to show that (a) implies (c). Suppose $T \in \mathcal{L}(V)$ such that T is invertible. Let $v_1, \dots, v_n$ be a basis of V . Let $a_1, \dots, a_n \in \mathbb{F}$ such that

$$0 = a_1Tv_1 + \dots + a_nTv_n$$

Hence,

$$\begin{aligned} T^{-1}(0) &= T^{-1}(a_1Tv_1 + \dots + a_nTv_n) \iff \\ 0 &= a_1T^{-1}Tv_1 + \dots + a_nT^{-1}Tv_n \iff \\ 0 &= a_1v_1 + \dots + a_nv_n. \end{aligned}$$

Therefore, $0 = a_1 = \cdots = a_n$, because all the v 's are linearly independent. We conclude that $Tv_1, \dots, Tv_n$ is a linearly independent list of length $n = \dim V$ and hence a basis of V . Thus, (a) implies (c).

Overall, we have shown that (b) $\implies$ (c), (c) $\implies$ (b), (c) $\implies$ (a), and (a) $\implies$ (c). So all three statements are equivalent. $\square$

- ⑤ Suppose V is finite-dimensional, U is a subspace of V , and $S \in \mathcal{L}(U, V)$. Prove that there exists an invertible linear map $T \in \mathcal{L}(U, V)$ such that $Tu = Su \quad \forall u \in U \iff S$ is injective.

PROOF: Suppose V is finite-dimensional, U is a subspace of V and $S \in \mathcal{L}(U, V)$.

$\implies$ **DIRECTION:** Suppose for every $u \in U$ we have that $Tu = Su$. Let $u_1, u_2 \in U$ such that $S(u_1) = S(u_2)$. This implies that

$$\begin{aligned} T(u_1) &= T(u_2) \\ T^{-1}T(u_1) &= T^{-1}T(u_2) \\ u_1 &= u_2 \end{aligned}$$

Hence, S is injective.

$\Leftarrow$ **DIRECTION:** Suppose S is injective. Let $u_1, \dots, u_m$ be a basis of U and $Su_1, \dots, Su_m$ be a basis of range S . We extend

$$u_1, \dots, u_m, v_{m+1}, \dots, v_n \tag{3.4}$$

to a basis of V , and

$$Su_1, \dots, Su_m, r_{m+1}, \dots, r_n \tag{3.5}$$

to a basis of V as well. We define $T \in \mathcal{L}(V)$ for each $v \in V$ as follows:

$$\begin{aligned} T(v) &= T(c_1u_1 + \cdots + c_mu_m + c_{m+1}v_{m+1} + \cdots + c_nr_n) \\ &:= c_1Su_1 + \cdots + c_mSu_m + c_{m+1}r_{m+1} + \cdots + c_nr_n. \end{aligned}$$

Clearly, $\forall u \in U : T(u) = S(u)$. Now we have to show that T is injective and surjective. If $v_1, v_2 \in V$ such that

$$y_1 = T(v_1) = T(v_2) = y_2$$

All the c coefficients of y_1 and y_2 , which are defined over the basis (3.5) match the c coefficients of v_1 and v_2 , which are defined over the basis (3.4). So these representations are unique and therefore $v_1 = v_2$. We conclude that T is injective. To prove that T is surjective, we have to show that for any given $y \in V$ there is a $v \in V$ such that $T(v) = y$. So let

$$y := c_1Su_1 + \cdots + c_mSu_m + c_{m+1}r_{m+1} + \cdots + c_nr_n.$$

Then there is exactly one $v \in V$ such that $T(v) = y$, namely

$$v := c_1u_1 + \cdots + c_mu_m + c_{m+1}v_{m+1} + \cdots + c_nv_n.$$

For this choice of y and v , we have $T(v) = y$. Hence, T is injective and surjective. This makes T invertible. $\square$

PRODUCTS AND QUOTIENTS OF VECTOR SPACES (SECTION 3E)

PRODUCTS OF VECTOR SPACES (SUB-SECTION 3E1)

DEF 3.87 (*product of vector spaces*):

The product, addition, and scalar multiplication of a list of vector spaces $V_1, \dots, V_m$ are defined as follows:

$$\begin{aligned} V_1 \times \dots \times V_m &:= \{(v_1, \dots, v_m) \mid v_1 \in V_1, \dots, v_m \in V_m\}, \\ (u_1, \dots, u_m) + (v_1, \dots, v_m) &:= (u_1 + v_1, \dots, u_m + v_m), \\ \lambda(v_1, \dots, v_m) &:= (\lambda v_1, \dots, \lambda v_m). \end{aligned}$$

THM 3.89 (*product of vector spaces is a vector space*):

$V_1 \times \dots \times V_m$ together with addition and scalar multiplication is a vector space over $\mathbb{F}$.

THM 3.92 (*dimension of a product is the sum of dimensions*):

Suppose $V_1, \dots, V_m$ are finite-dimensional vector spaces. Then $V_1 \times \dots \times V_m$ is finite-dimensional and $\dim(V_1 \times \dots \times V_m) = \dim V_1 + \dots + \dim V_m$.

PROOF: Choose a basis for each V_k . For each basis vector of each V_k , consider the element of $V_1 \times \dots \times V_m$ that equals that basis vector in the k^{th} slot and 0 in all other slots. The list of all such vectors is linearly independent and spans $V_1 \times \dots \times V_m$. Thus, it is a basis of $V_1 \times \dots \times V_m$, and the length of this basis is $\dim V_1 + \dots + \dim V_m$. $\square$

THM 3.93 (*products and direct sums*):

Let $V_1, \dots, V_m$ be subspaces of V , and define the linear map $\Gamma : V_1 \times \dots \times V_m \rightarrow V_1 + \dots + V_m$ by

$$\Gamma(v_1, \dots, v_m) := v_1 + \dots + v_m.$$

Then $V_1 + \dots + V_m$ is a direct sum if and only if Γ is injective (or equivalently, an isomorphism).

PROOF: By Theorem 3.15, Γ is injective if and only if $\text{null } \Gamma = \{0\}$. This means that the only way to write 0 as a sum $v_1 + \dots + v_m$, with each $v_k \in V_k$, is $v_1 = \dots = v_m = 0$. By Theorem 1.45, this is equivalent to $V_1 + \dots + V_m$ being a direct sum. $\square$

THM 3.94 (*a sum is a direct sum if and only if dimensions add up*):

Suppose V is finite-dimensional and $V_1, \dots, V_m$ are subspaces of V . Then $V_1 + \dots + V_m$ is a direct sum if and only if

$$\dim(V_1 + \dots + V_m) = \dim V_1 + \dots + \dim V_m.$$

PROOF: The map $\Gamma : V_1 \times \dots \times V_m \rightarrow V_1 + \dots + V_m$ is surjective. By Theorem 3.21,

$$\begin{aligned} \dim(V_1 \times \dots \times V_m) &= \dim \text{null } \Gamma + \dim \text{range } \Gamma \\ &= \dim \text{null } \Gamma + \dim(V_1 + \dots + V_m). \end{aligned}$$

By Theorem 3.92, $\dim(V_1 \times \dots \times V_m) = \dim V_1 + \dots + \dim V_m$. Thus, $\dim(V_1 + \dots + V_m) = \dim V_1 + \dots + \dim V_m$ if and only if $\dim \text{null } \Gamma = 0$, which holds if and only if Γ is injective. By Theorem 3.93, this occurs if and only if $V_1 + \dots + V_m$ is a direct sum. $\square$

QUOTIENT SPACES (SUB-SECTION 3E2)

DEF 3.95 (*notation: $v + U$*):

For $v \in V$ and a subspace U of V , define

$$v + U := \{v + u \mid u \in U\}.$$

DEF 3.97 (*translate*):

The set $v + U$ is called a “translate” of U , or an “affine subset” of V .

DEF 3.99 (*quotient space, V/U*):

Suppose U is a subspace of V . The “quotient space” V/U is the set of all translates of U :

$$V/U := \{v + U \mid v \in V\}.$$

EXAMPLE 3.100:

If $U = \{(x, 2x) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}$, then $\mathbb{R}^2/U$ is the set of all lines in $\mathbb{R}^2$ with slope 2. If U is a plane through the origin in $\mathbb{R}^3$, then $\mathbb{R}^3/U$ is the set of all planes parallel to U .

THM 3.101 (*two translates of a subspace are equal or disjoint*):

Let U be a subspace of V and let $v, w \in V$. Then the following are equivalent:

- (a) $v - w \in U$;
- (b) $v + U = w + U$;
- (c) $(v + U) \cap (w + U) \neq \emptyset$.

PROOF: **STEP 1:** First suppose (a) holds, so $v - w \in U$. If $u \in U$, then

$$v + u = w + (v - w + u) \in w + U.$$

Thus, $v + U \subseteq w + U$. Similarly, $w - v = -(v - w) \in U$, so $w + U \subseteq v + U$. Hence, $v + U = w + U$, showing that (a) implies (b).

STEP 2: If $v + U = w + U$, then $(v + U) \cap (w + U) = v + U \neq \emptyset$, so (b) implies (c).

STEP 3: Now suppose (c) holds, so there exists $x \in (v + U) \cap (w + U)$. Then $x = v + u_1 = w + u_2$ for some $u_1, u_2 \in U$. Rearranging gives $v - w = u_2 - u_1 \in U$, which proves (a). $\square$

DEF 3.102 (*addition and scalar multiplication on V/U*):

Suppose U is a subspace of V . Addition and scalar multiplication are defined on V/U by

$$\begin{aligned} (v + U) + (w + U) &:= (v + w) + U, \\ \lambda(v + U) &:= (\lambda v) + U \end{aligned}$$

for all $v, w \in V$ and all $\lambda \in \mathbb{F}$.

THM 3.103 (*quotient space is a vector space*):

V/U with the operations defined above is a vector space over $\mathbb{F}$, with additive identity $0 + U = U$ and additive inverse $-(v + U) = (-v) + U$.

DEF 3.104 (*quotient map, π*):

Suppose U is a subspace of V . The “quotient map” $\pi : V \rightarrow V/U$ is the linear map defined by

$$\pi(v) := v + U \quad \text{for all } v \in V.$$

THM 3.105 (*dimension of a quotient space*):

Suppose V is finite-dimensional and U is a subspace of V . Then

$$\dim(V/U) = \dim V - \dim U.$$

PROOF: The quotient map $\pi : V \rightarrow V/U$ is surjective, with $\text{range } \pi = V/U$. Its null space is $\text{null } \pi = \{v \in V \mid v + U = U\} = U$. By the Fundamental Theorem of Linear Maps (Theorem 3.21),

$$\dim V = \dim \text{null } \pi + \dim \text{range } \pi = \dim U + \dim(V/U).$$

Subtracting $\dim U$ from both sides gives the desired formula. $\square$

DEF 3.106 (*induced map on quotient, $\tilde{T}$*):

Suppose $T \in \mathcal{L}(V, W)$. Define $\tilde{T} : V/(\text{null } T) \rightarrow W$ by

$$\tilde{T}(v + \text{null } T) := Tv.$$

THM 3.107 (*properties of $\tilde{T}$*):

 Suppose $T \in \mathcal{L}(V, W)$. Then:

- (a) $\tilde{T} \circ \pi = T$, where $\pi : V \rightarrow V/(\text{null } T)$ is the quotient map;
- (b) $\tilde{T}$ is injective;
- (c) $\text{range } \tilde{T} = \text{range } T$;
- (d) $V/(\text{null } T)$ and $\text{range } T$ are isomorphic vector spaces.

EXERCISES ABOUT PRODUCTS AND QUOTIENTS OF VECTOR SPACES 3E

- ⑥ Suppose that $v, x \in V$ and that U, W are subspaces of V such that $v + U = x + W$. Prove that $U = W$. **PROOF:** Since $v \in v + U$ and $v + U = x + W$, it follows that $v \in x + W$. Hence, there is $w_0 \in W$ with $v = x + w_0$, so $(x - v) = -w_0 \in W$. Now,

$$U = (-v) + (v + U) = (-v) + (x + W) = (x - v) + W = W,$$

where the last equality uses $x - v \in W$ and the fact that translating a subspace by one of its own elements leaves it unchanged (i.e., $(x - v) + W = W$). Hence, $U = W$. $\square$

- ⑩ Suppose $A_1 = v + U_1$ and $A_2 = w + U_2$ for some $v, w \in V$ and some subspaces U_1, U_2 of V . Prove that the intersection $A_1 \cap A_2$ is either a translate of some subspace of V or is the empty set.

PROOF: By the definition of intersection, we get

$$A_1 \cap A_2 = \{x \in V : x = v + u_1 \text{ and } x = w + u_2, u_1 \in U_1, u_2 \in U_2\}.$$

The condition $v + u_1 = w + u_2$ is equivalent to

$$v - w = u_2 - u_1, u_1 \in U_1, u_2 \in U_2 \iff v - w \in U_1 + U_2.$$

If $v - w \notin U_1 + U_2$, there is no solution for u_1 and u_2 and $A_1 \cap A_2 = \emptyset$. If a solution exists, put

$$\begin{aligned} r^0 &:= v + u_1^0 \\ &:= w + u_2^0, \end{aligned}$$

where $r^0 \in A_1 \cap A_2$ and for appropriate $u_1^0 \in U_1, u_2^0 \in U_2$ (i.e., $v - w = u_2^0 - u_1^0$). Define

$$S := \{r^0 + u : u \in U_1 \cap U_2\} = r^0 + (U_1 \cap U_2).$$

We want to show that $A_1 \cap A_2 = S$. Therefore, take any $x \in A_1 \cap A_2$, so that $x = v + u_1$ and $x = w + u_2$ with $u_1 \in U_1, u_2 \in U_2$. Now, it follows that

$$x - r^0 = (v + u_1) - (v + u_1^0) = u_1 - u_1^0 \in U_1, \text{ and}$$

$$x - r^0 = (w + u_2) - (w + u_2^0) = u_2 - u_2^0 \in U_2,$$

and therefore, $x - r^0 \in U_1 \cap U_2$. Since $x = r^0 + (x - r^0)$, it is immediate that

$$x \in r^0 + (U_1 \cap U_2).$$

Hence, $A_1 \cap A_2 \subseteq S$. Conversely, pick any $u \in U_1 \cap U_2$, so that $r^0 + u \in S$. Then

$$r^0 + u = (v + u_1^0) + u = v + (u_1^0 + u) \in v + U_1 = A_1, \text{ and}$$

$$r^0 + u = (w + u_2^0) + u = w + (u_2^0 + u) \in w + U_2 = A_2.$$

Thus, $S \subseteq A_1 \cap A_2$. Therefore, we have shown that

$$A_1 \cap A_2 = \begin{cases} S, & \text{if } v - w \in U_1 + U_2, \\ \emptyset, & \text{otherwise.} \end{cases}$$

$\square$

DUALITY (SECTION 3F)**DUAL SPACE AND DUAL MAP** (SUB-SECTION 3F1)**DEF 3.108** (*linear functional*):

A “linear functional” on V is a linear map from V to $\mathbb{F}$, that is, an element of $\mathcal{L}(V, \mathbb{F})$.

EXAMPLE 3.109:

Examples of linear functionals:

$$\begin{aligned}\phi : \mathbb{R}^3 &\rightarrow \mathbb{R}, & \phi(x, y, z) &:= 4x - 5y - 2z, \\ \phi : \mathbb{F}^n &\rightarrow \mathbb{F}, & \phi(x_1, \dots, x_n) &:= c_1 x_1 + \dots + c_n x_n, \\ \phi : \mathcal{P}(\mathbb{R}) &\rightarrow \mathbb{R}, & \phi(p) &:= 3p''(5) + 7p(4), \\ \phi : \mathcal{P}(\mathbb{R}) &\rightarrow \mathbb{R}, & \phi(p) &:= \int_0^1 p(x) dx.\end{aligned}$$

DEF 3.110 (*dual space, V'*):

The “dual space” of V , denoted by V' , is the vector space of all linear functionals on V :

$$V' := \mathcal{L}(V, \mathbb{F}).$$

THM 3.111 ($\dim V' = \dim V$):

Suppose V is finite-dimensional. Then V' is also finite-dimensional and

$$\dim V' = \dim V.$$

PROOF: By Theorem 3.72, we have:

$$\dim V' = \dim \mathcal{L}(V, \mathbb{F}) = (\dim V)(\dim \mathbb{F}) = (\dim V)(1) = \dim V.$$

□

DEF 3.112 (*dual basis*):

If $v_1, \dots, v_n$ is a basis of V , the “dual basis” of $v_1, \dots, v_n$ is the list $\varphi_1, \dots, \varphi_n \in V'$ defined by

$$\varphi_j(v_k) := \delta_{j,k} = \begin{cases} 1, & \text{if } k = j, \\ 0, & \text{if } k \neq j. \end{cases}$$

THM 3.114 (*dual basis expansion*):

Let $v_1, \dots, v_n$ be a basis of V with dual basis $\varphi_1, \dots, \varphi_n$. For every $v \in V$,

$$v = \varphi_1(v)v_1 + \dots + \varphi_n(v)v_n.$$

PROOF: Let $v \in V$ be written as $v = c_1 v_1 + \dots + c_n v_n$. Applying φ_j to both sides gives $\varphi_j(v) = c_j$ for each $j \in \{1, \dots, n\}$. Hence, $v = \varphi_1(v)v_1 + \dots + \varphi_n(v)v_n$. □

THM 3.116 (*basis of the dual space*):

The dual basis of a basis of V is a basis of the dual space V' .

PROOF: Suppose $v_1, \dots, v_n$ is a basis of V , and let $\varphi_1, \dots, \varphi_n$ denote its dual basis. Suppose scalars $a_1, \dots, a_n \in \mathbb{F}$ satisfy:

$$a_1 \varphi_1 + \dots + a_n \varphi_n = 0.$$

Evaluating both sides at v_k yields:

$$(a_1 \varphi_1 + \dots + a_n \varphi_n)(v_k) = a_k = 0 \quad \text{for each } k \in \{1, \dots, n\}.$$

Thus, $a_1 = \dots = a_n = 0$, so $\varphi_1, \dots, \varphi_n$ is linearly independent in V' . Since its length n equals $\dim V'$ by Theorem 3.111, Theorem 2.38 implies that $\varphi_1, \dots, \varphi_n$ is a basis of V' . □

DEF 3.118 (*dual map, T'*):

The “dual map” of $T \in \mathcal{L}(V, W)$ is the linear map $T' \in \mathcal{L}(W', V')$ defined by

$$T'(\varphi) := \varphi \circ T \quad \text{for each } \varphi \in W'.$$

THM 3.120 (*algebraic properties of dual maps*):

Let $T \in \mathcal{L}(V, W)$. Then:

- (a) $(S + T)' = S' + T'$ for all $S \in \mathcal{L}(V, W)$;
- (b) $(\lambda T)' = \lambda T'$ for all $\lambda \in \mathbb{F}$;
- (c) $(ST)' = T'S'$ for all $S \in \mathcal{L}(W, U)$.

DUAL SPACE AND RANGE OF DUAL OF LINEAR MAP (SUB-SECTION 3F2)

DEF 3.121 (*annihilator, U^0*):

For a subset $U \subseteq V$, the “annihilator” of U , denoted by U^0 , is defined by

$$U^0 := \{\varphi \in V' \mid \varphi(u) = 0 \text{ for all } u \in U\}.$$

THM 3.124 (*the annihilator is a subspace*):

If $U \subseteq V$, then U^0 is a subspace of V' .

THM 3.125 (*dimension of the annihilator*):

If U is a subspace of a finite-dimensional vector space V , then

$$\dim U^0 = \dim V - \dim U.$$

THM 3.127 (*condition for the annihilator to equal $\{0\}$ or the whole space*):

If V is finite-dimensional and U is a subspace of V , then:

- (a) $U^0 = \{0\} \iff U = V$;
- (b) $U^0 = V' \iff U = \{0\}$.

THM 3.128 (*null space of T'*):

If V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$, then:

- (a) $\text{null } T' = (\text{range } T)^0$;
- (b) $\dim \text{null } T' = \dim \text{null } T + \dim W - \dim V$.

THM 3.129 (*surjectivity of T is equivalent to injectivity of T'*):

If V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$, then

$$T \text{ is surjective} \iff T' \text{ is injective}.$$

THM 3.130 (*range of T'*):

If V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$, then:

- (a) $\dim \text{range } T' = \dim \text{range } T$;
- (b) $\text{range } T' = (\text{null } T)^0$.

THM 3.131 (*injectivity of T is equivalent to surjectivity of T'*):

If V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$, then

$$T \text{ is injective} \iff T' \text{ is surjective}.$$

MATRIX OF DUAL OF LINEAR MAP (SUB-SECTION 3F3)**THM 3.132** (*matrix of dual map equals transpose of matrix*):Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then

$$\mathcal{M}(T') = (\mathcal{M}(T))^t,$$

where the matrix of T is with respect to bases of V and W , and the matrix of T' is with respect to the corresponding dual bases.

THM 3.133:If $A \in \mathbb{F}^{m,n}$, then the column rank of A equals the row rank of A .**EXERCISES ABOUT DUALITY 3F**

- ① Explain why each linear functional is surjective or the zero map.

SOLUTION: $\varphi \in V' = \mathcal{L}(V, \mathbb{F})$ implies that for every $\lambda \in \mathbb{F}$, and every $v \in V$,

$$\varphi(\lambda v) = \lambda \varphi(v).$$

If φ is not the zero map, then φ evaluated at $v \in V$ or $\lambda v \in V$ is a scalar in $\mathbb{F}$ for some $v \in V$ for which it holds that $\varphi(v) \neq 0$. In this case, we have $\varphi(v) \in \mathbb{F}$ and $\varphi(\lambda v) = \lambda \varphi(v) \in \mathbb{F}$. Hence, $\text{range } \varphi = \mathbb{F}$, since $\lambda \in \mathbb{F}$ was arbitrary.

- ③ Suppose V is finite-dimensional and $v \in V$ is nonzero. Prove that there exists $\varphi \in V'$ such that $\varphi(v) = 1$.

SOLUTION: Let $v_1, \dots, v_n$ be a basis of V . Let $v \in V$ be nonzero such that $v = a_1 v_1 + \dots + a_n v_n$, where $a_i \in \mathbb{F}$. Since v is nonzero, there exists a k such that $a_k \neq 0$. Define $\varphi : V \rightarrow \mathbb{R}$ by

$$\varphi(u) = \frac{c_k}{a_k},$$

whenever $u = c_1 v_1 + \dots + c_n v_n \in V$. Note that this definition corresponds to the use of the linear map lemma (3.4), where we define φ on each basis vector as follows:

$$\varphi(v_1) = 0, \dots, \varphi(v_{k-1}) = 0, \varphi(v_k) = \frac{1}{a_k}, \varphi(v_{k+1}) = 0, \dots, \varphi(v_n) = 0.$$

Hence, $\varphi(a_1 v_1 + \dots + a_n v_n) = \frac{a_k}{a_k} = 1$. It is easy to check that φ is a linear map.

- ⑤ Suppose V is finite dimensional and U is a subspace to V , where $U \neq V$. Prove that there exists a $\varphi \in V'$ such that $\varphi(u) = 0$ for every $u \in U$ but $\varphi \neq 0$.

SOLUTION: Choose $W \subset V$ such that $V = U \oplus W$. Hence, there is a basis

$$u_1, \dots, u_n, w_1, \dots, w_m$$

of V such that the u 's and w 's form a basis of U and W respectively. We define $\varphi \in V'$ such that for $v \in V$, written as

$$v = a_1 u_1 + \dots + a_n u_n + b_1 w_1 + \dots + b_m w_m,$$

we have

$$\varphi(v) = b_1 + \dots + b_m.$$

It is easy to see that $\varphi(u) = 0$ for every $u \in U$ (since $U \cap W = \{0\}$, by 1.46) and that φ is linear.

- ⑩ Suppose m is positive integer.

(a) Show that $1, x - 5, \dots, (x - 5)^m$ is a basis of $\mathcal{P}_m(\mathbb{R})$.

(b) What is the dual basis of the basis in (a)?

SOLUTION: We first start the exercise by solving it for $\mathcal{P}_4(\mathbb{R})$ straight forward before solving it

more abstract and generic for $\mathcal{P}_m(\mathbb{R})$. Recall that $e_0 = x^0, e_1 = x^1, \dots, e_4 = x^4$ is the standard basis of $\mathcal{P}_4(\mathbb{R})$. We want to show that $v_0 = 1, v_1 = x - 5, \dots, v_4 = (x - 5)^4$ is a basis as well. If we do some rewriting and expansion of the formulas, we get

$$\begin{aligned} v_0 &= (x^1 - 5x^0)^0 = 1x^0, \\ v_1 &= (x^1 - 5x^0)^1 = -5x^0 + x^1, \\ v_2 &= (x^1 - 5x^0)^2 = 25x^0 - 10x^1 + x^2, \\ v_3 &= (x^1 - 5x^0)^3 = -125x^0 + 75x^1 - 15x^2 + x^3, \\ v_4 &= (x^1 - 5x^0)^4 = 625x^0 - 500x^1 + 150x^2 - 20x^3 + x^4. \end{aligned}$$

Now, for a given $p \in \mathcal{P}_4(\mathbb{R})$ so that $p(x) = b_0x^0 + b_1x^1 + \dots + b_4x^4$ ($b_j \in \mathbb{F}$) we want to show that there exists $a_i \in \mathbb{F}$ such that $\sum_{j=0}^4 b_j e_j = \sum_{j=0}^4 a_j v_j$. So let's expand

$$\begin{aligned} \sum_{j=0}^4 a_j v_j &= a_0(1x^0) + a_1(-5x^0 + x^1) + a_2(25x^0 - 10x^1 + x^2) \\ &\quad + a_3(-125x^0 + 75x^1 - 15x^2 + x^3) + a_4(625x^0 - 500x^1 + 150x^2 + x^4) \\ &= (a_0 - 5a_1 + 25a_2 - 125a_3 + 625a_4) + (a_1 - 10a_2 + 75a_3 - 500a_4)x \\ &\quad + (a_2 - 15a_3 + 150a_4)x^2 + (a_3 - 20a_4)x^3 + a_4x^4 \\ &= b_0x^0 + b_1x^1 + \dots + b_4x^4 = \sum_{j=0}^4 b_j e_j. \end{aligned}$$

We can solve this for a_j by back substitution:

$$\begin{aligned} a_4 &= b_4, \\ a_3 &= b_3 + 20a_4, \\ a_2 &= b_2 + 15a_3 - 150a_4, \\ a_1 &= b_1 + 10a_2 + 75a_3 - 500a_4, \text{ and finally} \\ a_0 &= b_0 + 5a_1 - 25a_2 + 125a_3 - 625a_4. \end{aligned}$$

Hence, $v_1, \dots, v_4$ is indeed a basis of $\mathcal{P}_4(\mathbb{R})$. Now, we want to find the dual basis $\psi_0, \psi_1, \dots, \psi_4$ of $v_1, \dots, v_4$. Recall that the condition

$$\phi_j(v_k) = \delta_{j,k} = \begin{cases} 1, & j = k \\ 0, & j \neq k \end{cases}$$

must hold, and we thus want to define each ϕ_j on the standard basis $e_0, \dots, e_4$. So let's start with ψ_0 , where $\psi_0(v_0) = 1$ and zero on all other v 's.

$$\begin{aligned} \psi_0(v_0) &= 1\psi_0(x^0) = 1. \implies \psi_0(x^0) = 1. \\ \psi_0(v_1) &= -5\psi_0(x^0) + \psi_0(x^1) = 0. \implies \psi_0(x^1) = 5. \\ \psi_0(v_2) &= 25\psi_0(x^0) - 10\psi_0(x^1) + \psi_0(x^2) = 0. \implies \psi_0(x^2) = 25. \\ \psi_0(v_3) &= -125\psi_0(x^0) + 75\psi_0(x^1) - 15\psi_0(x^2) + \psi_0(x^3) = 0. \implies \psi_0(x^3) = 125. \\ \psi_0(v_4) &= 625\psi_0(x^0) - 500\psi_0(x^1) + 150\psi_0(x^2) - 20\psi_0(x^3) + \psi_0(x^4) = 0. \implies \psi_0(x^4) = 625. \end{aligned}$$

Now, the same for ψ_1 :

$$\begin{aligned} \psi_1(v_0) &= 1\psi_1(x^0) = 0. \implies \psi_1(x^0) = 0. \\ \psi_1(v_1) &= -5\psi_1(x^0) + \psi_1(x^1) = 1. \implies \psi_1(x^1) = 1. \\ \psi_1(v_2) &= 25\psi_1(x^0) - 10\psi_1(x^1) + \psi_1(x^2) = 0. \implies \psi_1(x^2) = 10. \\ \psi_1(v_3) &= -125\psi_1(x^0) + 75\psi_1(x^1) - 15\psi_1(x^2) + \psi_1(x^3) = 0. \implies \psi_1(x^3) = 75. \\ \psi_1(v_4) &= 625\psi_1(x^0) - 500\psi_1(x^1) + 150\psi_1(x^2) - 20\psi_1(x^3) + \psi_1(x^4) = 0. \implies \psi_1(x^4) = 500. \end{aligned}$$

Now, ψ_2 :

$$\psi_2(v_0) = 1\psi_2(x^0) = 0 \implies \psi_2(x^0) = 0.$$

$$\psi_2(v_1) = -5\psi_2(x^0) + \psi_2(x^1) = 0 \implies \psi_2(x^1) = 0.$$

$$\psi_2(v_2) = 25\psi_2(x^0) - 10\psi_2(x^1) + \psi_2(x^2) = 1 \implies \psi_2(x^2) = 1.$$

$$\psi_2(v_3) = -125\psi_2(x^0) + 75\psi_2(x^1) - 15\psi_2(x^2) + \psi_2(x^3) = 0 \implies \psi_2(x^3) = 15.$$

$$\psi_2(v_4) = 625\psi_2(x^0) - 500\psi_2(x^1) + 150\psi_2(x^2) - 20\psi_2(x^3) + \psi_2(x^4) = 0 \implies \psi_2(x^4) = 150.$$

For ψ_3 we get $\psi_3(x^0) = \psi_3(x_1) = \psi_3(x^2) = 0$ and $\psi_3(x^3) = 1$. Now what is left to calculate is

$$\psi_3(v_4) = 625\psi_2(x^0) - 500\psi_2(x^1) + 150\psi_2(x^2) - 20\psi_2(x^3) + \psi_2(x^4) = 0.$$

Hence, $\psi_4(v_4) = 20$. For ψ_4 we get $\psi_4(x^0) = \psi_4(x_1) = \dots = \psi_4(x^3) = 0$, and $\psi_4(v_4) = 1$, and thus, $\psi_4(x^4) = 1$. Now, let's take a more generic approach making extensive use of the binomic formula

$$(a+b)^n = \sum_{i=0}^n \binom{n}{i} a^i b^{n-i}, \quad (a, b \in \mathbb{R}, n \in \mathbb{N}),$$

and Pascal's identity

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \quad (n, k \in \mathbb{N}).$$

We start by rewriting once again

$$v_0 = (x^1 - 5x^0)^0 = \binom{0}{0}(-5)^0 e_0 = e^0,$$

$$v_1 = (x^1 - 5x^0)^1 = \binom{1}{0}(-5)^1 e_0 + \binom{1}{1}(-5)^0 e_1,$$

$$v_2 = (x^1 - 5x^0)^2 = \binom{2}{0}(-5)^2 e_0 + \binom{2}{1}(-5)^1 e_1 + \binom{2}{2}(-5)^0 e_2,$$

$\vdots$

$$v_m = (x^1 - 5x^0)^m = \binom{m}{0}(-5)^m e_0 + \binom{m}{1}(-5)^{m-1} e_1 + \dots + \binom{m}{m}(-5)^0 e_m = \sum_{i=0}^m \binom{m}{i}(-5)^{m-i} e_i.$$

Now, $\psi_j(e_k) = \binom{k}{j} 5^{k-j}$, taylor polynomials. (why?).

ChatGTP: Now, why is $\psi_j(e_k) = \binom{k}{j} \cdot 5^{k-j}$? We want to define the dual basis $\{\psi_j\}_{j=0}^m$ of $\{v_k\}_{k=0}^m$, meaning we require

$$\psi_j((x-5)^k) = \delta_{j,k}.$$

Using the binomial theorem, we can expand each $(x-5)^k$ as

$$(x-5)^k = \sum_{i=0}^k \binom{k}{i} (-5)^{k-i} e_i.$$

To satisfy the condition $\psi_j((x-5)^k) = \delta_{j,k}$, it is enough to define ψ_j on the standard basis $\{e_k\}$ by

$$\psi_j(e_k) = \binom{k}{j} \cdot 5^{k-j}.$$

This works because ψ_j corresponds to applying the j -th derivative at $x = 5$, divided by $j!$, to polynomials:

$$\psi_j(p(x)) = \frac{1}{j!} p^{(j)}(5),$$

and

$$\frac{1}{j!} \cdot \frac{d^j}{dx^j} x^k \Big|_{x=5} = \binom{k}{j} \cdot 5^{k-j}.$$

Thus, $\{\psi_j\}_{j=0}^m$ is indeed the dual basis of $\{v_k\}_{k=0}^m$.

- 11** Suppose $v_1, \dots, v_n$ is a basis of V and $\varphi_1, \dots, \varphi_n$ is the corresponding dual basis of V' . Suppose $\psi \in V'$. Prove that

$$\psi = \psi(v_1)\varphi_1 + \dots + \psi(v_n)\varphi_n. \quad (3.6)$$

SOLUTION: For a given $\psi \in V'$, let $\Gamma \in V'$ defined by

$$\Gamma := \psi(v_1)\varphi_1 + \dots + \psi(v_n)\varphi_n$$

just like in (3.6). We want to show that $\Gamma = \psi$. Observe that Γ is a linear functional, since it is the linear combination of some linear functionals $\varphi_1, \dots, \varphi_n$ (which are the dual basis of V). In this case, $\psi(v_1), \dots, \psi(v_n)$ are the scalars of this sum, which do not change for a given ψ and a given basis of V . If we now apply Γ to each v_k (for $k = 1, \dots, n$), we get

$$\begin{aligned} \Gamma(v_k) &= (\psi(v_1)\varphi_1 + \dots + \psi(v_n)\varphi_n)(v_k) \\ &= \psi(v_1)\varphi_1(v_k) + \dots + \psi(v_k)\varphi_n(v_k) + \dots + \psi(v_n)\varphi_n(v_k) \\ &= \psi(v_1) \cdot 0 + \dots + \psi(v_k) \cdot 1 + \dots + \psi(v_k) \cdot 0 \\ &= \psi(v_k). \end{aligned}$$

Hence, Γ and ψ agree on every basis vector v_k . Therefore, $\Gamma = \psi$.

- 14** Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by

$$T(x, y, z) := (4x + 5y + 6z, 7x + 8y + 9z).$$

Suppose φ_1, φ_2 denotes the dual basis of the standard basis of $\mathbb{R}^2$ and ψ_1, ψ_2, ψ_3 denotes the dual basis of the standard basis of $\mathbb{R}^3$.

(a) Describe the linear functionals $T'(\varphi_1)$ and $T'(\varphi_2)$.

(b) Write $T'(\varphi_1)$ and $T'(\varphi_2)$ as a linear combination of ψ_1, ψ_2, ψ_3 .

SOLUTION: Let's start by getting comfortable about the notation. Let e_1, e_2, e_3 denote the standard basis of $\mathbb{R}^3$ and e_1, e_2 denote the standard basis of $\mathbb{R}^2$. In this case, $T(x, y, z) = (4x + 5y + 6z, 7x + 8y + 9z)$ actually means

$$T(xe_1 + ye_2 + ze_3) = (4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2$$

Thus, we have $Te_1 = 4e_1 + 7e_2$, $Te_2 = 5e_1 + 8e_2$, and $Te_3 = 6e_1 + 9e_2$. The Matrix $\mathcal{M}(T)$ of T would look like this:

$$\mathcal{M}(T) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 \end{matrix} \\ \begin{matrix} e_1 \\ e_2 \end{matrix} & \begin{pmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \end{matrix}$$

Now, $T'(\varphi_1) = \varphi_1 \circ T$ and $T'(\varphi_2) = \varphi_2 \circ T$. Hence,

$$T'(\varphi_1)(xe_1 + ye_2 + ze_3) = \varphi_1((4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2) = 4x + 5y + 6z, \text{ and}$$

$$T'(\varphi_2)(xe_1 + ye_2 + ze_3) = \varphi_2((4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2) = 7x + 8y + 9z,$$

which answers question (a). For question (b), we use the result (3.6) from question 11 and write

$$\begin{aligned} T'(\varphi_1) &= \sum_{k=1}^3 T'(\varphi_1)(e_k)\psi_k = 4\psi_1 + 5\psi_2 + 6\psi_3, \text{ and} \\ T'(\varphi_2) &= \sum_{k=1}^3 T'(\varphi_2)(e_k)\psi_k = 7\psi_1 + 8\psi_2 + 9\psi_3 \end{aligned}$$

VARIATION OF QUESTION 14: The question becomes more interesting if we use another basis for $\mathbb{R}^2$ and $\mathbb{R}^3$, say $v_1 = (1, 1), v_2 = (2, 1)$ for $\mathbb{R}^2$ and $w_1 = (2, 2, 0), w_2 = (4, 2, 0), w_3 = (0, 0, 1)$ for $\mathbb{R}^3$. In this case, let ϕ_1, ϕ_2 denote the dual basis of v_1, v_2 and ψ_1, ψ_2, ψ_3 denote the dual basis of

w_1, w_2, w_3 . Now, we have to calculate ϕ and ψ first. Per definition, we have

$$\phi(v) = \phi(a_1 v_1 + a_2 v_2) = a_1 \phi_1(v_1) + a_2 \phi_2(v_2).$$

And we must assure that $\phi_1(v_1) = 1, \phi(v_2) = 0$ and $\phi_2(v_2) = 0$ and $\phi_2(v_2) = 1$. Using the facts that $v_1 = e_1 + e_2$ and $v_2 = 2e_1 + e_2$, we get

$$\begin{cases} 1\phi_1(e_1) + 1\phi_1(e_2) = 1 \\ 2\phi_1(e_1) + 1\phi_1(e_2) = 0 \end{cases} \implies \begin{cases} 1\phi_1(e_1) + 1\phi_1(e_2) = 1 \\ -2\phi_1(e_2) = -2 \end{cases},$$

and hence, $\phi_1(e_1) = -1$ and $\phi_1(e_2) = 2$. Likewise,

$$\begin{cases} 1\phi_2(e_1) + 1\phi_2(e_2) = 0 \\ 2\phi_2(e_1) + 1\phi_2(e_2) = 1 \end{cases} \implies \begin{cases} 1\phi_2(e_1) + 1\phi_2(e_2) = 0 \\ -2\phi_2(e_2) = 1 \end{cases}.$$

Therefore, $\phi_2(e_1) = 1$ and $\phi_2(e_2) = -1$. To write ϕ_1 and ϕ_2 as members of $\mathbb{R}^2 \rightarrow \mathbb{R}$, we have $\phi_1(x, y) = -x + 2y$ and $\phi_2(x, y) = x - y$. Now,

$$T'(\phi_1) = \phi_1 \circ T = -(4x + 5y + 6z) + 2(7x + 8y + 9z) = 10x + 11y + 12z, \text{ and}$$

$$T'(\phi_2) = \phi_2 \circ T = 4x + 5y + 6z - (7x + 8y + 9z) = -3x - 3y - 3z = -3(x + y + z).$$

With the same approach, we can calculate the dual basis for w_1, w_2, w_3 . We get $\psi_1(x, y, z) = -\frac{1}{2}x + y$, $\psi_2(x, y, z) = \frac{1}{2}x - \frac{1}{2}y$ and $\psi_3(x, y, z) = z$. To answer question (b), we use the theorem (3.6) from question 11 and write

$$\begin{aligned} T'(\varphi_1) &= \sum_{k=1}^3 T'(\varphi_1)(w_k) \psi_k = (20 + 22 + 0)\psi_1 + (40 + 22 + 0)\psi_2 + (0 + 0 + 12)\psi_3 \\ &= 42\psi_1 + 62\psi_2 + 12\psi_3, \text{ and} \end{aligned}$$

$$T'(\varphi_2) = \sum_{k=1}^3 T'(\varphi_2)(w_k) \psi_k = -12\psi_1 - 18\psi_2 - 3\psi_3.$$

16 Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$T' = 0 \iff T = 0.$$

PROOF: $\implies$ **DIRECTION:** Assume $T' = 0$. By definition, $T' = \varphi \circ T = 0$ for every $\varphi \in W'$. If $W = \{0\}$, then necessarily $T = 0$, so $T' = 0$ holds trivially. For this reason, assume for now that $W \neq 0$ and let $w_1, \dots, w_n$ be a basis of W . Suppose for the sake of contradiction that $T \neq 0$. Hence, there exists a nonzero $v \in V$ and a nonzero $w \in W$ such that $w = T(v)$. Express w as $w = a_1 w_1 + \dots + a_n w_n$. Since $w \neq 0$, pick k such that $a_k \neq 0$. Define $\varphi \in W'$ by

$$\varphi(b_1 w_1 + \dots + b_n w_n) = \frac{b_k}{a_k},$$

where b_k is the k -th coordinate of φ 's input. Hence, $\varphi(w) = 1$. But then

$$T'(\varphi)(v) = \varphi(T(v)) = \varphi(w) = 1 \neq 0.$$

Thus, $T'(\varphi) \neq 0$, contradicting $T' = 0$. Hence, $T = 0$.

$\Leftarrow$ **DIRECTION:** If $T = 0$, then for each $\varphi \in W'$ and $T' \in \mathcal{L}(W', V')$ we have

$$T'(\varphi)(v) = \varphi(T(v)) = \varphi(0) = 0.$$

Hence, $T'(\varphi)$ is always sent to a map in W' that is the zero map. □

16 Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$T \text{ is invertible} \iff T' \in \mathcal{L}(W', V') \text{ is invertible.}$$

PROOF: $\implies$ **DIRECTION:** Let $T \in \mathcal{L}(V, W)$ be invertible and $T^{-1} \in \mathcal{L}(W, V)$ be the inverse of T . We want first want to show that $(T^{-1})'$ is the inverse of T' , that is

$$(T^{-1})' = (T')^{-1}.$$

For $(T^{-1})' \in \mathcal{L}(V', W')$ to be an inverse of $T' \in \mathcal{L}(W', V')$, it must hold that $(T^{-1})' T' = I_{W'}$ and $T' (T^{-1})' = I_{V'}$. Note that $I_W \neq I_{W'}$ and $I_V \neq I_{V'}$. Therefore, let's compute for $\psi \in V'$:

$$\left((T^{-1})' \circ T' \right) (\psi) = (T^{-1})' (\psi \circ T^{-1}) = (\psi \circ T) \circ T^{-1} = \psi \circ (T T^{-1}) = \psi.$$

Thus, $T' (T^{-1})' = I_{V'}$. The other composition works by the same argument. Let $\psi \in W'$. So,

$$\left(T' \circ (T^{-1})' \right) (\varphi) = T' (\varphi \circ T^{-1}) = (\varphi \circ T^{-1}) \circ T = \varphi \circ (T^{-1} T) = \varphi,$$

for that $(T^{-1})' T' = I_{W'}$. Hence, T' is invertible with $(T^{-1})' = (T')^{-1}$.

$\Leftarrow$ **DIRECTION:** If T' is invertible, then it is surjective (and also injective) by 3.65. Take any nonzero $v \in V$ and $\varphi \in V'$ with $\varphi(v) \neq 0$. Surjectivity of T' yields a $\psi \in W'$ with $T'(\psi) = \varphi$, or $\varphi = \psi \circ T$. Therefore,

$$0 \neq \varphi(v) = \psi(Tv) = (\psi \circ T)(v).$$

This implies that $Tv \neq 0$, and thus, $\text{null } T = \{0\}$ (since v was arbitrary and nonzero). This forces T to be injective. Because T' is invertible, we have $\dim W' = \dim V'$, and hence, $\dim W = \dim V$ (see 3.111). An injective linear map between finite-dimensional vector spaces of the same dimension is surjective, so T is bijective, and therefore, invertible. $\square$

POLYNOMIALS (CHAPTER 4)

Throughout this chapter, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

ZEROS OF POLYNOMIALS (SECTION 4A)

DEF 4.5 (*zero of a polynomial*):

A number $\lambda \in \mathbb{F}$ is called a “zero” (or “root”) of a polynomial $p \in \mathcal{P}(\mathbb{F})$ if $p(\lambda) = 0$.

THM 4.6 (*each zero corresponds to a degree-one factor*):

Let $p \in \mathcal{P}_m(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then

$$p(\lambda) = 0 \iff \exists q \in \mathcal{P}_{m-1}(\mathbb{F}) \text{ such that } p(z) = (z - \lambda)q(z) \text{ for all } z \in \mathbb{F}.$$

PROOF: $\Leftarrow$ **DIRECTION:** Suppose there exists $q \in \mathcal{P}_{m-1}(\mathbb{F})$ such that $p(z) = (z - \lambda)q(z)$ for all $z \in \mathbb{F}$. Then

$$p(\lambda) = (\lambda - \lambda)q(\lambda) = 0.$$

$\Rightarrow$ **DIRECTION:** Now suppose $p(\lambda) = 0$. Write $p(z) = a_0 + a_1z + \cdots + a_mz^m$. Then

$$p(z) = p(z) - p(\lambda) = a_1(z - \lambda) + a_2(z^2 - \lambda^2) + \cdots + a_m(z^m - \lambda^m).$$

Using the identity $z^k - \lambda^k = (z - \lambda) \sum_{j=1}^k \lambda^{j-1} z^{k-j}$ for each $k \in \{1, \dots, m\}$, we obtain:

$$p(z) = (z - \lambda) \left(a_1 + a_2(z + \lambda) + \cdots + a_m \sum_{j=1}^m \lambda^{j-1} z^{m-j} \right).$$

The expression inside parentheses is a polynomial $q \in \mathcal{P}_{m-1}(\mathbb{F})$, which completes the proof. $\square$

THM 4.8 (*degree m implies at most m zeros*):

A polynomial $p \in \mathcal{P}(\mathbb{F})$ of degree $m \geq 0$ has at most m distinct zeros in $\mathbb{F}$.

PROOF: We use induction on m . If $m = 0$, then p is a nonzero constant polynomial and has 0 zeros.

Now suppose $m \geq 1$ and the result holds for polynomials of degree $m - 1$. If p has no zeros in $\mathbb{F}$, then the number of zeros is $0 \leq m$. If p has a zero $\lambda \in \mathbb{F}$, then by Theorem 4.6, there exists $q \in \mathcal{P}_{m-1}(\mathbb{F})$ such that $p(z) = (z - \lambda)q(z)$. If $\tau \in \mathbb{F}$ is any zero of p with $\tau \neq \lambda$, then $0 = p(\tau) = (\tau - \lambda)q(\tau)$, so $q(\tau) = 0$. By the induction hypothesis, q has at most $m - 1$ distinct zeros. Therefore, p has at most $1 + (m - 1) = m$ distinct zeros. $\square$

DIVISION ALGORITHM FOR POLYNOMIALS (SECTION 4B)

THM 4.9 (*division algorithm for polynomials*):

Suppose $p, s \in \mathcal{P}(\mathbb{F})$ with $s \neq 0$. Then there exist unique polynomials $q, r \in \mathcal{P}(\mathbb{F})$ such that

$$p = sq + r \quad \text{and} \quad \deg r < \deg s.$$

PROOF: Let $n := \deg p$ and $m := \deg s$. If $n < m$, we can take $q = 0$ and $r = p$. Now assume $n \geq m$. The list

$$1, z, \dots, z^{m-1}, s, zs, \dots, z^{n-m}s \tag{4.1}$$

is linearly independent because each polynomial in the list has a distinct degree. The length of this list is $m + (n - m + 1) = n + 1 = \dim \mathcal{P}_n(\mathbb{F})$. Hence, by Theorem 2.38, the list (4.1) is a basis of $\mathcal{P}_n(\mathbb{F})$. Thus, there exist unique scalars $a_0, \dots, a_{m-1} \in \mathbb{F}$ and $b_0, \dots, b_{n-m} \in \mathbb{F}$ such that

$$p = \underbrace{(a_0 + a_1z + \cdots + a_{m-1}z^{m-1})}_r + s \cdot \underbrace{(b_0 + b_1z + \cdots + b_{n-m}z^{n-m})}_{sq},$$

with $\deg r \leq m - 1 < \deg s$. The uniqueness of q and r follows from the uniqueness of the

coefficients with respect to the basis. □

FACTORIZATION OF POLYNOMIALS OVER $\mathbb{C}$ (SECTION 4C)

THM 4.12 (FUNDAMENTAL THEOREM OF ALGEBRA, first version):

Every non-constant polynomial with complex coefficients has a zero in $\mathbb{C}$.

THM 4.13 (FUNDAMENTAL THEOREM OF ALGEBRA, second version):

If $p \in \mathcal{P}(\mathbb{C})$ is a non-constant polynomial of degree $m \geq 1$, then p has a unique factorization

$$p(z) = c(z - \lambda_1) \cdots (z - \lambda_m),$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbb{C}$. This factorization is unique up to the order of the factors.

PROOF: Let $p \in \mathcal{P}(\mathbb{C})$ with $m := \deg p \geq 1$. We use induction on m . If $m = 1$, then $p(z) = a_0 + a_1 z = a_1(z - (-a_0/a_1))$, which gives the desired factorization with $c = a_1$ and $\lambda_1 = -a_0/a_1$.

Now suppose $m > 1$ and the result holds for all polynomials of degree $m - 1$. By Theorem 4.12, p has a zero $\lambda \in \mathbb{C}$. By Theorem 4.6, there exists a polynomial $q \in \mathcal{P}_{m-1}(\mathbb{C})$ such that $p(z) = (z - \lambda)q(z)$. By the induction hypothesis, q has a factorization $q(z) = c(z - \lambda_1) \cdots (z - \lambda_{m-1})$. Substituting this into $p(z) = (z - \lambda)q(z)$ gives the desired factorization of p .

UNIQUENESS: The constant c is uniquely determined as the leading coefficient of z^m in p . Suppose $c(z - \lambda_1) \cdots (z - \lambda_m) = c(z - \tau_1) \cdots (z - \tau_m)$ for all $z \in \mathbb{C}$. Substituting $z = \lambda_1$ makes the left side zero, so the right side must also be zero, which implies $\lambda_1 = \tau_k$ for some k . Relabeling so that $\tau_1 = \lambda_1$, we have $(z - \lambda_1)((z - \lambda_2) \cdots (z - \lambda_m) - (z - \tau_2) \cdots (z - \tau_m)) = 0$ for all $z \in \mathbb{C}$. Since a nonzero polynomial can only have finitely many zeros by Theorem 4.8, the factor $(z - \lambda_2) \cdots (z - \lambda_m) - (z - \tau_2) \cdots (z - \tau_m)$ must be the zero polynomial. By the induction hypothesis, the lists $\lambda_2, \dots, \lambda_m$ and $\tau_2, \dots, \tau_m$ are identical up to permutation, completing the proof. □

FACTORIZATION OF POLYNOMIALS OVER $\mathbb{R}$ (SECTION 4D)

THM 4.14 (polynomials with real coefficients have nonreal zeros in pairs):

Suppose $p \in \mathcal{P}(\mathbb{C})$ is a polynomial with real coefficients. If $\lambda \in \mathbb{C}$ is a zero of p , then $\bar{\lambda}$ is also a zero of p .

PROOF: Let $p(z) = a_0 + a_1 z + \cdots + a_m z^m$, where $a_0, \dots, a_m \in \mathbb{R}$. Suppose $\lambda \in \mathbb{C}$ is a zero of p . Taking the complex conjugate of both sides gives:

$$0 = \bar{0} = \overline{p(\lambda)} = \overline{a_0 + a_1 \lambda + \cdots + a_m \lambda^m} = \overline{a_0} + \overline{a_1} \bar{\lambda} + \cdots + \overline{a_m} \bar{\lambda}^m.$$

Since each $a_k \in \mathbb{R}$, $\overline{a_k} = a_k$, so $0 = a_0 + a_1 \bar{\lambda} + \cdots + a_m \bar{\lambda}^m = p(\bar{\lambda})$. Thus, $\bar{\lambda}$ is also a zero of p . □

THM 4.15 (factorization of a quadratic polynomial):

Let $p(x) = x^2 + bx + c$ with $b, c \in \mathbb{R}$. Then p can be written as $(x - \lambda_1)(x - \lambda_2)$ with $\lambda_1, \lambda_2 \in \mathbb{R}$ if and only if $b^2 \geq 4c$.

PROOF: Completing the square gives:

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right).$$

If $b^2 < 4c$, then $c - b^2/4 > 0$, so $p(x) > 0$ for all $x \in \mathbb{R}$, meaning p has no real zeros and cannot be factored over $\mathbb{R}$. If $b^2 \geq 4c$, set $d := \sqrt{b^2/4 - c}$. Then

$$p(x) = \left(x + \frac{b}{2}\right)^2 - d^2 = \left(x + \frac{b}{2} - d\right)\left(x + \frac{b}{2} + d\right) = (x - \lambda_1)(x - \lambda_2),$$

where $\lambda_1 := -b/2 + d \in \mathbb{R}$ and $\lambda_2 := -b/2 - d \in \mathbb{R}$. □

THM 4.16 (*factorization of a polynomial over $\mathbb{R}$*):

Suppose $p \in \mathcal{P}(\mathbb{R})$ is non-constant. Then p has a unique factorization:

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)(x^2 + b_1x + c_1) \cdots (x^2 + b_Mx + c_M),$$

where $c, \lambda_1, \dots, \lambda_m, b_1, \dots, b_M, c_1, \dots, c_M \in \mathbb{R}$, with $b_k^2 < 4c_k$ for each $k \in \{1, \dots, M\}$.

PROOF: We view p as an element of $\mathcal{P}(\mathbb{C})$ with real coefficients. If all complex zeros of p are real, the result follows immediately from Theorem 4.13. If p has a nonreal zero $\lambda \in \mathbb{C} \setminus \mathbb{R}$, then by Theorem 4.14, $\bar{\lambda}$ is also a zero of p . Then $(x - \lambda)(x - \bar{\lambda}) = x^2 - 2(\operatorname{Re} \lambda)x + |\lambda|^2$ is a quadratic with real coefficients and discriminant $(-2\operatorname{Re} \lambda)^2 - 4|\lambda|^2 = 4((\operatorname{Re} \lambda)^2 - |\lambda|^2) = -4(\operatorname{Im} \lambda)^2 < 0$. Factoring out this quadratic reduces the problem to a polynomial q of degree $\deg p - 2$ with real coefficients. By induction on $\deg p$, the desired factorization exists.

UNIQUENESS: Each quadratic factor $x^2 + b_kx + c_k$ with $b_k^2 < 4c_k$ factors uniquely over $\mathbb{C}$ as $(x - \lambda_k)(x - \bar{\lambda}_k)$. Uniqueness of the real factorization then follows directly from the uniqueness of the complex factorization in Theorem 4.13. $\square$

EXERCISES ABOUT POLYNOMIALS 4D

- ④ Suppose m is a positive integer. Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) : \deg p = m\} = S$$

a subspace of $\mathcal{P}(\mathbb{F})$?

SOLUTION: The set in question is not a subspace of $\mathcal{P}(\mathbb{F})$, because it is not closed under addition. Take $m = 4$, then $p = 1 + 2x^2 + 3x^3 + 4x^4$ and $s = 4x^4$ are both in S , but $p - s$ is not.

- ⑤ Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) : \deg p \text{ is even}\} = S$$

a subspace of $\mathcal{P}(\mathbb{F})$?

SOLUTION: The same counter-example as in the previous question applies to this one.

- ⑥ Suppose that m and n are positive integers with $m \leq n$, and suppose $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ are distinct. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{F})$ with $\deg p = n$ such that $0 = p(\lambda_1) = \dots = p(\lambda_m)$ and such that p has no other zeros.

PROOF: Define $p \in \mathcal{P}_n(\mathbb{F})$ by

$$\begin{aligned} p(z) &= (z - \lambda_1)(z - \lambda_2) \cdots (z - \lambda_{m-1}) \underbrace{(z - \lambda_m)(z - \lambda_m) \cdots (z - \lambda_m)}_{n-m+1 \text{ times}} \\ &= \left(\prod_{k=1}^{m-1} (z - \lambda_k) \right) (z - \lambda_m)^{n-m+1} \quad \forall z \in \mathbb{F}. \end{aligned}$$

Clearly, $\deg p = (m - 1) + (n - m + 1) = n$, and $p(\lambda_k) = 0$ for each $k \in \{1, \dots, m\}$. Furthermore, if $\lambda \in \mathbb{F}$ satisfies $p(\lambda) = 0$, then since $\mathbb{F}$ is a field (with no zero divisors), at least one factor $(\lambda - \lambda_k)$ must be zero. Hence $\lambda \in \{\lambda_1, \dots, \lambda_m\}$, meaning p has no other zeros. $\square$

EIGENVALUES AND EIGENVECTORS (CHAPTER 5)

INVARIANT SUBSPACES (SECTION 5A)

EIGENVALUES (SUB-SECTION 5A1)

DEF 5.1 (operator):

A linear map from a vector space to itself is called an “operator”.

DEF 5.2 (invariant subspace):

Let $T \in \mathcal{L}(V)$. A subspace U of V is called “invariant under T ” if $Tu \in U$ for all $u \in U$. Equivalently, U is invariant under T if the restriction $T|_U$ is an operator on U .

EXAMPLE 5.3:

Let $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ be the differentiation operator $Tp := p'$. The subspace $U = \mathcal{P}_4(\mathbb{R})$ is invariant under T because $\deg p' < \deg p \leq 4$ for all $p \in U$.

EXAMPLE 5.4:

For every operator $T \in \mathcal{L}(V)$, the subspaces $\{0\}$, V , $\text{null } T$, and $\text{range } T$ are all invariant under T .

INVARIANT SUBSPACES OF DIMENSION ONE: Let $v \in V$ with $v \neq 0$, and consider the one-dimensional subspace $U := \text{span}(v)$. If U is invariant under $T \in \mathcal{L}(V)$, then $Tv \in U$, which means there exists $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$. Conversely, if $Tv = \lambda v$ for some $\lambda \in \mathbb{F}$, then $\text{span}(v)$ is invariant under T .

DEF 5.5 (eigenvalue):

Suppose $T \in \mathcal{L}(V)$. A scalar $\lambda \in \mathbb{F}$ is called an “eigenvalue” of T if there exists a nonzero vector $v \in V$ such that

$$Tv = \lambda v.$$

THM 5.7 (equivalent conditions to be an eigenvalue):

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. The following are equivalent:

- (a) λ is an eigenvalue of T ;
- (b) $T - \lambda I$ is not injective;
- (c) $T - \lambda I$ is not surjective;
- (d) $T - \lambda I$ is not invertible.

PROOF: Condition (a) holds if and only if there exists $v \neq 0$ such that $(T - \lambda I)v = 0$, which is equivalent to $\text{null}(T - \lambda I) \neq \{0\}$, proving (a) $\iff$ (b). The equivalence of (b), (c), and (d) follows directly from Theorem 3.65. $\square$

DEF 5.8 (eigenvector):

Let $T \in \mathcal{L}(V)$. A vector $v \in V$ is called an “eigenvector” of T corresponding to $\lambda \in \mathbb{F}$ if $v \neq 0$ and $Tv = \lambda v$. In other words, $v \neq 0$ is an eigenvector corresponding to λ if and only if $v \in \text{null}(T - \lambda I)$.

THM 5.11 (linearly independent eigenvectors):

Every list of eigenvectors of T corresponding to distinct eigenvalues of T is linearly independent.

PROOF: Suppose the result is false. Let $v_1, \dots, v_m$ be a shortest list of linearly dependent eigenvectors of T corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Since eigenvectors are nonzero, $m \geq 2$.

Linear dependence implies that there exist scalars $a_1, \dots, a_m$, not all zero, such that

$$a_1 v_1 + \dots + a_m v_m = 0. \quad (5.1)$$

Applying $(T - \lambda_m I)$ to both sides gives:

$$a_1(\lambda_1 - \lambda_m)v_1 + \dots + a_{m-1}(\lambda_{m-1} - \lambda_m)v_{m-1} = 0.$$

By the minimality of m , the list $v_1, \dots, v_{m-1}$ is linearly independent, so

$$a_1(\lambda_1 - \lambda_m) = \dots = a_{m-1}(\lambda_{m-1} - \lambda_m) = 0.$$

Since the eigenvalues are distinct, $\lambda_k - \lambda_m \neq 0$ for each $k < m$, which forces $a_1 = \dots = a_{m-1} = 0$.

Substituting this back into (5.1) gives $a_m v_m = 0$, so $a_m = 0$ (since $v_m \neq 0$). This contradicts that not all coefficients were zero, completing the proof. $\square$

THM 5.12 (*number of eigenvalues*):

Each operator on a finite-dimensional vector space V has at most $\dim V$ distinct eigenvalues.

PROOF: Let $T \in \mathcal{L}(V)$, and suppose $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues with corresponding eigenvectors $v_1, \dots, v_m$. By Theorem 5.11, the list $v_1, \dots, v_m$ is linearly independent in V . By Theorem 2.22, $m \leq \dim V$. $\square$

POLYNOMIALS APPLIED TO OPERATORS (SUB-SECTION 5A2)

DEF 5.13 (*notation: T^m*):

Let $T \in \mathcal{L}(V)$ and $m \in \mathbb{N}$.

- $T^m := \underbrace{T \dots T}_{m \text{ times}}$ for $m \geq 1$;
- $T^0 := I_V$;
- If T is invertible, then for $m \geq 1$, $T^{-m} := (T^{-1})^m$.

DEF 5.14 (*notation: $p(T)$*):

For $p(z) = a_0 + a_1 z + \dots + a_m z^m \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$, define:

$$p(T) := a_0 I + a_1 T + a_2 T^2 + \dots + a_m T^m \in \mathcal{L}(V).$$

THM 5.17 (*multiplicative properties of polynomial evaluation*):

Suppose $p, q \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$. Then

$$(pq)(T) = p(T)q(T) = q(T)p(T).$$

PROOF: Suppose $p(z) = \sum_{j=0}^m a_j z^j$ and $q(z) = \sum_{k=0}^n b_k z^k$. Then $(pq)(z) = \sum_{j=0}^m \sum_{k=0}^n a_j b_k z^{j+k}$. Evaluating at T gives:

$$(pq)(T) = \sum_{j=0}^m \sum_{k=0}^n a_j b_k T^{j+k} = \left(\sum_{j=0}^m a_j T^j \right) \left(\sum_{k=0}^n b_k T^k \right) = p(T)q(T).$$

Since polynomial multiplication is commutative ($pq = qp$), we also have $(pq)(T) = q(T)p(T)$. $\square$

THM 5.18 (*null space and range of $p(T)$ are invariant under T*):

For every $T \in \mathcal{L}(V)$ and every $p \in \mathcal{P}(\mathbb{F})$, the subspaces $\text{null } p(T)$ and $\text{range } p(T)$ are invariant under T .

PROOF: STEP 1: Let $u \in \text{null } p(T)$. Then $p(T)u = 0$. Since $p(T)$ and T commute by Theorem 5.17, we have:

$$p(T)(Tu) = (p(T)T)u = (Tp(T))u = T(p(T)u) = T(0) = 0.$$

Hence, $Tu \in \text{null } p(T)$, which proves that $\text{null } p(T)$ is invariant under T .

STEP 2: Let $u \in \text{range } p(T)$. Then $u = p(T)v$ for some $v \in V$. Thus,

$$Tu = T(p(T)v) = p(T)(Tv) \in \text{range } p(T),$$

which proves that $\text{range } p(T)$ is invariant under T . $\square$

EXERCISES ABOUT INVARIANT SUBSPACES 5A

(1) Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V . Then

(a) $U \subseteq \text{null } T \implies U$ is invariant under T .

(b) $\text{range } T \subseteq U \implies U$ is invariant under T .

PROOF: (a). If $v \in U$, then $Tv = 0 \in U$, since U is a subspace and therefore $0 \in U$.

(b). Let $w \in U \subseteq V$. $\implies Tw \in \text{range } T$, and since $\text{range } T \subseteq U$, $Tw \in U$. $\square$

(2) Suppose that $T \in \mathcal{L}(V)$ and $V_1, \dots, V_m$ are subspaces of V invariant under T . Prove that $V_1 + \dots + V_m$ is invariant under T .

PROOF: Let $w \in V_1 + \dots + V_m$ such that $w = v_1 + \dots + v_m$ where $v_1 \in V_1, \dots, v_m \in V_m$. Then $Tw = Tv_1 + \dots + Tv_m$ such that $Tv_1 \in V_1, \dots, Tv_m \in V_m$ because $V_1, \dots, V_m$ are invariant under T . Therefore we have $Tw \in V_1 + \dots + V_m$. So $V_1 + \dots + V_m$ is invariant under T as well. $\square$

(3) Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of every collection of subspaces of V invariant under T is invariant under T .

PROOF: Let $w \in V_1 \cap \dots \cap V_m \subseteq V$ where each V_i is invariant under T . Then $w \in V_1, \dots, w \in V_m$, and thus, $Tw \in V_1, \dots, Tw \in V_m$. Hence, $Tw \in V_1 \cap \dots \cap V_m$. $\square$

(12) Suppose $V = U \oplus W$, where U and W are nonzero subspaces of V . Define $P \in \mathcal{L}(V)$ by $P(u+w) = u$ for each $u \in U$ and each $w \in W$. Find all eigenvalues and eigenvectors of P .

SOLUTION: We have an eigenvalue $\lambda_1 = 1$ for all vectors $u \in U$, because P acts as the identity on U . We also have an eigenvalue $\lambda_0 = 0$ for all $w \in W$, because P acts as the zero operator there.

(15) Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. Show that λ is an eigenvalue of $T \iff \lambda$ is an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$.

SOLUTION: The definition of T' states that

$$T'(\varphi) = \varphi \circ T \quad \forall \varphi \in V'.$$

Hence, being an eigenvalue λ and an eigenvector $\varphi \in V' = \mathcal{L}(V, \mathbb{F})$ of T' means

$$T'(\varphi) = \lambda \varphi.$$

$\implies$ **DIRECTION:** If $u \in V$ is an eigenvector of T with an eigenvalue λ such that $T(u) = \lambda u$, we can extend u to a basis $u, v_2, \dots, v_n$ of V ($n = \dim V$). Define $\varphi \in V'$ by

$$\varphi(u) = 1, \quad \varphi(v_i) = 0 \quad (i = 2, \dots, n),$$

on the basis vectors by the linear map lemma 3.4.

then we have for every $\varphi \in V'$:

$$\begin{aligned} T'(\varphi)(u) &= (\varphi \circ T)(u) \\ &= \varphi(T(u)) \\ &= \varphi(\lambda u) \\ &= \lambda \varphi(u) \end{aligned}$$

So, if we construct ψ such that for every $v \in V, v = au + \sum_{i \geq 2} b_i v_i$ we have

$$(T'\varphi)(v) = (\varphi \circ T)(u) = \varphi(aTu + \sum_{i \geq 2} b_i Tv_i) = \varphi(a\lambda u) + \sum_{i \geq 2} b_i Tv_i$$

$$\psi(v) = \begin{cases} \varphi(v), & v \in E(\lambda, T) \text{ and } v \notin \text{null } T \\ 0, & \text{otherwise} \end{cases}$$

we have found an eigenvector ψ with the same eigenvalue λ .

Or: λ is eigenvalue of $T \iff T - \lambda I$ is not injective $\iff (T - \lambda I)'$ is not surjective $\iff T' - \lambda I$ is not invertible $\iff \lambda$ is an eigenvalue of T' .

(21) Suppose $T \in \mathcal{L}(V)$ is invertible

(a) Suppose $\lambda \in \mathbb{F}$ with $\lambda \neq 0$. Prove that λ is an eigenvalue of $T \iff \frac{1}{\lambda}$ is an eigenvalue of T^{-1} .

(b) Prove that T and T^{-1} have the same eigenvectors.

SOLUTION: Let $v \in V$ be such that $T(v) = \lambda v$. This is the case $\iff T^{-1}T(v) = T^{-1}(\lambda v) \iff T^{-1}(v) = \frac{1}{\lambda}v$. Note that in a field $\mathbb{F}$, we can “replace” $\frac{1}{\lambda} \in \mathbb{F}$ with its multiplicative inverse $\alpha \in \mathbb{F}$. Since $(T^{-1})^{-1} = T$, we have proven (a) and (b).

(22) Suppose $T \in \mathcal{L}(V)$ and there exist nonzero vectors $u, w \in V$ such that

$$Tu = 3w \quad \text{and} \quad Tw = 3u$$

Prove that 3 or -3 is an eigenvalue of T

SOLUTION: If T is defined as above, then $T(u+w) = 3w+3u = 3(u+w)$. Hence, $u+w$ gets sent to $3(u+w)$, which makes 3 an eigenvalue of T . If we change the definition of T to

$$T(u) = 3(-w) \quad \text{and} \quad T(-w) = 3u,$$

then we have

$$T(u+w) = T(u+(-1)(-1)w) = T(u) - T(-w) = -3w - 3u = -3(u+w).$$

Thus, $u+w$ gets sent to $-3(u+w)$ and -3 is an eigenvalue of T as well. Note that we didn't prove that those are the only eigenvalues.

(23) Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an eigenvalue $\iff$ there exists a subspace of V of dimension $\dim V - 1$ that is invariant under T .

KI Bemerkung: Hier standen zwei fehlerhafte bzw. unvollständige Beweisversuche. Ich habe diese durch einen korrekten und eleganten Beweis über den dualen Operator ersetzt, der auch das Ergebnis von Aufgabe 15 direkt ausnutzt.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose T has an eigenvalue $\lambda \in \mathbb{F}$. By Exercise 15, λ is also an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$. Thus, there exists a nonzero linear functional $\varphi \in V'$ such that $T'(\varphi) = \lambda\varphi$. This means that for every $u \in V$, we have $\varphi(Tu) = (T'\varphi)(u) = (\lambda\varphi)(u) = \lambda\varphi(u)$. Let $U = \text{null } \varphi$. Because $\varphi \neq 0$, the Fundamental Theorem of Linear Maps tells us that $\dim U = \dim V - \dim(\text{range } \varphi) = \dim V - 1$. If $u \in U$, then $\varphi(u) = 0$, so $\varphi(Tu) = \lambda\varphi(u) = 0$, which implies $Tu \in \text{null } \varphi = U$. Hence, U is a subspace of dimension $\dim V - 1$ that is invariant under T .

$\Leftarrow$ **DIRECTION:** Suppose there exists a subspace $U \subseteq V$ of dimension $\dim V - 1$ that is invariant under T . Let $U^0 \subseteq V'$ be the annihilator of U . Since $\dim U = \dim V - 1$, we know that $\dim U^0 = \dim V - \dim U = 1$. Because U is invariant under T , its annihilator U^0 is invariant under the dual operator T' . Since U^0 is a one-dimensional subspace invariant under T' , any nonzero functional $\varphi \in U^0$ is an eigenvector of T' . Hence, T' has an eigenvalue. By Exercise 15, T must also have an eigenvalue. $\square$

(25) Suppose $T \in \mathcal{L}(V)$ and u, w are eigenvectors of T such that $u+w$ is also an eigenvector of T .

Prove that u and w are eigenvectors of T corresponding to the same eigenvalue.

PROOF: Let $u, w \in V \setminus \{0\}$ such that $Tu = \lambda_u u$, $Tw = \lambda_w w$, and $T(u + w) = \lambda_{u+w}(u + w)$. Since

$$T(u + w) = Tu + Tw = \lambda_u u + \lambda_w w = \lambda_{u+w}(u + w) = \lambda_{u+w}u + \lambda_{u+w}w,$$

it follows that

$$(\lambda_u - \lambda_{u+w})u + (\lambda_w - \lambda_{u+w})w = 0.$$

If u and w are linearly independent, the coefficients must vanish, so $\lambda_u = \lambda_{u+w} = \lambda_w$. Otherwise, if u and w are linearly dependent, we can write $u = \alpha w$ for some $\alpha \neq 0$. Applying T to u in two ways gives

$$Tu = T(\alpha w) = \alpha Tw = \alpha \lambda_w w \text{ and}$$

$$Tu = \lambda_u u = \lambda_u \alpha w.$$

Hence, $\lambda_u = \lambda_w$. □

26 Let u, v

33 Suppose $T \in \mathcal{L}(V)$ and m is a positive integer.

(a) Prove that T is injective $\iff T^m$ is injective.

(b) Prove that T is surjective $\iff T^m$ is surjective.

PROOF: We show (a) and (b) both with induction, first in the $\Rightarrow$ direction, and then in the $\Leftarrow$ direction.

(a) $\Rightarrow$ **DIRECTION:** Suppose T is injective. For $m = 1$ the statement is trivial ($T^1 = T$). Fix $m \geq 1$ and suppose as induction hypothesis that T^m is injective. If $T^{m+1}(v_1) = T^{m+1}(v_2)$, then

$$T(T^m(v_1)) = T(T^m(v_2)).$$

Since T is injective, $T^m(v_1) = T^m(v_2)$. The induction hypothesis forces $v_1 = v_2$. Therefore, T^{m+1} is injective as well. (In general, function composition is injective. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are injective, so is $g \circ f : X \rightarrow Z$. So, a finite composition of injective maps is injective.)

$\Leftarrow$ **DIRECTION:** Assume T^m is injective for some fixed $m \geq 1$. If $m = 1$ we are done. If $m > 1$, take $v_1, v_2 \in V$ with $T(v_1) = T(v_2)$. Apply T^{m-1} to both sides to get $T^{m-1}(T(v_1)) = T^{m-1}(T(v_2))$. Injectivity of T^m forces $v_1 = v_2$. Thus, T is injective.

(b) $\Rightarrow$ **DIRECTION:** Suppose T is surjective. For $m = 1$ the statement is trivial. Fix $m \geq 1$ and assume that T^m is surjective. Let $y \in V$ be arbitrary. Since T is surjective, there exists $z \in V$ with $T(z) = y$. By the induction hypothesis, there exists $v \in V$ with $T^m(v) = z$. Thus,

$$T^{m+1}(v) = T(T^m(v)) = T(z) = y.$$

Hence, T^{m+1} is surjective.

(Or: A finite composition of surjective maps (not necessarily linear) is surjective.)

$\Leftarrow$ **DIRECTION:** Assume T^m is surjective for some fixed $m \geq 1$. Let $y \in V$ be arbitrary. Choose $v \in V$ with $T^m(v) = y$ (always possible, since T^m is surjective). Put $x := T^{m-1}(v)$. Then

$$y = T(x) = T(T^{m-1}(v)) = T^m(v).$$

Hence, T is surjective. □

40 Suppose $S, T \in \mathcal{L}(V)$, and S is invertible. Suppose $p \in \mathcal{P}(\mathbb{F})$ is a polynomial. Prove that

$$p(STS^{-1}) = Sp(T)S^{-1}.$$

PROOF: We first want to prove that $(STS^{-1})^m = ST^mS^{-1}$ for every $m \in \mathbb{N}_{\geq 0}$. To this end, we use

induction over m . For $m = 0$, the statement holds because

$$(STS^{-1})^0 = I = SIS^{-1} = ST^0S^{-1}.$$

If $m = 1$ the claim is immediate. Hence, assume $m \geq 1$. Then

$$\begin{aligned} (STS^{-1})^{m+1} &= STS^{-1}(STS^{-1})^m \\ &= ST \underbrace{S^{-1}S}_{I} T^m S^{-1} \\ &= ST^{m+1}S^{-1}, \end{aligned}$$

This completes the induction. Let $p(z) = a_0z^0 + a_1z^1 + \cdots + a_mz^m$, $a_j \in \mathbb{F}$. Then

$$\begin{aligned} p(STS^{-1}) &= a_0I + a_1(STS^{-1}) + a_2(STS^{-1})^2 + \cdots + a_m(STS^{-1})^m = \sum_{j=0}^m a_j (STS^{-1})^j \\ &= a_0I \underbrace{SS^{-1}}_{=I=(SIS^{-1})^0} + a_1(STS^{-1}) + a_2(ST^2S^{-1}) + \cdots + a_m(ST^mS^{-1}) = \sum_{j=0}^m a_j (ST^jS^{-1}) \\ &= S(a_0I)S^{-1} + S(a_1T)S^{-1} + S(a_2T^2)S^{-1} + \cdots + S(a_mT^m)S^{-1} = \sum_{j=0}^m S(a_jT^j)S^{-1} \\ &= S(a_0I + a_1T + a_2T^2 + \cdots + a_mT^m)S^{-1} = S\left(\sum_{j=0}^m a_jT^j\right)S^{-1} \\ &= Sp(T)S^{-1}. \end{aligned}$$

Here we have been making use of the distributive property of linear maps, see 3.8. And we used that scalar multiples of the identity (i.e. a_0I) commute with every operator. $\square$

THE MINIMAL POLYNOMIAL (SECTION 5B)

EXISTENCE OF EIGENVALUES ON COMPLEX VECTOR SPACES (SUB-SECTION 5B1)

THM 5.19 (*existence of eigenvalues on complex vector spaces*):

Every operator on a finite-dimensional nonzero complex vector space has an eigenvalue.

PROOF: Suppose V is a finite-dimensional complex vector space with $\dim V = n \geq 1$, and let $T \in \mathcal{L}(V)$.

Let $v \in V$ be a nonzero vector. Then the list

$$v, Tv, T^2v, \dots, T^n v$$

has length $n + 1$, so it is linearly dependent. Thus, there exist scalars $c_0, c_1, \dots, c_n \in \mathbb{C}$, not all zero, such that

$$c_0 v + c_1 Tv + \dots + c_n T^n v = 0.$$

Let $p(z) := c_0 + c_1 z + \dots + c_n z^n \in \mathcal{P}(\mathbb{C})$. Since the coefficients are not all zero, p is not the zero polynomial, and $p(T)v = 0$. By the Fundamental Theorem of Algebra (Theorem 4.13), p factors as

$$p(z) = c(z - \lambda_1) \cdots (z - \lambda_m),$$

where $c \in \mathbb{C}$ is nonzero and $\lambda_1, \dots, \lambda_m \in \mathbb{C}$. Applying this factorization to T yields:

$$0 = p(T)v = c(T - \lambda_1 I) \cdots (T - \lambda_m I)v.$$

This implies that at least one of the operators $T - \lambda_j I$ is not injective. By Theorem 5.7, λ_j is an eigenvalue of T . $\square$

EIGENVALUES AND THE MINIMAL POLYNOMIAL (SUB-SECTION 5B2)

DEF 5.20 (*monic polynomial*):

A polynomial is called “*monic*” if its highest-degree coefficient equals 1.

EXAMPLE 5.21:

$p(z) = 2 + 9z^2 + z^7$ is a monic polynomial of degree 7.

THM 5.22 (*existence, uniqueness, and degree of minimal polynomial*):

Suppose $T \in \mathcal{L}(V)$ and V is finite-dimensional. Then there exists a unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that

$$p(T) = 0.$$

Furthermore, $\deg p \leq \dim V$.

PROOF: Before beginning our inductive proof, we examine some base cases. We first prove the existence of a monic polynomial that annihilates T and later establish its uniqueness; these two parts are independent.

BASE CASE: If $\dim V = 0$, then $V = \{\vec{0}\}$, and the identity operator I is the zero-operator on V . We let $p(z) := 1$ so that

$$p(T) = I = 0.$$

ANOTHER BASE CASE: Now suppose that $\dim V = 1$. For any nonzero $v \in V$, the list v, Tv is linearly dependent (since it has length 2), so there exists a scalar $c_0 \in \mathbb{F}$ such that

$$c_0 v + Tv = 0.$$

This follows from the linear dependence lemma Theorem 2.19. If we choose $q(z) := c_0 + z \quad \forall z \in \mathbb{F}$

$\mathbb{F}$, then

$$q(T)v = (c_0I + T)(v) = c_0v + Tv = 0.$$

In the case where T is the zero operator, one may instead choose $q(z) = z$. So we have $\dim \text{null } q(T) = 1$ and therefore $\dim \text{range } q(T) = \dim V - \dim \text{null } q(T) = 1 - 1 = 0$. Thus, every vector in this one-dimensional space is a scalar multiple of v and maps to zero by $q(T)$. Thus, $q(z)$ serves as a minimal polynomial for T .

YET ANOTHER BASE CASE: Suppose $\dim V = 2$. For any nonzero $v \in V$, the list v, Tv, T^2v is linearly dependent (since it has length 3). According to the linear dependence lemma Theorem 2.19, we either have

$$c_0v + Tv = 0 \quad \text{or} \quad c_0v + c_1Tv + T^2v = 0,$$

for some scalars $c_0, c_1 \in \mathbb{F}$. If we define $q(z) := c_0 + z$ in the first case or $q(z) := c_0 + c_1z + z^2$ in the second case accordingly, then $q(T)v = 0$.

In the first case where $(-1)c_0v = Tv$ and $q(z) = c_0 + z$, the vector v by itself is linearly independent and thus $\dim \text{null } q(T) \geq 1$. Therefore

$$\dim \text{range } q(T) \leq \dim V - \dim \text{null } q(T) \leq 1.$$

Since $\text{range } q(T)$ is at most one-dimensional and is invariant under T (by Theorem 5.18), we can restrict T to this subspace. In this restricted setting $\text{range } q(T)$, we know how to construct the desired polynomial; denote it by s , defined such that

$$s\left(T|_{\text{range } q(T)}\right) = 0.$$

Just for clarity, the equation above means that s is the zero-operator of $T|_{\text{range } q(T)}$. Therefore,

$$\forall u \in V : ((sq)(T))(u) = s(T)(q(T)u) = 0.$$

Thus, sq is a candidate for our minimal polynomial of T .

A similar reasoning goes for the case where $(-1)c_0v + (-1)c_1Tv = T^2v$ and $q(z) = c_0 + c_1z + z^2$. Here we have v and Tv as linearly independent vectors. Note that if $k \in \mathbb{N}$, then

$$q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0. \quad (5.2)$$

Hence the equation (5.2) above tells us that $v \in \text{null } q(T)$ and $Tv \in \text{null } q(T)$. Here in this case we conclude that $\dim \text{null } q(T) \geq 2$ and therefore $\dim \text{range } q(T) = 0$. Since $\dim \text{null } q(T) = 2$, we have that

$$\text{null } q(T) = V.$$

This means that every vector of V gets mapped to 0 if $q(T)$ is applied. So we have found a minimal monic polynomial!

GENERAL CASE: As we can see a pattern now, we can use induction on $\dim V$ and assume $\dim V > 2$ and that the theorem holds for all vector spaces of smaller dimension than $\dim V$. Let $v \in V$ be nonzero. Then the list

$$v, Tv, \dots, T^{\dim V} v \quad (5.3)$$

(which has length $1 + \dim V$) is linearly dependent. By the linear dependence lemma (Theorem 2.19), there is a smallest positive integer $m \leq \dim V$ and some $c_0, \dots, c_{m-1} \in \mathbb{F}$ such that

$$c_0v + c_1Tv + \dots + c_{m-1}T^{m-1}v + T^m v = 0.$$

Let us define $q \in \mathcal{P}_m(\mathbb{F})$ as

$$q(z) := c_0 + c_1z + \dots + c_{m-1}z^{m-1} + z^m$$

accordingly. Thus, we have

$$q(T)v = 0.$$

Note that $q(z)$ is a monic polynomial and that $\deg q = m$. Observe that if $k \in \mathbb{N}$, then

$$q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0. \quad (5.4)$$

By the linear dependence lemma (Theorem 2.19) and the definition of m , the list

$$v, Tv, \dots, T^{m-1}v \quad (5.5)$$

from before is linearly independent (This is not the same list as (5.3) since $m \leq \dim V$). This fact together with equation (5.4) imply that the vectors of the list (5.5) all belong to $\text{null } q(T)$. Hence,

$$\dim \text{null } q(T) \geq m.$$

Using the Rank-Nullity Theorem, this implies:

$$\dim \text{range } q(T) = \dim V - \dim \text{null } q(T) \leq \dim V - m.$$

Because $\text{range } q(T)$ is invariant under T by Theorem 5.18, we can apply our induction hypothesis to the operator $T|_{\text{range } q(T)}$. Hence, there exists a monic polynomial $s \in \mathcal{P}(\mathbb{F})$ such that $\deg s \leq \dim V - m$ and

$$s\left(T|_{\text{range } q(T)}\right) = 0.$$

In other words, $\forall u \in \text{range } q(T) : s(T)(u) = 0$. So we can safely say that

$$s(T)|_{\text{range } q(T)} = s\left(T|_{\text{range } q(T)}\right).$$

Because $\forall u \in V : q(T)u \in \text{range } q(T)$, we conclude:

$$\forall u \in V : (sq)(T)(u) = s(T)(q(T)u) = 0.$$

Therefore, sq is a monic polynomial such that

$$\deg sq \leq (\dim V - m) + m = \dim V$$

and $(sq)(T) = 0$.

PROOF OF UNIQUENESS: Let $p \in \mathcal{P}(\mathbb{F})$ be a monic polynomial of smallest degree such that

$$p(T) = 0.$$

Let $r \in \mathcal{P}(\mathbb{F})$ be another monic polynomial of the same degree such that $r(T) = 0$. This implies

$$(p - r)(T) = 0.$$

Moreover, $\deg(p - r) < \deg p = \deg r$. If $p - r \neq 0$, we could divide $p - r$ by the coefficient of the highest order term in $p - r$ to obtain a monic polynomial that, when applied to T , yields the zero-operator. This polynomial would have a smaller degree than p or r , which would be a contradiction. Therefore, $p - r = 0 \iff p = r$. $\square$

DEF 5.24 (minimal polynomial):

Let $T \in \mathcal{L}(V)$ on a finite-dimensional vector space V . The “minimal polynomial” of T is the unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$.

COMPUTATION OF THE MINIMAL POLYNOMIAL: Find the smallest $m \in \mathbb{N}$ such that

$$c_0 I + c_1 T + \dots + c_{m-1} T^{m-1} = -T^m$$

has a solution $c_0, \dots, c_{m-1} \in \mathbb{F}$. Solve for $m = 1, 2, \dots, \dim V$.

THM 5.27 (eigenvalues are the zeros of the minimal polynomial):

Let $T \in \mathcal{L}(V)$ and suppose V is finite-dimensional. Then:

- (a) The zeros of the minimal polynomial of T are precisely the eigenvalues of T .

(b) If V is a complex vector space, the minimal polynomial of T has the form

$$(z - \lambda_1) \cdots (z - \lambda_m),$$

where $\lambda_1, \dots, \lambda_m$ are the eigenvalues of T (possibly with repetitions).

PROOF: Let p be the minimal polynomial of T .

(a) $\Rightarrow$ **DIRECTION:** Suppose $\lambda \in \mathbb{F}$ is a zero of p . By Theorem 4.6, there exists a monic polynomial $q \in \mathcal{P}(\mathbb{F})$ such that $p(z) = (z - \lambda)q(z)$. Since $\deg q < \deg p$ and p is minimal, $q(T) \neq 0$. Thus, there exists $u \in V$ such that $q(T)u \neq 0$. Then

$$(T - \lambda I)(q(T)u) = p(T)u = 0,$$

so $q(T)u$ is an eigenvector of T with eigenvalue λ .

$\Leftarrow$ **DIRECTION:** Conversely, suppose $\lambda \in \mathbb{F}$ is an eigenvalue of T , so $Tu = \lambda u$ for some nonzero $u \in V$. Then $T^k u = \lambda^k u$ for all $k \in \mathbb{N}$, which implies $p(T)u = p(\lambda)u$. Since $p(T) = 0$, we have $0 = p(\lambda)u$. Because $u \neq 0$, it follows that $p(\lambda) = 0$.

(b) This follows directly from part (a) and the Fundamental Theorem of Algebra (Theorem 4.13). $\square$

THM 5.29 (*every annihilating polynomial is a multiple of the minimal polynomial*):

Let $T \in \mathcal{L}(V)$ on a finite-dimensional vector space V , and let $q \in \mathcal{P}(\mathbb{F})$. Then $q(T) = 0$ if and only if q is a polynomial multiple of the minimal polynomial of T .

PROOF: Let p denote the minimal polynomial of T .

$\Rightarrow$ **DIRECTION:** Suppose $q(T) = 0$. By the division algorithm (Theorem 4.9), there exist $s, r \in \mathcal{P}(\mathbb{F})$ such that

$$q = ps + r \quad \text{with } \deg r < \deg p.$$

Applying both sides to T gives:

$$0 = q(T) = p(T)s(T) + r(T) = r(T).$$

If $r \neq 0$, dividing r by its leading coefficient would give a monic polynomial of degree strictly less than $\deg p$ that annihilates T , contradicting the minimality of p . Thus, $r = 0$, so $q = ps$.

$\Leftarrow$ **DIRECTION:** If $q = ps$, then $q(T) = p(T)s(T) = 0 \cdot s(T) = 0$. $\square$

THM 5.31 (*minimal polynomial of a restriction operator*):

If U is a subspace of V that is invariant under $T \in \mathcal{L}(V)$, then the minimal polynomial of T is a polynomial multiple of the minimal polynomial of $T|_U$.

PROOF: Let p be the minimal polynomial of T . For all $u \in U$, we have $p(T|_U)u = p(T)u = 0$, so $p(T|_U) = 0$. By Theorem 5.29, p is a multiple of the minimal polynomial of $T|_U$. $\square$

THM 5.32 (*invertibility and the constant term of the minimal polynomial*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T is invertible if and only if the constant term of the minimal polynomial of T is nonzero.

PROOF: T is invertible $\iff 0$ is not an eigenvalue of $T \iff 0$ is not a zero of the minimal polynomial p of T (by Theorem 5.27) $\iff p(0) \neq 0$, which means the constant term of p is nonzero. $\square$

EIGENVALUES ON ODD-DIMENSIONAL REAL VECTOR SPACES (SUB-SECTION 5B3)

THM 5.33 (*even-dimensional null space*):

Suppose $\mathbb{F} = \mathbb{R}$, V is finite-dimensional, $T \in \mathcal{L}(V)$, and $b, c \in \mathbb{R}$ with $b^2 < 4c$. Then $\dim \text{null}(T^2 + bT + cI)$

is an even number.

PROOF: Let $N := \text{null}(T^2 + bT + cI)$, and let $S := T|_N$. By Theorem 5.18, N is invariant under T , so $S \in \mathcal{L}(N)$ with $S^2 + bS + cI = 0$. If $\lambda \in \mathbb{R}$ were an eigenvalue of S with eigenvector $v \neq 0$, then $(\lambda^2 + b\lambda + c)v = 0$. Since $b^2 < 4c$, $\lambda^2 + b\lambda + c = (\lambda + b/2)^2 + (c - b^2/4) > 0$, forcing $v = 0$, a contradiction. Thus, S has no real eigenvalues.

Let U be an even-dimensional S -invariant subspace of N of maximal dimension. If $U = N$, we are done. Otherwise, choose $w \in N \setminus U$, and set $W := \text{span}(w, Sw)$. Since $S^2w = -bSw - cw$, W is invariant under S . Because S has no eigenvalues, w and Sw are linearly independent, so $\dim W = 2$. Furthermore, $U \cap W$ is an S -invariant subspace of W . Since S has no eigenvectors, $\dim(U \cap W)$ cannot be 1, and since $w \notin U$, $\dim(U \cap W) \neq 2$, so $U \cap W = \{0\}$. Then $\dim(U + W) = \dim U + 2$, contradicting the maximality of U . Therefore, $U = N$, and $\dim N$ is even. $\square$

THM 5.34 (*operators on odd-dimensional vector spaces have eigenvalues*):

Every operator on an odd-dimensional real vector space has an eigenvalue.

PROOF: Let V be a finite-dimensional real vector space of odd dimension $n := \dim V$. We use induction on n (in steps of 2). If $n = 1$, every nonzero vector is an eigenvector. Suppose $n \geq 3$ and the result holds for all odd-dimensional spaces of dimension less than n . Let p be the minimal polynomial of T . If p has a degree-one factor $x - \lambda$, then $\lambda \in \mathbb{R}$ is an eigenvalue of T by Theorem 5.27. Otherwise, by Theorem 4.16, p has a factor $x^2 + bx + c$ with $b^2 < 4c$, so $p(x) = q(x)(x^2 + bx + c)$. Then $\text{range}(T^2 + bT + cI)$ is invariant under T , and by Theorem 3.21 and Theorem 5.33,

$$\dim \text{range}(T^2 + bT + cI) = \dim V - \dim \text{null}(T^2 + bT + cI)$$

is an odd number strictly less than $\dim V$. By the induction hypothesis, $T|_{\text{range}(T^2 + bT + cI)}$ has an eigenvalue, which is also an eigenvalue of T . $\square$

EXERCISES ABOUT THE MINIMAL POLYNOMIAL 5B

- ① Suppose $T \in \mathcal{L}(V)$. Prove that 9 is an eigenvalue of $T^2 \iff 3$ or -3 is an eigenvalue of T .

PROOF: $\Leftarrow$ **DIRECTION:** Suppose -3 is an eigenvalue of T . Thus, there exists $v \in V$ such that $Tv = -3v$. If we apply T a second time, we get $TTv = T(-3v) = -3Tv = (-3)(-3)v = 9v$. Therefore, 9 is an eigenvalue of T^2 . The case where we start with 3 as an eigenvalue of T works analogously.

$\Rightarrow$ **DIRECTION:** Let $v \neq 0$ satisfy $T^2v = 9v$. Then

$$0 = (T^2 - 9I)v = (T - 3I)(T + 3I)v.$$

Set $w := (T + 3I)v$. Then $(T - 3I)w = 0$, so $Tw = 3w$ (because $w \in \text{null}(T - 3I)$). There are two cases:

- If $w = 0$, then $(T + 3I)v = 0$. Hence, $Tv = -3v$ and -3 is an eigenvalue of T .
- If $w \neq 0$, then w is an eigenvector of T with eigenvalue 3 .

Thus, T has eigenvalue 3 or -3 , as required. $\square$

- ③ Suppose $n \in \mathbb{N}_{>0}$ and $T \in \mathcal{L}(\mathbb{F}^n)$ is defined by $T(x_1, \dots, x_n) := (x_1 + \dots + x_n, \dots, x_1 + \dots + x_n)$. The

matrix $\mathcal{M}(T)$ of T looks like this

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}$$

(a) Find all eigenvalues and eigenvectors of T .

(b) Find the minimal polynomial of T .

SOLUTION: We want to find all $x \in V$ such that $Tx = \lambda x$. Therefore,

$$x_1 + \cdots + x_n = \lambda x_1$$

$$\vdots$$

$$x_1 + \cdots + x_n = \lambda x_n$$

If we define $S := x_1 + \cdots + x_n$, we can rewrite the system of equations as

$$S = \lambda x_1, \dots, S = \lambda x_n, \quad (5.6)$$

or equivalently: $S = \lambda x_1 = \lambda x_2 = \cdots = \lambda x_n$. Solving by x_i for $\lambda \neq 0$ gives

$$x_1 = \frac{S}{\lambda}, x_2 = \frac{S}{\lambda}, \dots, x_n = \frac{S}{\lambda}$$

Summing up all equations for S in (5.6) gives:

$$nS = \lambda x_1 + \cdots + \lambda x_n = \lambda S$$

Hence,

$$nS - \lambda S = 0$$

$$S(n - \lambda) = 0. \quad (\text{which is the same as } (S - 0)(n - \lambda) = 0)$$

If $\lambda = n$, then $x_1 = x_2 = \cdots = x_n$. Meaning all eigenvectors corresponding to the eigenvalue n are scalar multiples of $(1, 1, \dots, 1) \in \mathbb{F}^n$.

If $\lambda \neq n$, then $S(\lambda - n) = 0$ implies that $S = 0$ and thus, $\lambda = 0$. This means that x belongs to the eigenspace of 0, if $x_1 + x_2 + x_3 + \cdots + x_n = 0$.

THE MINIMAL POLYNOMIAL OF T :

$$T(e_1) = e_1 + \cdots + e_n$$

$$T(e_2) = e_1 + \cdots + e_n$$

$$\vdots$$

$$T(e_n) = e_1 + \cdots + e_n$$

Define $v := e_1 + \cdots + e_n$. Note that $\text{range } T = \text{span}(v)$. Applying T to v gives

$$T(v) = T(e_1 + \cdots + e_n) = T(e_1) + \cdots + T(e_n) = nv$$

$$T^2(v) = nT(v)$$

The last line tells us that $T^2 = nT$ or equivalently $T(T - nI) = 0$. Therefore, the minimal polynomial is

$$p(z) = z(z - n).$$

Because $\text{range } T = \text{span}(v)$, the polynomial is in fact minimal.

- ④ Suppose $\mathbb{F} = \mathbb{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbb{C})$, and $\alpha \in \mathbb{C}$. Prove that α is an eigenvalue of $p(T) \iff \alpha = p(\lambda)$ for some eigenvalue λ of T .

PROOF:

$\Leftarrow$ **DIRECTION:** Suppose $\lambda \in \mathbb{C}$ is an eigenvalue of T with $\alpha = p(\lambda)$. Then there exists a nonzero $v \in V$ such that $Tv = \lambda v$. Repeated application of T yields $T^k v = \lambda^k v$ for all $k \in \mathbb{N}$, and therefore $p(T)v = p(\lambda)v = \alpha v$. Since $v \neq 0$, α is an eigenvalue of $p(T)$.

$\Rightarrow$ **DIRECTION:** Conversely, suppose α is an eigenvalue of $p(T)$. Consider the polynomial $q(z) := p(z) - \alpha \in \mathcal{P}(\mathbb{C})$. Since $\mathbb{F} = \mathbb{C}$, by the Fundamental Theorem of Algebra (second version), $q(z)$ factors completely into linear factors over $\mathbb{C}$:

$$p(z) - \alpha = c(z - \lambda_1)(z - \lambda_2) \cdots (z - \lambda_m),$$

for some non-zero scalar $c \in \mathbb{C}$ and $\lambda_1, \dots, \lambda_m \in \mathbb{C}$. Substituting T into this polynomial equation gives:

$$p(T) - \alpha I = c(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_m I).$$

Since α is an eigenvalue of $p(T)$, the operator $p(T) - \alpha I$ is not injective (and hence not invertible). A product of linear maps is non-invertible if and only if at least one factor is non-invertible. Therefore, there exists some $j \in \{1, \dots, m\}$ such that $T - \lambda_j I$ is not injective. This means λ_j is an eigenvalue of T . Evaluating the polynomial identity at $z = \lambda_j$ yields:

$$p(\lambda_j) - \alpha = c(\lambda_j - \lambda_1) \cdots (\lambda_j - \lambda_j) \cdots (\lambda_j - \lambda_m) = 0 \implies \alpha = p(\lambda_j).$$

Setting $\lambda := \lambda_j$ completes the proof. $\square$

- 12** Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $T(x_1, x_2, x_3, \dots, x_n) = (x_1, 2x_2, 3x_3, \dots, nx_n)$. Find the minimal polynomial of T .

SOLUTION: For a standard basis $e_1, \dots, e_n$ we have $T(e_1 + \cdots + e_n) = e_1 + 2e_2 + \cdots + ne_n$. Or equivalently:

$$T(e_1) = e_1$$

$$T(e_2) = 2e_2$$

$$\vdots$$

$$T(e_n) = ne_n.$$

Hence, $e_1, \dots, e_n$ are $\dim \mathbb{F}^n = n$ distinct eigenvectors with corresponding eigenvalues $1, \dots, n$. This makes $Te_1, \dots, Te_n$ linearly independent as well. Using Theorem 5.27, we conclude that the minimal polynomial of T looks like this:

$$p(z) = (z - 1)(z - 2) \cdots (z - n) \quad \forall z \in \mathbb{F}. \quad (5.7)$$

We can check if $\forall v \in \mathbb{F}^n$, given as $v = a_1 e_1 + \cdots + a_n e_n$ ($a_i \in \mathbb{F}$), the following equation holds:

$$(T - I)(T - 2I) \cdots (T - nI)v = 0.$$

This is true because $a_k e_k \in \text{null}(T - kI)$ for $k = 1, \dots, n$. Therefore,

$$\begin{aligned} (T - I)(T - 2I) \cdots (T - nI)v &= (T - I)(T - 2I)(T - 3I) \cdots (T - nI)(a_1 e_1) \\ &\quad + (T - I)(T - 2I)(T - 3I) \cdots (T - nI)(a_2 e_2) \\ &\quad + (T - I)(T - 2I)(T - 3I) \cdots (T - nI)(a_3 e_3) \\ &\quad + \cdots \\ &\quad + (T - I)(T - 2I)(T - 3I) \cdots (T - nI)(a_n e_n) = 0. \end{aligned}$$

Thus, we have found our minimal polynomial given in equation (5.7). It is easy to conclude that there exists no polynomial of smaller degree that distributes to every basis vector of the sum $a_1 e_1 + \cdots + a_n e_n$.

- 16** Suppose $a_0, a_1, \dots, a_{n-1} \in \mathbb{F}$. Let T be the operator on $\mathbb{F}^n$ whose matrix (with respect to the

standard basis) is

$$\mathcal{M}(T) = \begin{matrix} & e_1 & e_2 & \cdots & e_{n-1} & e_n \\ \begin{matrix} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_{n-1} \\ e_n \end{matrix} & \begin{pmatrix} 0 & & & & -a_0 \\ 1 & 0 & & & -a_1 \\ & 1 & \ddots & & -a_2 \\ & & \ddots & & \vdots \\ & & & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{pmatrix} \end{matrix}.$$

Note that all the other entries of this matrix are zero. Show that the minimal polynomial of T is the polynomial

$$a_0 + a_1z + \cdots + a_{n-1}z^{n-1} + z^n.$$

SOLUTION: Looking at the matrix $\mathcal{M}(T)$ and having a glance at Theorem 3.31, we observe that

$$\begin{aligned} Te_1 &= e_2 \\ Te_2 &= e_3 = T(Te_1) = T^2e_1 \\ Te_3 &= e_4 = T(Te_2) = T^3e_1 \\ Te_4 &= e_5 = T(Te_3) = T^4e_1 \\ &\vdots \\ Te_{n-1} &= e_n = T(Te_{n-2}) = T^{n-1}e_1 \\ Te_n &= -a_0e_1 - a_1e_2 - a_2e_3 - \cdots - a_{n-2}e_{n-1} - a_{n-1}e_n = T(T^{n-1}e_1) = T^ne_1. \end{aligned}$$

We can rewrite the last expression as

$$-T^n(e_1) = a_0e_1 + a_1Te_1 + a_2T^2e_1 + \cdots + a_{n-2}T^{n-2}e_1 + a_{n-1}T^{n-1}e_1$$

The list $e_1, Te_1, T^2e_1, T^3e_1, \dots, T^{n-1}e_1$, which equals the list $e_1, e_2, e_3, \dots, e_n$, is linearly independent. Therefore, no non-zero polynomial of degree strictly less than n can annihilate e_1 . Hence, the minimal polynomial of T is

$$a_0 + a_1z + \cdots + a_{n-1}z^{n-1} + z^n.$$

UPPER-TRIANGULAR MATRICES (SECTION 5C)

THM 5.39 (conditions for upper-triangular matrix):

Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . The following are equivalent:

- (a) The matrix $\mathcal{M}(T)$ of T with respect to $v_1, \dots, v_n$ is upper triangular.
- (b) $\text{span}(v_1, \dots, v_k)$ is invariant under T for every $k \in \{1, \dots, n\}$.
- (c) $Tv_k \in \text{span}(v_1, \dots, v_k)$ for every $k \in \{1, \dots, n\}$.

PROOF: We prove (a) $\implies$ (b) $\implies$ (c) $\implies$ (a).

STEP 1: Suppose (a) holds, and let $k \in \{1, \dots, n\}$. For each $j \in \{1, \dots, k\}$, the definition of the matrix of a linear map (Theorem 3.31) and the upper-triangular form of $\mathcal{M}(T)$ give:

$$Tv_j = \sum_{\ell=1}^j A_{\ell,j} v_\ell \in \text{span}(v_1, \dots, v_j) \subseteq \text{span}(v_1, \dots, v_k).$$

Since each basis vector v_j for $j \leq k$ is mapped into $\text{span}(v_1, \dots, v_k)$, the subspace $\text{span}(v_1, \dots, v_k)$ is invariant under T , proving (b).

STEP 2: If (b) holds, then $Tv_k \in \text{span}(v_1, \dots, v_k)$ for each $k \in \{1, \dots, n\}$, which is (c).

STEP 3: If (c) holds, then for each $k \in \{1, \dots, n\}$, $Tv_k = \sum_{j=1}^k a_j v_j = \sum_{j=1}^k a_j v_j + \sum_{j=k+1}^n 0v_j$. Thus, all entries in column k below the k -th row are zero. Hence, $\mathcal{M}(T)$ is upper triangular, which proves (a). $\square$

THM 5.40 (equation satisfied by operator with upper-triangular matrix):

Suppose $T \in \mathcal{L}(V)$ and V has a basis with respect to which T has an upper-triangular matrix with diagonal entries $\lambda_1, \dots, \lambda_n$. Then

$$(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0.$$

PROOF: Let $v_1, \dots, v_n$ be a basis of V with respect to which $\mathcal{M}(T)$ is upper triangular with diagonal entries $\lambda_1, \dots, \lambda_n$. Then $Tv_1 = \lambda_1 v_1$, so $(T - \lambda_1 I)v_1 = 0$. For $k \geq 2$, $(T - \lambda_k I)v_k \in \text{span}(v_1, \dots, v_{k-1})$. By induction on k , it follows that

$$(T - \lambda_1 I) \cdots (T - \lambda_k I)v_k = 0.$$

Since the polynomial factors $(T - \lambda_j I)$ commute with each other by Theorem 5.17, we have

$$(T - \lambda_1 I) \cdots (T - \lambda_n I)v_k = 0 \quad \text{for each } k \in \{1, \dots, n\}.$$

Since $(T - \lambda_1 I) \cdots (T - \lambda_n I)$ evaluates to 0 on each vector in a basis of V , it must be the zero operator on V . $\square$

THM 5.41 (determination of eigenvalues from upper-triangular matrix):

Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V . Then the eigenvalues of T are precisely the entries on the diagonal of that upper-triangular matrix.

PROOF: Let $v_1, \dots, v_n$ be a basis of V with respect to which $\mathcal{M}(T)$ is upper triangular with diagonal entries $\lambda_1, \dots, \lambda_n$. First, $Tv_1 = \lambda_1 v_1$, so λ_1 is an eigenvalue. For $k \in \{2, \dots, n\}$, $(T - \lambda_k I)$ maps $\text{span}(v_1, \dots, v_k)$ into $\text{span}(v_1, \dots, v_{k-1})$. Since $\dim \text{span}(v_1, \dots, v_{k-1}) = k-1 < k = \dim \text{span}(v_1, \dots, v_k)$, the restriction of $T - \lambda_k I$ to $\text{span}(v_1, \dots, v_k)$ is not injective by Theorem 3.22. Thus, there exists a nonzero $v \in \text{span}(v_1, \dots, v_k)$ such that $(T - \lambda_k I)v = 0$, so each λ_k is an eigenvalue of T . Conversely, let $q(z) := (z - \lambda_1) \cdots (z - \lambda_n)$. By Theorem 5.40, $q(T) = 0$. By Theorem 5.29, the minimal polynomial of T divides q . Since the eigenvalues of T are the zeros of its minimal polynomial (Theorem 5.27), every eigenvalue of T must be one of the zeros of q , that is, an element of $\{\lambda_1, \dots, \lambda_n\}$. $\square$

THM 5.44 (*characterization of operators with upper-triangular matrices*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some basis of V if and only if the minimal polynomial of T factors into degree-one polynomials:

$$(z - \lambda_1) \cdots (z - \lambda_m) \quad \text{for some } \lambda_1, \dots, \lambda_m \in \mathbb{F}.$$

PROOF: $\Rightarrow$ **DIRECTION:** Suppose T has an upper-triangular matrix with diagonal entries $\alpha_1, \dots, \alpha_n \in \mathbb{F}$. Let $q(z) := (z - \alpha_1) \cdots (z - \alpha_n)$. By Theorem 5.40, $q(T) = 0$. By Theorem 5.29, the minimal polynomial p of T divides q , so $p(z) = (z - \lambda_1) \cdots (z - \lambda_m)$ with $\{\lambda_1, \dots, \lambda_m\} \subseteq \{\alpha_1, \dots, \alpha_n\} \subseteq \mathbb{F}$.

$\Leftarrow$ **DIRECTION:** Suppose the minimal polynomial of T is $p(z) = (z - \lambda_1) \cdots (z - \lambda_m)$ with $\lambda_1, \dots, \lambda_m \in \mathbb{F}$. We use induction on m . If $m = 1$, then $T = \lambda_1 I$, whose matrix with respect to any basis is $\lambda_1 I$, which is diagonal (hence upper triangular). Now suppose $m > 1$ and the result holds for all operators whose minimal polynomial factors into fewer than m linear terms. Let $U := \text{range}(T - \lambda_m I)$. By Theorem 5.18, U is invariant under T . For all $u = (T - \lambda_m I)v \in U$, we have:

$$(T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)u = p(T)v = 0.$$

Thus, the minimal polynomial of $T|_U$ divides $(z - \lambda_1) \cdots (z - \lambda_{m-1})$. By the induction hypothesis, there exists a basis $u_1, \dots, u_M$ of U with respect to which $T|_U$ has an upper-triangular matrix. Extend $u_1, \dots, u_M$ to a basis $u_1, \dots, u_M, v_1, \dots, v_N$ of V . For each $k \in \{1, \dots, N\}$,

$$Tv_k = (T - \lambda_m I)v_k + \lambda_m v_k \in U + \text{span}(v_k) = \text{span}(u_1, \dots, u_M, v_k).$$

By Theorem 5.39, T has an upper-triangular matrix with respect to the basis $u_1, \dots, u_M, v_1, \dots, v_N$.

□

THM 5.47 (*upper-triangular matrix over $\mathbb{C}$*):

Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some basis of V .

PROOF: By the Fundamental Theorem of Algebra (Theorem 4.13), the minimal polynomial of T factors completely into degree-one polynomials over $\mathbb{C}$. The result then follows immediately from Theorem 5.44. □

EXERCISES ABOUT UPPER-TRIANGULAR MATRICES 5C

10 Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Show that the following are equivalent.

- (a) The matrix $\mathcal{M}(T)$ of T with respect to $v_1, \dots, v_n$ is lower triangular.
- (b) $\text{span}(v_k, \dots, v_n)$ is invariant under T for each $k = 1, \dots, n$.
- (c) $Tv_k \in \text{span}(v_k, \dots, v_n)$ for each $k = 1, \dots, n$.

PROOF: Like in 5.39, we will prove that (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (a).

STEP 1: Suppose (a) holds and fix $k \in \{1, \dots, n\}$. Consider $j \in \{k, \dots, n\}$ (i.e., $k \leq j$). The matrix $\mathcal{M}(T)$ of T looks as follows.

$$\mathcal{M}(T) = \begin{array}{c} v_1 \quad \cdots \quad v_j \quad \cdots \quad v_n \\ \begin{pmatrix} A_{1,1} & & & & \\ \vdots & \ddots & & & \\ A_{j,1} & \cdots & A_{j,j} & & \\ \vdots & \vdots & \ddots & \ddots & \\ A_{n,1} & \cdots & A_{n,j} & \cdots & A_{n,n} \end{pmatrix} \end{array}.$$

We have

$$Tv_j = A_{j,j}v_j + A_{j+1,j}v_{j+1} + \cdots + A_{n,j}v_n = \sum_{\ell=1}^n A_{\ell,j}v_\ell.$$

Therefore, $Tv_j \in \text{span}(v_j, \dots, v_n)$. Because

$$\text{span}(v_j, \dots, v_n) \subseteq \text{span}(v_k, \dots, v_n),$$

it's immediate that $Tv_j \in \text{span}(v_k, \dots, v_n)$. Since v_j could be any basis vector in the list $v_k, \dots, v_n$, $\text{span}(v_k, \dots, v_n)$ is invariant under T , completing the proof that (a) implies (b).

STEP 2: Now, suppose (b) holds. Hence, we assume $\text{span}(v_k, \dots, v_n)$ is invariant under T for each $k = 1, \dots, n$. In particular,

$$Tv_j \in \text{span}(v_j, \dots, v_n) \quad \forall j \in \{1, \dots, n\}.$$

Thus, (b) implies (c)

STEP 3: Finally, suppose (c) holds. Therefore, $Tv_k \in \text{span}(v_k, \dots, v_n)$ for $k = 1, \dots, n$. This implies that for a fixed k , there exist scalars $a_k, \dots, a_n \in \mathbb{F}$ such that

$$Tv_k = a_kv_k + \cdots + a_nv_n = 0v_1 + 0v_2 + \cdots + 0v_{k-1} + a_kv_k + \cdots + a_nv_n.$$

This implies that all the entries above the diagonal of $\mathcal{M}(T)$ are 0, because the list $v_1, \dots, v_n$ is linearly independent and we cannot choose another representation for Tv_k in respect to that list. Therefore, the scalars $a_k, \dots, a_n$ coincide with the entries of $\mathcal{M}(T)$. Thus, $\mathcal{M}(T)$ is a lower-triangular matrix, completing the proof that (c) implies (a). $\square$

DIAGONALIZABLE OPERATORS (SECTION 5D)

DIAGONAL MATRICES (SUB-SECTION 5D1)

DEF 5.48 (*diagonal matrix*):

A “*diagonal matrix*” is a square matrix that is 0 everywhere except possibly on the main diagonal.

DEF 5.50 (*diagonalizable operator*):

An operator $T \in \mathcal{L}(V)$ is called “*diagonalizable*” if T has a diagonal matrix with respect to some basis of V .

DEF 5.52 (*eigenspace, $E(\lambda, T)$*):

Let $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. The “*eigenspace*” of T corresponding to λ is the subspace $E(\lambda, T)$ of V defined by

$$E(\lambda, T) := \text{null}(T - \lambda I) = \{v \in V \mid Tv = \lambda v\}.$$

THM 5.54 (*sum of eigenspaces is a direct sum*):

Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of T . Then

$$E(\lambda_1, T) + \dots + E(\lambda_m, T)$$

is a direct sum. Furthermore, if V is finite-dimensional, then

$$\dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T) = \dim(E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)) \leq \dim V.$$

PROOF: Suppose $v_1 + \dots + v_m = 0$, where each $v_k \in E(\lambda_k, T)$. By Theorem 5.11, eigenvectors corresponding to distinct eigenvalues are linearly independent. Thus, each $v_k = 0$, which proves that the sum is a direct sum by Theorem 1.45. The dimension inequality follows from Theorem 3.94 and Theorem 2.37. $\square$

CONDITIONS FOR DIAGONALIZABILITY (SUB-SECTION 5D2)

THM 5.55 (*conditions equivalent to diagonalizability*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T . The following are equivalent:

- (a) T is diagonalizable;
- (b) V has a basis consisting of eigenvectors of T ;
- (c) $V = E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)$;
- (d) $\dim V = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T)$.

PROOF: We prove (a) $\iff$ (b), (b) $\implies$ (c), (c) $\implies$ (d), and (d) $\implies$ (b).

STEP 1: (a) $\iff$ (b): The matrix of T with respect to a basis $v_1, \dots, v_n$ is diagonal if and only if each Tv_k is a scalar multiple of v_k , which means each v_k is an eigenvector of T .

STEP 2: (b) $\implies$ (c): If V has a basis of eigenvectors, then every vector in V is in the sum $E(\lambda_1, T) + \dots + E(\lambda_m, T)$. By Theorem 5.54, this sum is direct, so $V = E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)$.

STEP 3: (c) $\implies$ (d): Follows directly from Theorem 3.94.

STEP 4: (d) $\implies$ (b): Choosing a basis of each eigenspace $E(\lambda_k, T)$ and combining them yields a list of $\dim V$ eigenvectors. By Theorem 5.54, this list is linearly independent, and by Theorem 2.38, it is a basis of V . $\square$

THM 5.58 (*enough eigenvalues implies diagonalizability*):

If $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then T is diagonalizable.

PROOF: Let $\lambda_1, \dots, \lambda_n$ (where $n = \dim V$) be distinct eigenvalues of T , with corresponding eigenvectors $v_1, \dots, v_n$. By Theorem 5.11, $v_1, \dots, v_n$ is linearly independent. By Theorem 2.38, it is a basis of V consisting of eigenvectors, so T is diagonalizable by Theorem 5.55. $\square$

THM 5.62 (*characterization of diagonalizability via minimal polynomial*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T is diagonalizable if and only if the minimal polynomial of T equals

$$(z - \lambda_1) \cdots (z - \lambda_m)$$

for distinct $\lambda_1, \dots, \lambda_m \in \mathbb{F}$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose T is diagonalizable with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Then V has a basis of eigenvectors of T . For any basis vector v , we have $(T - \lambda_k I)v = 0$ for some $k \in \{1, \dots, m\}$. Since the factors $(T - \lambda_j I)$ commute,

$$(T - \lambda_1 I) \cdots (T - \lambda_m I)v = 0.$$

Thus, $(T - \lambda_1 I) \cdots (T - \lambda_m I) = 0$. By Theorem 5.29, the minimal polynomial of T divides $(z - \lambda_1) \cdots (z - \lambda_m)$. Since each λ_k is an eigenvalue and thus a zero of the minimal polynomial (Theorem 5.27), the minimal polynomial must be precisely $(z - \lambda_1) \cdots (z - \lambda_m)$.

$\Leftarrow$ **DIRECTION:** Suppose the minimal polynomial of T is $p(z) = (z - \lambda_1) \cdots (z - \lambda_m)$ with $\lambda_1, \dots, \lambda_m$ distinct. We use induction on m . If $m = 1$, then $T = \lambda_1 I$, which is clearly diagonalizable. Now suppose $m > 1$ and the result holds for smaller m . Let $U := \text{range}(T - \lambda_m I)$. By Theorem 5.18, U is invariant under T . For all $u = (T - \lambda_m I)v \in U$, we have $(T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)u = p(T)v = 0$. Thus, the minimal polynomial of $T|_U$ divides $(z - \lambda_1) \cdots (z - \lambda_{m-1})$. By the induction hypothesis, $T|_U$ is diagonalizable, so U has a basis consisting of eigenvectors of T .

Next, suppose $x \in U \cap \text{null}(T - \lambda_m I)$. Then $Tx = \lambda_m x$. Since $x \in U$, we also have

$$0 = (T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)x = (\lambda_m - \lambda_1) \cdots (\lambda_m - \lambda_{m-1})x.$$

Since the λ_j are distinct, this forces $x = 0$. Thus, $U \cap \text{null}(T - \lambda_m I) = \{0\}$, so $U + \text{null}(T - \lambda_m I) = U \oplus \text{null}(T - \lambda_m I)$. By the Rank-Nullity Theorem (Theorem 3.21),

$$\dim(U \oplus \text{null}(T - \lambda_m I)) = \dim U + \dim \text{null}(T - \lambda_m I) = \dim V.$$

Hence, $V = U \oplus \text{null}(T - \lambda_m I) = U \oplus E(\lambda_m, T)$. Combining a basis of eigenvectors of U with a basis of $E(\lambda_m, T)$ yields a basis of V consisting of eigenvectors of T , which proves that T is diagonalizable. $\square$

THM 5.65 (*restriction of diagonalizable operator to invariant subspace*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$ is diagonalizable. If U is a subspace of V that is invariant under T , then $T|_U$ is a diagonalizable operator on U .

PROOF: Since T is diagonalizable, by Theorem 5.62, the minimal polynomial of T is of the form $(z - \lambda_1) \cdots (z - \lambda_m)$ with distinct $\lambda_j \in \mathbb{F}$. By Theorem 5.31, the minimal polynomial of $T|_U$ divides the minimal polynomial of T . Thus, the minimal polynomial of $T|_U$ also factors into distinct linear factors over $\mathbb{F}$. By Theorem 5.62, $T|_U$ is diagonalizable. $\square$

EXERCISES ABOUT DIAGONALIZABLE OPERATORS 5D

- ② Suppose $T \in \mathcal{L}(V)$ has a diagonal matrix $A = \mathcal{M}(T)$ with respect to some basis of V . Prove that if $\lambda \in \mathbb{F}$, then λ appears on the diagonal of A precisely $\dim E(\lambda, T)$ times.

KI Bemerkung: Es gab hier drei verschiedene, teilweise fehlerhafte oder unvollständige Beweise für dieselbe Aufgabenstellung (die zudem als Exercise 2 und 3 dupliziert war). Ich habe

diese zusammengefasst und den besten, ausführlichsten Beweis behalten sowie die Makros (wie $\backslash\text{linmap}$ statt $\backslash\text{op}$) vereinheitlicht.

PROOF: If λ^* is not an eigenvalue of T , then

$$\dim E(\lambda^*, T) = \dim\{v \in V : Tv = \lambda^*v\} = \dim\{0\} = 0,$$

and therefore λ^* does not appear among the diagonal entries of A .

Now assume λ^* is an eigenvalue. Let $\mathcal{B} = \{v_1, \dots, v_\ell\}$ be a basis of V with respect to which $A = \mathcal{M}(T)$ is diagonal; write

$$A = \text{diag}(\mu_1, \dots, \mu_\ell),$$

so that for each j we have $Tv_j = \mu_j v_j$. In particular every v_j is an eigenvector of T (possibly for different eigenvalues).

Define the subset of basis vectors corresponding to the eigenvalue λ^* by

$$S = \{v_j \in \mathcal{B} \mid \mu_j = \lambda^*\}.$$

We first observe that $\text{span}(S) \subseteq E(\lambda^*, T)$, since each vector in S is by definition a λ^* -eigenvector and linear combinations of λ^* -eigenvectors remain in the λ^* -eigenspace.

For the reverse inclusion, let $w \in E(\lambda^*, T)$. Express w in the basis $\mathcal{B}$:

$$w = \sum_{j=1}^{\ell} c_j v_j \quad (c_j \in \mathbb{F}).$$

Applying T to both sides and using $Tv_j = \mu_j v_j$ gives

$$\lambda^* \sum_{j=1}^{\ell} c_j v_j = Tw = \sum_{j=1}^{\ell} c_j \mu_j v_j.$$

By uniqueness of coordinates with respect to the basis $\mathcal{B}$, the coefficient of v_j on the left must equal the coefficient of v_j on the right for each j , hence

$$(\lambda^* - \mu_j)c_j = 0 \quad \text{for every } j = 1, \dots, \ell.$$

Consequently $c_j = 0$ whenever $\mu_j \neq \lambda^*$, so every nonzero coefficient c_j corresponds to some $v_j \in S$. Thus w is a linear combination of vectors from S , meaning $w \in \text{span}(S)$. This proves the reverse inclusion and therefore

$$\text{span}(S) = E(\lambda^*, T).$$

Since S is a subset of the basis $\mathcal{B}$, its vectors are linearly independent; hence the number of elements of S equals the dimension of its span. Therefore

$$|S| = \dim(\text{span}(S)) = \dim E(\lambda^*, T).$$

Finally, by definition $|S|$ is exactly the number of diagonal entries of A equal to λ^* . We conclude that the scalar λ^* appears on the diagonal of A precisely $\dim E(\lambda^*, T)$ times. $\square$

COMMUTING OPERATORS (SECTION 5E)

DEF 5.66 (*commuting operators*):Two operators $S, T \in \mathcal{L}(V)$ “commute” if $ST = TS$.**THM 5.67** (*eigenspace is invariant under commuting operator*):Suppose $S, T \in \mathcal{L}(V)$ commute and $\lambda \in \mathbb{F}$. Then the eigenspace $E(\lambda, S)$ is invariant under T .**PROOF:** Let $v \in E(\lambda, S)$. Then $Sv = \lambda v$. Since S and T commute,

$$S(Tv) = (ST)v = (TS)v = T(Sv) = T(\lambda v) = \lambda(Tv).$$

Hence, $Tv \in E(\lambda, S)$, which shows that $E(\lambda, S)$ is invariant under T . □

INNER PRODUCT SPACES (CHAPTER 6)

INNER PRODUCTS AND NORMS (SECTION 6A)

INNER PRODUCTS (SUB-SECTION 6A1)

For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, the Euclidean norm is defined by

$$\|x\| := \sqrt{x_1^2 + \dots + x_n^2}.$$

DEF 6.1 (*dot product*):

The “dot product” of $x, y \in \mathbb{R}^n$ is defined by

$$x \cdot y := x_1 y_1 + \dots + x_n y_n.$$

For $z = (z_1, \dots, z_n) \in \mathbb{C}^n$, the norm is defined by

$$\|z\| := \sqrt{|z_1|^2 + \dots + |z_n|^2} = \sqrt{z_1 \bar{z}_1 + \dots + z_n \bar{z}_n}.$$

DEF 6.2 (*inner product*):

An “inner product” on V is a function that assigns to each ordered pair of vectors $u, v \in V$ a scalar $\langle u, v \rangle \in \mathbb{F}$ satisfying the following properties:

- **POSITIVITY:** $\langle v, v \rangle \geq 0$ for all $v \in V$;
- **DEFINITENESS:** $\langle v, v \rangle = 0 \iff v = 0$;
- **ADDITIVITY IN FIRST SLOT:** $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$;
- **HOMOGENEITY IN FIRST SLOT:** $\langle \lambda u, v \rangle = \lambda \langle u, v \rangle$ for all $\lambda \in \mathbb{F}$ and all $u, v \in V$;
- **CONJUGATE SYMMETRY:** $\langle u, v \rangle = \overline{\langle v, u \rangle}$ for all $u, v \in V$.

DEF 6.3 (*inner product space*):

An “inner product space” is a vector space V equipped with an inner product on V .

THM 6.6 (*basic properties of an inner product*):

Let $u, v, w \in V$ and $\lambda \in \mathbb{F}$. Then:

- (a) For each fixed $v \in V$, the map $u \mapsto \langle u, v \rangle$ is linear from V to $\mathbb{F}$;
- (b) $\langle 0, v \rangle = 0$;
- (c) $\langle v, 0 \rangle = 0$;
- (d) $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$;
- (e) $\langle u, \lambda v \rangle = \bar{\lambda} \langle u, v \rangle$.

NORMS (SUB-SECTION 6A2)

DEF 6.7 (*norm*):

For $v \in V$, the “norm” of v , denoted by $\|v\|$, is defined by

$$\|v\| := \sqrt{\langle v, v \rangle}.$$

THM 6.8 (*basic properties of the norm*):

Suppose $v \in V$ and $\lambda \in \mathbb{F}$. Then:

- (a) $\|v\| = 0 \iff v = 0$;
- (b) $\|\lambda v\| = |\lambda| \|v\|$.

PROOF: (a) $\|v\| = 0 \iff \langle v, v \rangle = 0 \iff v = 0$ by definiteness.

(b) $\|\lambda v\|^2 = \langle \lambda v, \lambda v \rangle = \lambda \bar{\lambda} \langle v, v \rangle = |\lambda|^2 \|v\|^2$. Taking square roots gives $\|\lambda v\| = |\lambda| \|v\|$.

□

DEF 6.10 (*orthogonal*):

Two vectors $u, v \in V$ are called “orthogonal” (written $u \perp v$) if

$$\langle u, v \rangle = 0.$$

THM 6.11 (*orthogonality and 0*):

- (a) The zero vector 0 is orthogonal to every vector in V .
- (b) The zero vector 0 is the only vector that is orthogonal to itself.

THM 6.12 (*Pythagorean theorem*):

If $u, v \in V$ are orthogonal, then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

PROOF: $\|u + v\|^2 = \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle = \|u\|^2 + 0 + 0 + \|v\|^2 = \|u\|^2 + \|v\|^2.$ □

THM 6.13 (*an orthogonal decomposition*):

Suppose $u, v \in V$ with $v \neq 0$. Set $c := \frac{\langle u, v \rangle}{\|v\|^2}$ and $w := u - \frac{\langle u, v \rangle}{\|v\|^2}v$. Then

$$u = cv + w \quad \text{and} \quad \langle w, v \rangle = 0.$$

PROOF: Clearly $u = cv + w$. Furthermore,

$$\langle w, v \rangle = \left\langle u - \frac{\langle u, v \rangle}{\|v\|^2}v, v \right\rangle = \langle u, v \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \langle v, v \rangle = \langle u, v \rangle - \langle u, v \rangle = 0.$$

□

THM 6.14 (*Cauchy-Schwarz inequality*):

Suppose $u, v \in V$. Then

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Equality holds if and only if one of u, v is a scalar multiple of the other.

PROOF: If $v = 0$, both sides equal 0, so the inequality holds. Now suppose $v \neq 0$. By Theorem 6.13,

$u = \frac{\langle u, v \rangle}{\|v\|^2}v + w$ with $\langle w, v \rangle = 0$. By the Pythagorean theorem (Theorem 6.12),

$$\|u\|^2 = \left\| \frac{\langle u, v \rangle}{\|v\|^2}v \right\|^2 + \|w\|^2 = \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \|w\|^2 \geq \frac{|\langle u, v \rangle|^2}{\|v\|^2}.$$

Multiplying both sides by $\|v\|^2$ and taking square roots gives $|\langle u, v \rangle| \leq \|u\| \|v\|$. Equality holds if and only if $w = 0$, which means $u = cv$ for some $c \in \mathbb{F}$. □

THM 6.17 (*Triangle inequality*):

Suppose $u, v \in V$. Then

$$\|u + v\| \leq \|u\| + \|v\|.$$

Equality holds if and only if one of u, v is a non-negative real multiple of the other.

PROOF: Using Cauchy-Schwarz (Theorem 6.14), we compute:

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle = \|u\|^2 + \langle u, v \rangle + \langle v, u \rangle + \|v\|^2 \\ &= \|u\|^2 + 2\operatorname{Re}\langle u, v \rangle + \|v\|^2 \\ &\leq \|u\|^2 + 2|\langle u, v \rangle| + \|v\|^2 \\ &\leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2. \end{aligned}$$

Taking square roots yields $\|u + v\| \leq \|u\| + \|v\|$. Equality holds if and only if $\operatorname{Re}\langle u, v \rangle = |\langle u, v \rangle| =$

$\|u\|\|v\|$, which occurs if and only if one of u, v is a non-negative scalar multiple of the other. $\square$

ORTHONORMAL BASES (SECTION 6B)

ORTHONORMAL LISTS AND THE GRAM-SCHMIDT PROCEDURE (SUB-SECTION 6B1)

DEF 6.18 (*orthonormal list*):

A list of vectors in V is called “*orthonormal*” if each vector in the list has norm 1 and is orthogonal to all other vectors in the list. In other words, a list $e_1, \dots, e_m$ of vectors in V is orthonormal if

$$\langle e_j, e_k \rangle = \delta_{j,k} = \begin{cases} 1, & \text{if } j = k, \\ 0, & \text{if } j \neq k. \end{cases}$$

THM 6.19 (*norm of an orthonormal linear combination*):

If $e_1, \dots, e_m$ is an orthonormal list of vectors in V , then

$$\|a_1 e_1 + \dots + a_m e_m\|^2 = |a_1|^2 + \dots + |a_m|^2 \quad \text{for all } a_1, \dots, a_m \in \mathbb{F}.$$

PROOF: Since $\langle e_j, e_k \rangle = 0$ for $j \neq k$ and $\|e_k\| = 1$, by the Pythagorean theorem (Theorem 6.12),

$$\left\| \sum_{j=1}^m a_j e_j \right\|^2 = \sum_{j=1}^m \|a_j e_j\|^2 = \sum_{j=1}^m |a_j|^2 \|e_j\|^2 = \sum_{j=1}^m |a_j|^2.$$

$\square$

THM 6.20 (*linear independence of orthonormal lists*):

Every orthonormal list of vectors is linearly independent.

PROOF: Suppose $e_1, \dots, e_m$ is an orthonormal list in V and scalars $a_1, \dots, a_m \in \mathbb{F}$ satisfy:

$$a_1 e_1 + \dots + a_m e_m = 0.$$

Taking norms and using Theorem 6.19 gives $|a_1|^2 + \dots + |a_m|^2 = \|0\|^2 = 0$, which forces $a_1 = \dots = a_m = 0$. $\square$

THM 6.21 (*Bessel's inequality*):

Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in V . If $v \in V$, then

$$\sum_{j=1}^m |\langle v, e_j \rangle|^2 \leq \|v\|^2.$$

PROOF: Let $u := \sum_{j=1}^m \langle v, e_j \rangle e_j$ and $w := v - u$. For each $k \in \{1, \dots, m\}$,

$$\langle w, e_k \rangle = \langle v, e_k \rangle - \sum_{j=1}^m \langle v, e_j \rangle \langle e_j, e_k \rangle = \langle v, e_k \rangle - \langle v, e_k \rangle = 0.$$

Thus, $\langle w, u \rangle = \sum_{j=1}^m \overline{\langle v, e_j \rangle} \langle w, e_j \rangle = 0$, so $u \perp w$. By the Pythagorean theorem (Theorem 6.12) and Theorem 6.19,

$$\|v\|^2 = \|u + w\|^2 = \|u\|^2 + \|w\|^2 \geq \|u\|^2 = \sum_{j=1}^m |\langle v, e_j \rangle|^2.$$

$\square$

DEF 6.22 (*orthonormal basis*):

An “*orthonormal basis*” of V is an orthonormal list of vectors in V that is also a basis of V .

THM 6.23 (*orthonormal lists of the right length are orthonormal bases*):

Suppose V is finite-dimensional. Then every orthonormal list of vectors in V of length $\dim V$ is an

orthonormal basis of V .

PROOF: By Theorem 6.20, every orthonormal list is linearly independent. By Theorem 2.38, a linearly independent list of length $\dim V$ is a basis of V . $\square$

THM 6.24 (*writing a vector as a linear combination of an orthonormal basis*):

Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $u, v \in V$. Then:

- (a) $v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n$;
- (b) $\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2$;
- (c) $\langle u, v \rangle = \langle u, e_1 \rangle \overline{\langle v, e_1 \rangle} + \dots + \langle u, e_n \rangle \overline{\langle v, e_n \rangle}$.

PROOF: Since $e_1, \dots, e_n$ is a basis, $v = a_1 e_1 + \dots + a_n e_n$ for some $a_j \in \mathbb{F}$. Taking the inner product with e_k gives $\langle v, e_k \rangle = \sum_{j=1}^n a_j \langle e_j, e_k \rangle = a_k$, proving (a). Part (b) follows from (a) and Theorem 6.19. Part (c) follows from evaluating $\langle u, v \rangle = \langle \sum_j \langle u, e_j \rangle e_j, \sum_k \langle v, e_k \rangle e_k \rangle$ using linearity and conjugate symmetry. $\square$

LINEAR FUNCTIONALS ON INNER PRODUCT SPACES (SUB-SECTION 6B2)

THM 6.25 (*Riesz representation theorem*):

Suppose V is finite-dimensional and $\phi \in V'$. Then there exists a unique vector $u \in V$ such that

$$\phi(v) = \langle v, u \rangle \quad \text{for all } v \in V.$$

PROOF: Let $e_1, \dots, e_n$ be an orthonormal basis of V . For every $v \in V$, Theorem 6.24 gives $v = \sum_{j=1}^n \langle v, e_j \rangle e_j$. Applying ϕ yields:

$$\phi(v) = \sum_{j=1}^n \langle v, e_j \rangle \phi(e_j) = \left\langle v, \sum_{j=1}^n \overline{\phi(e_j)} e_j \right\rangle.$$

Setting $u := \sum_{j=1}^n \overline{\phi(e_j)} e_j$ proves the existence of u .

For uniqueness, suppose $u_1, u_2 \in V$ satisfy $\langle v, u_1 \rangle = \langle v, u_2 \rangle = \phi(v)$ for all $v \in V$. Then $\langle v, u_1 - u_2 \rangle = 0$ for all $v \in V$. Setting $v = u_1 - u_2$ gives $\|u_1 - u_2\|^2 = 0$, so $u_1 = u_2$. $\square$

ORTHOGONAL COMPLEMENTS AND MINIMIZATION PROBLEMS (SECTION 6C)

ORTHOGONAL COMPLEMENTS (SUB-SECTION 6C1)

DEF 6.26 (*orthogonal complement, $U^\perp$*):

If $U \subseteq V$, the “orthogonal complement” of U , denoted by $U^\perp$, is defined by

$$U^\perp := \{v \in V \mid \langle u, v \rangle = 0 \text{ for all } u \in U\}.$$

THM 6.27 (*properties of the orthogonal complement*):

- (a) If $U \subseteq V$, then $U^\perp$ is a subspace of V .
- (b) $\{0\}^\perp = V$.
- (c) $V^\perp = \{0\}$.
- (d) If $U \subseteq V$, then $U \cap U^\perp \subseteq \{0\}$.
- (e) If $G \subseteq H \subseteq V$, then $H^\perp \subseteq G^\perp$.

PROOF: (a) We check the subspace properties:

- $\langle u, 0 \rangle = 0$ for all $u \in U$, so $0 \in U^\perp$.
- If $v, w \in U^\perp$, then $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle = 0 + 0 = 0$ for all $u \in U$, so $v + w \in U^\perp$.

- If $\lambda \in \mathbb{F}$ and $w \in U^\perp$, then $\langle u, \lambda w \rangle = \overline{\lambda} \langle u, w \rangle = \overline{\lambda}(0) = 0$ for all $u \in U$, so $\lambda w \in U^\perp$.

Thus, $U^\perp$ is a subspace of V .

- (b) For every $v \in V$, $\langle 0, v \rangle = 0$, so $v \in \{0\}^\perp$, which means $\{0\}^\perp = V$.
- (c) If $v \in V^\perp$, then $\langle v, v \rangle = 0$, which implies $v = 0$ by definiteness. Thus, $V^\perp = \{0\}$.
- (d) If $u \in U \cap U^\perp$, then $\langle u, u \rangle = 0$, so $u = 0$.
- (e) If $v \in H^\perp$, then $\langle u, v \rangle = 0$ for all $u \in H$. Since $G \subseteq H$, $\langle u, v \rangle = 0$ for all $u \in G$, so $v \in G^\perp$.

□

THM 6.28 (*direct sum of a subspace and its orthogonal complement*):

Suppose U is a finite-dimensional subspace of V . Then

$$V = U \oplus U^\perp.$$

PROOF: Let $e_1, \dots, e_m$ be an orthonormal basis of U . For any $v \in V$, define

$$u := \sum_{j=1}^m \langle v, e_j \rangle e_j \in U \quad \text{and} \quad w := v - u.$$

For each $k \in \{1, \dots, m\}$, $\langle w, e_k \rangle = \langle v, e_k \rangle - \langle v, e_k \rangle = 0$. Since w is orthogonal to every basis vector of U , linearity gives $\langle x, w \rangle = 0$ for all $x \in U$, so $w \in U^\perp$. Thus, $v = u + w \in U + U^\perp$. Since $U \cap U^\perp = \{0\}$ by Theorem 6.27(d), the sum is direct: $V = U \oplus U^\perp$. □

THM 6.29 (*dimension of orthogonal complement*):

Suppose V is finite-dimensional and U is a subspace of V . Then

$$\dim U^\perp = \dim V - \dim U.$$

PROOF: This follows immediately from $V = U \oplus U^\perp$ (Theorem 6.28) and Theorem 3.94. □

THM 6.30 (*orthogonal complement of the orthogonal complement*):

Suppose U is a finite-dimensional subspace of V . Then

$$U = (U^\perp)^\perp.$$

PROOF: First, if $u \in U$, then for all $w \in U^\perp$, $\langle w, u \rangle = \overline{\langle u, w \rangle} = \overline{0} = 0$, so $u \in (U^\perp)^\perp$. Thus, $U \subseteq (U^\perp)^\perp$.

Second, let $v \in (U^\perp)^\perp$. By Theorem 6.28, write $v = u + w$ with $u \in U$ and $w \in U^\perp$. Then $w = v - u \in (U^\perp)^\perp$ because $v \in (U^\perp)^\perp$ and $u \in U \subseteq (U^\perp)^\perp$. Thus, $w \in U^\perp \cap (U^\perp)^\perp = \{0\}$, so $w = 0$ and $v = u \in U$. Hence, $(U^\perp)^\perp \subseteq U$, completing the proof. □

THM 6.31 (*condition for $U^\perp = \{0\}$*):

Suppose U is a finite-dimensional subspace of V . Then

$$U^\perp = \{0\} \iff U = V.$$

PROOF: $\Rightarrow$ **DIRECTION:** If $U^\perp = \{0\}$, then $U = (U^\perp)^\perp = \{0\}^\perp = V$ by Theorem 6.30 and Theorem 6.27(b).

$\Leftarrow$ **DIRECTION:** If $U = V$, then $U^\perp = V^\perp = \{0\}$ by Theorem 6.27(c). □

DEF 6.32 (*orthogonal projection, P_U*):

Suppose U is a finite-dimensional subspace of V . The “orthogonal projection” of V onto U is the operator $P_U \in \mathcal{L}(V)$ defined as follows: for each $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp$, and set

$$P_U v := u.$$

EXAMPLE 6.33 (*orthogonal projection onto a one-dimensional subspace*):

Suppose $u \in V$ with $u \neq 0$, and let $U := \text{span}(u)$. For any $v \in V$,

$$P_U v = \frac{\langle v, u \rangle}{\|u\|^2} u.$$

THM 6.34 (*properties of orthogonal projection P_U*):

Suppose U is a finite-dimensional subspace of V . Then:

- (a) $P_U \in \mathcal{L}(V)$;
- (b) $P_U u = u$ for all $u \in U$;
- (c) $P_U w = 0$ for all $w \in U^\perp$;
- (d) $\text{range } P_U = U$;
- (e) $\text{null } P_U = U^\perp$;
- (f) $v - P_U v \in U^\perp$ for all $v \in V$;
- (g) $P_U^2 = P_U$;
- (h) $\|P_U v\| \leq \|v\|$ for all $v \in V$;
- (i) If $e_1, \dots, e_m$ is an orthonormal basis of U , then for every $v \in V$,

$$P_U v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m.$$

MINIMIZATION PROBLEMS (SUB-SECTION 6C2)

THM 6.35 (*minimizing distance to a subspace*):

Suppose U is a finite-dimensional subspace of V , $v \in V$, and $u \in U$. Then

$$\|v - P_U v\| \leq \|v - u\|.$$

Furthermore, equality holds if and only if $u = P_U v$.

PROOF: We write $v - u = (v - P_U v) + (P_U v - u)$. Since $v - P_U v \in U^\perp$ and $P_U v - u \in U$, these two vectors are orthogonal. By the Pythagorean theorem (Theorem 6.12),

$$\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \geq \|v - P_U v\|^2.$$

Taking square roots gives $\|v - P_U v\| \leq \|v - u\|$, with equality if and only if $\|P_U v - u\| = 0$, which is equivalent to $u = P_U v$. $\square$

PSEUDOINVERSE (SUB-SECTION 6C3)

THM 6.36 (*restriction of a linear map to $(\text{null } T)^\perp$*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $T|_{(\text{null } T)^\perp}$ is an isomorphism of $(\text{null } T)^\perp$ onto $\text{range } T$.

PROOF: Let $S := T|_{(\text{null } T)^\perp}$. If $v \in (\text{null } T)^\perp$ and $Sv = 0$, then $Tv = 0$, so $v \in \text{null } T$. Thus, $v \in (\text{null } T) \cap (\text{null } T)^\perp = \{0\}$, so $v = 0$. Hence, S is injective. For surjectivity onto $\text{range } T$, let $w \in \text{range } T$, so $w = Tv$ for some $v \in V$. By Theorem 6.28, $v = u + x$ with $u \in \text{null } T$ and $x \in (\text{null } T)^\perp$. Then $w = Tv = Tu + Tx = 0 + Sx = Sx$, so $w \in \text{range } S$. Thus, S is both injective and surjective onto $\text{range } T$. $\square$

DEF 6.37 (*pseudoinverse, $T^\dagger$*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. The “pseudoinverse” $T^\dagger \in \mathcal{L}(W, V)$ of T is defined by

$$T^\dagger w := \left(T|_{(\text{null } T)^\perp} \right)^{-1} P_{\text{range } T} w \quad \text{for all } w \in W.$$

THM 6.38 (*algebraic properties of the pseudoinverse*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then:

- (a) If T is invertible, then $T^\dagger = T^{-1}$;
- (b) $TT^\dagger = P_{\text{range } T}$;

(c) $T^\dagger T = P_{(\text{null } T)^\perp}$.

PROOF: (a) If T is invertible, then $\text{null } T = \{0\}$, so $(\text{null } T)^\perp = V$ and $\text{range } T = W$. Thus, $P_{\text{range } T} = I_W$ and $T|_{(\text{null } T)^\perp} = T$, giving $T^\dagger = T^{-1}$.

(b) For $w \in \text{range } T$, $P_{\text{range } T} w = w$, so $TT^\dagger w = T \left(T|_{(\text{null } T)^\perp} \right)^{-1} w = w = P_{\text{range } T} w$. For $w \in (\text{range } T)^\perp$, $P_{\text{range } T} w = 0$, so $TT^\dagger w = 0 = P_{\text{range } T} w$. Since $W = \text{range } T \oplus (\text{range } T)^\perp$, $TT^\dagger = P_{\text{range } T}$.

(c) For $v \in (\text{null } T)^\perp$, $Tv \in \text{range } T$, so $P_{\text{range } T}(Tv) = Tv$, giving $T^\dagger Tv = \left(T|_{(\text{null } T)^\perp} \right)^{-1}(Tv) = v = P_{(\text{null } T)^\perp} v$. For $v \in \text{null } T$, $Tv = 0$, so $T^\dagger Tv = 0 = P_{(\text{null } T)^\perp} v$. Since $V = \text{null } T \oplus (\text{null } T)^\perp$, $T^\dagger T = P_{(\text{null } T)^\perp}$.

□

THM 6.39 (*pseudoinverse provides best approximate solution*):

Suppose V is finite-dimensional, $T \in \mathcal{L}(V, W)$, and $w \in W$.

(a) For all $v \in V$,

$$\|T(T^\dagger w) - w\| \leq \|Tv - w\|,$$

with equality if and only if $v \in T^\dagger w + \text{null } T$.

(b) If $v \in T^\dagger w + \text{null } T$, then

$$\|T^\dagger w\| \leq \|v\|,$$

with equality if and only if $v = T^\dagger w$.

PROOF: (a) By Theorem 6.38(b), $T(T^\dagger w) = P_{\text{range } T} w$. By Theorem 6.35, $P_{\text{range } T} w$ is the unique vector in $\text{range } T$ closest to w . Thus, $\|T(T^\dagger w) - w\| \leq \|Tv - w\|$, with equality if and only if $Tv = T(T^\dagger w)$, which is equivalent to $v - T^\dagger w \in \text{null } T$, i.e., $v \in T^\dagger w + \text{null } T$.

(b) If $v \in T^\dagger w + \text{null } T$, write $v = T^\dagger w + (v - T^\dagger w)$. Since $T^\dagger w \in (\text{null } T)^\perp$ and $v - T^\dagger w \in \text{null } T$, these vectors are orthogonal. By the Pythagorean theorem (Theorem 6.12),

$$\|v\|^2 = \|T^\dagger w\|^2 + \|v - T^\dagger w\|^2 \geq \|T^\dagger w\|^2,$$

with equality if and only if $v - T^\dagger w = 0$, i.e., $v = T^\dagger w$.

□

OPERATORS ON INNER PRODUCT SPACES (CHAPTER 7)

Throughout this chapter, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$, and V, W are finite-dimensional nonzero inner product spaces over $\mathbb{F}$.

SELF-ADJOINT AND NORMAL OPERATORS (SECTION 7A)

ADJOINTS (SUB-SECTION 7A1)

DEF 7.1 (*adjoint, T^**):

Suppose $T \in \mathcal{L}(V, W)$. The “adjoint” of T is the function $T^* : W \rightarrow V$ defined by the property that

$$\langle Tv, w \rangle = \langle v, T^*w \rangle \quad \text{for all } v \in V \text{ and all } w \in W.$$

EXAMPLE 7.2 (*adjoint of a linear map from $\mathbb{R}^3$ to $\mathbb{R}^2$*):

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x_1, x_2, x_3) := (x_2 + 3x_3, 2x_1)$. Then

$$\langle T(x_1, x_2, x_3), (y_1, y_2) \rangle = (x_2 + 3x_3)y_1 + 2x_1y_2 = \langle (x_1, x_2, x_3), (2y_2, y_1, 3y_1) \rangle.$$

Thus, $T^*(y_1, y_2) = (2y_2, y_1, 3y_1)$.

EXAMPLE 7.3 (*adjoint of a rank-one operator*):

Fix $u \in V$ and $x \in W$, and define $T \in \mathcal{L}(V, W)$ by $Tv := \langle v, u \rangle x$. Then

$$\langle Tv, w \rangle = \langle \langle v, u \rangle x, w \rangle = \langle v, u \rangle \langle x, w \rangle = \langle v, \overline{\langle x, w \rangle} u \rangle = \langle v, \langle w, x \rangle u \rangle.$$

Thus, $T^*w = \langle w, x \rangle u$.

THM 7.4 (*the adjoint is a linear map*):

If $T \in \mathcal{L}(V, W)$, then $T^* \in \mathcal{L}(W, V)$.

PROOF: Let $T \in \mathcal{L}(V, W)$. For all $v \in V$ and $w_1, w_2 \in W$,

$$\langle v, T^*(w_1 + w_2) \rangle = \langle Tv, w_1 + w_2 \rangle = \langle Tv, w_1 \rangle + \langle Tv, w_2 \rangle = \langle v, T^*w_1 \rangle + \langle v, T^*w_2 \rangle = \langle v, T^*w_1 + T^*w_2 \rangle.$$

Thus, $T^*(w_1 + w_2) = T^*w_1 + T^*w_2$. Similarly, for all $\lambda \in \mathbb{F}$ and $w \in W$,

$$\langle v, T^*(\lambda w) \rangle = \langle Tv, \lambda w \rangle = \overline{\lambda} \langle Tv, w \rangle = \overline{\lambda} \langle v, T^*w \rangle = \langle v, \lambda T^*w \rangle.$$

Thus, $T^*(\lambda w) = \lambda T^*w$, proving that $T^* \in \mathcal{L}(W, V)$. □

THM 7.5 (*properties of the adjoint*):

Suppose $T \in \mathcal{L}(V, W)$. Then:

- (a) $(S + T)^* = S^* + T^*$ for all $S \in \mathcal{L}(V, W)$;
- (b) $(\lambda T)^* = \overline{\lambda} T^*$ for all $\lambda \in \mathbb{F}$;
- (c) $(T^*)^* = T$;
- (d) $(ST)^* = T^*S^*$ for all $S \in \mathcal{L}(W, U)$, where U is an inner product space;
- (e) $I_V^* = I_V$;
- (f) If T is invertible, then T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$.

PROOF: Let $v \in V, w \in W$, and $\lambda \in \mathbb{F}$.

(a) $\langle v, (S + T)^*w \rangle = \langle (S + T)v, w \rangle = \langle Sv, w \rangle + \langle Tv, w \rangle = \langle v, S^*w \rangle + \langle v, T^*w \rangle = \langle v, (S^* + T^*)w \rangle$, so $(S + T)^* = S^* + T^*$.

(b) $\langle v, (\lambda T)^*w \rangle = \langle \lambda Tv, w \rangle = \lambda \langle Tv, w \rangle = \lambda \langle v, T^*w \rangle = \langle v, \overline{\lambda} T^*w \rangle$, so $(\lambda T)^* = \overline{\lambda} T^*$.

(c) $\langle w, (T^*)^*v \rangle = \langle T^*w, v \rangle = \overline{\langle v, T^*w \rangle} = \overline{\langle Tv, w \rangle} = \langle w, Tv \rangle$, so $(T^*)^* = T$.

- (d) For $u \in U$, $\langle v, (ST)^*u \rangle = \langle STv, u \rangle = \langle Tv, S^*u \rangle = \langle v, T^*S^*u \rangle$, so $(ST)^* = T^*S^*$.
 (e) $\langle v_1, I_V^*v_2 \rangle = \langle I_Vv_1, v_2 \rangle = \langle v_1, v_2 \rangle = \langle v_1, I_Vv_2 \rangle$, so $I_V^* = I_V$.
 (f) From $T^{-1}T = I_V$ and $TT^{-1} = I_W$, taking adjoints gives $T^*(T^{-1})^* = I_V^* = I_V$ and $(T^{-1})^*T^* = I_W^* = I_W$. Thus, $(T^*)^{-1} = (T^{-1})^*$.

□

THM 7.6 (*null space and range of T^**):

 If $T \in \mathcal{L}(V, W)$, then:

- (a) $\text{null } T^* = (\text{range } T)^\perp$;
 (b) $\text{range } T^* = (\text{null } T)^\perp$;
 (c) $\text{null } T = (\text{range } T^*)^\perp$;
 (d) $\text{range } T = (\text{null } T^*)^\perp$.

PROOF: For (a): $w \in \text{null } T^* \iff T^*w = 0 \iff \langle v, T^*w \rangle = 0$ for all $v \in V \iff \langle Tv, w \rangle = 0$ for all $v \in V \iff w \in (\text{range } T)^\perp$. Taking orthogonal complements gives (d): $\text{range } T = ((\text{range } T)^\perp)^\perp = (\text{null } T^*)^\perp$. Replacing T with T^* and using $(T^*)^* = T$ gives (c) and (b). □

DEF 7.7 (*conjugate transpose, A^**):

 For $A \in \mathbb{F}^{m,n}$, the “conjugate transpose” of A , denoted by $A^* \in \mathbb{F}^{n,m}$, is defined by

$$(A^*)_{j,k} := \overline{A_{k,j}}.$$

THM 7.8 (*matrix of T^* equals conjugate transpose of matrix of T*):

 Let $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Then

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*.$$

PROOF: By Theorem 6.24, the entry in row j , column k of $\mathcal{M}(T)$ is $\langle Te_k, f_j \rangle$. The entry in row k , column j of $\mathcal{M}(T^*)$ is $\langle T^*f_j, e_k \rangle = \langle f_j, Te_k \rangle = \overline{\langle Te_k, f_j \rangle}$. Thus, $\mathcal{M}(T^*) = (\mathcal{M}(T))^*$. □

SELF-ADJOINT OPERATORS (SUB-SECTION 7A2)

DEF 7.9 (*self-adjoint operator*):

 An operator $T \in \mathcal{L}(V)$ is called “self-adjoint” if $T = T^*$, that is,

$$\langle Tv, w \rangle = \langle v, Tw \rangle \quad \text{for all } v, w \in V.$$

THM 7.10 (*eigenvalues of self-adjoint operators are real*):

Every eigenvalue of a self-adjoint operator is real.

PROOF: Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Let λ be an eigenvalue of T with eigenvector $v \neq 0$. Then

$$\lambda \|v\|^2 = \langle \lambda v, v \rangle = \langle Tv, v \rangle = \langle v, Tv \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \|v\|^2.$$

 Since $v \neq 0$, $\|v\|^2 > 0$, so $\lambda = \bar{\lambda}$, meaning $\lambda \in \mathbb{R}$. □

THM 7.11 (*$\langle Tv, v \rangle = 0$ for all v on complex spaces*):

 Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$. Then

$$\langle Tv, v \rangle = 0 \text{ for all } v \in V \iff T = 0.$$

PROOF: $\Rightarrow$ **DIRECTION:** By polarization, for all $u, w \in V$,

$$\langle Tu, w \rangle = \frac{\langle T(u+w), u+w \rangle - \langle T(u-w), u-w \rangle}{4} + \frac{\langle T(u+iw), u+iw \rangle - \langle T(u-iw), u-iw \rangle}{4}i.$$

 If $\langle Tv, v \rangle = 0$ for all $v \in V$, each term in the identity above is zero, so $\langle Tu, w \rangle = 0$ for all $u, w \in V$.

 Taking $w = Tu$ gives $\|Tu\|^2 = 0$, so $Tu = 0$ for all $u \in V$, which means $T = 0$.

$\Leftarrow$ **DIRECTION:** If $T = 0$, then $\langle Tv, v \rangle = 0$ trivially. □

THM 7.12 (*characterization of self-adjoint operators over $\mathbb{C}$*):

Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$. Then

T is self-adjoint $\iff \langle Tv, v \rangle \in \mathbb{R}$ for all $v \in V$.

PROOF: For any $v \in V$, $\langle T^*v, v \rangle = \overline{\langle Tv, v \rangle}$. Thus,

$$\begin{aligned} T \text{ is self-adjoint} &\iff T - T^* = 0 \\ &\iff \langle (T - T^*)v, v \rangle = 0 \quad \text{for all } v \in V \quad (\text{by Theorem 7.11}) \\ &\iff \langle Tv, v \rangle - \overline{\langle Tv, v \rangle} = 0 \quad \text{for all } v \in V \\ &\iff \langle Tv, v \rangle \in \mathbb{R} \quad \text{for all } v \in V. \end{aligned}$$

□

THM 7.13 ($\langle Tv, v \rangle = 0$ for all v implies $T = 0$ for self-adjoint operators):

Suppose $T \in \mathcal{L}(V)$ is a self-adjoint operator. Then

$\langle Tv, v \rangle = 0$ for all $v \in V \iff T = 0$.

PROOF: If $\mathbb{F} = \mathbb{C}$, this is Theorem 7.11. If $\mathbb{F} = \mathbb{R}$, since T is self-adjoint, we have the real polarization identity:

$$\langle Tu, w \rangle = \frac{\langle T(u+w), u+w \rangle - \langle T(u-w), u-w \rangle}{4} \quad \text{for all } u, w \in V.$$

If $\langle Tv, v \rangle = 0$ for all $v \in V$, then $\langle Tu, w \rangle = 0$ for all $u, w \in V$. Taking $w = Tu$ yields $Tu = 0$ for all u , so $T = 0$. □

NORMAL OPERATORS (SUB-SECTION 7A3)

DEF 7.14 (*normal operator*):

An operator $T \in \mathcal{L}(V)$ is called “normal” if it commutes with its adjoint, that is,

$$TT^* = T^*T.$$

THM 7.15 (*characterization of normal operators*):

An operator $T \in \mathcal{L}(V)$ is normal if and only if

$$\|Tv\| = \|T^*v\| \quad \text{for all } v \in V.$$

PROOF: $T^*T - TT^*$ is self-adjoint because $(T^*T - TT^*)^* = T^*T - TT^*$. By Theorem 7.13,

$$\begin{aligned} T \text{ is normal} &\iff T^*T - TT^* = 0 \\ &\iff \langle (T^*T - TT^*)v, v \rangle = 0 \quad \text{for all } v \in V \\ &\iff \langle Tv, Tv \rangle = \langle T^*v, T^*v \rangle \quad \text{for all } v \in V \\ &\iff \|Tv\|^2 = \|T^*v\|^2 \quad \text{for all } v \in V \\ &\iff \|Tv\| = \|T^*v\| \quad \text{for all } v \in V. \end{aligned}$$

□

THM 7.16 (*properties of normal operators*):

Suppose $T \in \mathcal{L}(V)$ is normal. Then:

- (a) $\text{null } T = \text{null } T^*$;
- (b) $\text{range } T = \text{range } T^*$;
- (c) $V = \text{null } T \oplus \text{range } T$;
- (d) $T - \lambda I$ is normal for every $\lambda \in \mathbb{F}$;

(e) If $v \in V$ and $\lambda \in \mathbb{F}$, then $Tv = \lambda v \iff T^*v = \bar{\lambda}v$.

PROOF: (a) $v \in \text{null } T \iff \|Tv\| = 0 \iff \|T^*v\| = 0 \iff v \in \text{null } T^*$ by Theorem 7.15.

(b) $\text{range } T = (\text{null } T^*)^\perp = (\text{null } T)^\perp = \text{range } T^*$ by Theorem 7.6 and part (a).

(c) $V = \text{null } T \oplus (\text{null } T)^\perp = \text{null } T \oplus \text{range } T^*$ by Theorem 6.28 and Theorem 7.6(b). By (b), $V = \text{null } T \oplus \text{range } T$.

(d) $(T - \lambda I)^*(T - \lambda I) = (T^* - \bar{\lambda}I)(T - \lambda I) = T^*T - \lambda T^* - \bar{\lambda}T + |\lambda|^2 I = TT^* - \bar{\lambda}T - \lambda T^* + |\lambda|^2 I = (T - \lambda I)(T - \lambda I)^*$.

(e) By (d), $T - \lambda I$ is normal. Applying Theorem 7.15 to $T - \lambda I$ gives $\|(T - \lambda I)v\| = \|(T^* - \bar{\lambda}I)v\|$, so $(T - \lambda I)v = 0 \iff (T^* - \bar{\lambda}I)v = 0$. □

THM 7.17 (*orthogonal eigenvectors for normal operators*):

Suppose $T \in \mathcal{L}(V)$ is normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

PROOF: Suppose $\alpha, \beta \in \mathbb{F}$ are distinct eigenvalues of T with eigenvectors $u, v \in V$. By Theorem 7.16(e), $T^*v = \bar{\beta}v$. Thus,

$$(\alpha - \beta)\langle u, v \rangle = \langle \alpha u, v \rangle - \langle u, \beta v \rangle = \langle Tu, v \rangle - \langle u, T^*v \rangle = 0.$$

Since $\alpha \neq \beta$, $\alpha - \beta \neq 0$, which implies $\langle u, v \rangle = 0$. □

THM 7.18 (*characterization of normal operators via real and imaginary parts*):

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then T is normal if and only if there exist commuting self-adjoint operators $A, B \in \mathcal{L}(V)$ such that $T = A + iB$.

PROOF: $\Rightarrow$ **DIRECTION:** If T is normal, define $A := \frac{T+T^*}{2}$ and $B := \frac{T-T^*}{2i}$. Then $A^* = A$ and $B^* = B$, so A and B are self-adjoint, and $T = A + iB$. A direct calculation shows that $AB - BA = \frac{T^*T - TT^*}{2i} = 0$, so A and B commute.

$\Leftarrow$ **DIRECTION:** If $T = A + iB$ with A, B commuting and self-adjoint, then $T^* = A - iB$. Then $TT^* = (A + iB)(A - iB) = A^2 + B^2 = (A - iB)(A + iB) = T^*T$, so T is normal. □

SPECTRAL THEOREM (SECTION 7B)

REAL SPECTRAL THEOREM (SUB-SECTION 7B1)

If $x, b, c \in \mathbb{R}$ and $b^2 < 4c$, then

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) > 0.$$

THM 7.19 (*invertible quadratic expressions*):

Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbb{R}$ satisfy $b^2 < 4c$. Then $T^2 + bT + cI$ is an invertible operator.

PROOF: Let $v \in V$ with $v \neq 0$. Since T is self-adjoint, $\langle T^2v, v \rangle = \langle Tv, Tv \rangle = \|Tv\|^2$. Using the Cauchy-Schwarz inequality (Theorem 6.14),

$$\begin{aligned} \langle (T^2 + bT + cI)v, v \rangle &= \|Tv\|^2 + b\langle Tv, v \rangle + c\|v\|^2 \\ &\geq \|Tv\|^2 - |b|\|Tv\|\|v\| + c\|v\|^2 \\ &= \left(\|Tv\| - \frac{|b|\|v\|}{2}\right)^2 + \left(c - \frac{b^2}{4}\right)\|v\|^2 \\ &> 0. \end{aligned}$$

Thus, $(T^2 + bT + cI)v \neq 0$ whenever $v \neq 0$. Hence, $\text{null}(T^2 + bT + cI) = \{0\}$, so $T^2 + bT + cI$ is

injective and therefore invertible. $\square$

THM 7.20 (*minimal polynomial of a self-adjoint operator*):

Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Then the minimal polynomial of T equals $(z - \lambda_1) \dots (z - \lambda_m)$ for some distinct $\lambda_1, \dots, \lambda_m \in \mathbb{R}$.

PROOF: First suppose $\mathbb{F} = \mathbb{C}$. The zeros of the minimal polynomial of T are the eigenvalues of T by Theorem 5.27. Because T is self-adjoint, all eigenvalues of T are real by Theorem 7.10. Thus, by the Fundamental Theorem of Algebra, the minimal polynomial of T splits completely into linear factors with real zeros.

Now suppose $\mathbb{F} = \mathbb{R}$. By Theorem 4.16, the minimal polynomial of T factors as

$$p(z) = (z - \lambda_1) \dots (z - \lambda_m)(z^2 + b_1z + c_1) \dots (z^2 + b_Nz + c_N),$$

where $\lambda_j \in \mathbb{R}$ and $b_k, c_k \in \mathbb{R}$ with $b_k^2 < 4c_k$. Evaluating this polynomial at T gives $p(T) = 0$. If $N > 0$, by Theorem 7.19, each factor $T^2 + b_kT + c_kI$ is invertible. Multiplying $p(T) = 0$ by $(T^2 + b_NT + c_NI)^{-1}$ yields an annihilating polynomial of lower degree, contradicting the minimality of the minimal polynomial. Thus, $N = 0$, so $p(z) = (z - \lambda_1) \dots (z - \lambda_m)$, as desired. $\square$

THM 7.21 (*Real Spectral Theorem*):

Suppose $\mathbb{F} = \mathbb{R}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent:

- (a) T is self-adjoint;
- (b) T has a diagonal matrix with respect to some orthonormal basis of V ;
- (c) V has an orthonormal basis consisting of eigenvectors of T .

PROOF: STEP 1: (a) $\implies$ (b): By Theorem 7.20, the minimal polynomial of T factors into distinct linear terms, which means T is diagonalizable by Theorem 5.62. By Schur's Theorem, T has an upper-triangular matrix $\mathcal{M}(T)$ with respect to some orthonormal basis of V . With respect to this orthonormal basis, $\mathcal{M}(T^*) = \mathcal{M}(T)^t$. Since $T = T^*$, $\mathcal{M}(T) = \mathcal{M}(T)^t$. An upper-triangular matrix that equals its transpose is diagonal, so $\mathcal{M}(T)$ is diagonal.

STEP 2: (b) $\implies$ (a): If $\mathcal{M}(T)$ is diagonal with respect to an orthonormal basis, then $\mathcal{M}(T^*) = \mathcal{M}(T)^* = \mathcal{M}(T)^t = \mathcal{M}(T)$, so $T^* = T$.

STEP 3: The equivalence of (b) and (c) follows directly from the definition of the matrix of an operator with respect to a basis. $\square$

COMPLEX SPECTRAL THEOREM (SUB-SECTION 7B2)

THM 7.22 (*Complex Spectral Theorem*):

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent:

- (a) T is normal;
- (b) T has a diagonal matrix with respect to some orthonormal basis of V ;
- (c) V has an orthonormal basis consisting of eigenvectors of T .

PROOF: STEP 1: (a) $\implies$ (b): By Schur's Theorem, there is an orthonormal basis $e_1, \dots, e_n$ of V with respect to which T has an upper-triangular matrix:

$$\mathcal{M}(T) = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ & \ddots & \vdots \\ 0 & & a_{n,n} \end{pmatrix}.$$

Then $\|Te_1\|^2 = |a_{1,1}|^2$ and $\|T^*e_1\|^2 = |a_{1,1}|^2 + |a_{1,2}|^2 + \cdots + |a_{1,n}|^2$. Since T is normal, $\|Te_1\| = \|T^*e_1\|$ by Theorem 7.15, which forces $a_{1,2} = \cdots = a_{1,n} = 0$. Continuing inductively for $k = 2, \dots, n$, we

find that all off-diagonal entries are zero. Thus, $\mathcal{M}(T)$ is diagonal.

STEP 2: (b) $\implies$ (a): If $\mathcal{M}(T)$ is diagonal with respect to an orthonormal basis, then $\mathcal{M}(T^*)$ is also diagonal. Since diagonal matrices commute, $TT^* = T^*T$, so T is normal.

STEP 3: The equivalence of (b) and (c) follows from the definition of a diagonal matrix with respect to a basis. $\square$

POSITIVE OPERATORS (SECTION 7C)

DEF 7.23 (*positive operator*):

An operator $T \in \mathcal{L}(V)$ is called “*positive*” (or “*positive semidefinite*”) if T is self-adjoint and

$$\langle Tv, v \rangle \geq 0 \quad \text{for all } v \in V.$$

DEF 7.24 (*square root*):

An operator $R \in \mathcal{L}(V)$ is called a “*square root*” of an operator $T \in \mathcal{L}(V)$ if $R^2 = T$.

THM 7.25 (*characterizations of positive operators*):

Let $T \in \mathcal{L}(V)$. Then the following are equivalent:

- (a) T is a positive operator;
- (b) T is self-adjoint and all eigenvalues of T are non-negative;
- (c) with respect to some orthonormal basis of V , the matrix $\mathcal{M}(T)$ is a diagonal matrix with non-negative entries on the diagonal;
- (d) T has a positive square root;
- (e) T has a self-adjoint square root;
- (f) $T = R^*R$ for some $R \in \mathcal{L}(V)$.

PROOF: We prove (a) $\implies$ (b) $\implies$ (c) $\implies$ (d) $\implies$ (e) $\implies$ (f) $\implies$ (a).

STEP 1: (a) $\implies$ (b): If T is positive, then T is self-adjoint. Let λ be an eigenvalue of T with eigenvector $v \neq 0$. Then

$$0 \leq \langle Tv, v \rangle = \langle \lambda v, v \rangle = \lambda \|v\|^2 \implies \lambda \geq 0.$$

STEP 2: (b) $\implies$ (c): By the Spectral Theorem (Theorems 7.21 and 7.22), there is an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T . The diagonal entries are the eigenvalues λ_k , which are non-negative by (b).

STEP 3: (c) $\implies$ (d): Suppose $\mathcal{M}(T) = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_k \geq 0$ with respect to an orthonormal basis $e_1, \dots, e_n$. Define $R \in \mathcal{L}(V)$ by $Re_k := \sqrt{\lambda_k} e_k$. Then $R^2 e_k = \lambda_k e_k = Te_k$, so $R^2 = T$. Moreover, R is positive because its matrix with respect to an orthonormal basis is diagonal with non-negative entries.

STEP 4: (d) $\implies$ (e): Every positive operator is self-adjoint by definition.

STEP 5: (e) $\implies$ (f): If $R = R^*$ and $R^2 = T$, then $T = R^*R$.

STEP 6: (f) $\implies$ (a): If $T = R^*R$, then $T^* = (R^*R)^* = R^*R = T$, so T is self-adjoint. Furthermore, for all $v \in V$,

$$\langle Tv, v \rangle = \langle R^*Rv, v \rangle = \langle Rv, Rv \rangle = \|Rv\|^2 \geq 0.$$

Thus, T is positive. $\square$

THM 7.26 (*uniqueness of positive square root*):

Each positive operator on V has a unique positive square root.

DEF 7.27 (*notation: $\sqrt{T}$*):

For a positive operator $T \in \mathcal{L}(V)$, $\sqrt{T}$ denotes the unique positive square root of T .

THM 7.28 ($\langle Tv, v \rangle = 0$ implies $Tv = 0$ for positive operators):

Suppose $T \in \mathcal{L}(V)$ is a positive operator and $v \in V$ satisfies $\langle Tv, v \rangle = 0$. Then $Tv = 0$.

PROOF: Since T is positive, it has a positive square root $\sqrt{T}$. Then $0 = \langle Tv, v \rangle = \langle \sqrt{T}\sqrt{T}v, v \rangle = \langle \sqrt{T}v, \sqrt{T}v \rangle = \|\sqrt{T}v\|^2$, which implies $\sqrt{T}v = 0$. Therefore, $Tv = \sqrt{T}(\sqrt{T}v) = \sqrt{T}(0) = 0$. $\square$

ISOMETRIES, UNITARY OPERATORS, AND MATRIX FACTORIZATION (SECTION 7D)

ISOMETRIES (SUB-SECTION 7D1)

DEF 7.29 (isometry):

A linear map $S \in \mathcal{L}(V, W)$ is called an “isometry” if

$$\|Sv\| = \|v\| \quad \text{for all } v \in V.$$

In other words, an isometry is a linear map that preserves norms.

THM 7.30 (characterizations of isometries):

Suppose $S \in \mathcal{L}(V, W)$, $e_1, \dots, e_n$ is an orthonormal basis of V , and $f_1, \dots, f_m$ is an orthonormal basis of W . Then the following are equivalent:

- (a) S is an isometry;
- (b) $S^*S = I_V$;
- (c) $\langle Su, Sv \rangle = \langle u, v \rangle$ for all $u, v \in V$;
- (d) $Se_1, \dots, Se_n$ is an orthonormal list in W ;
- (e) the columns of $\mathcal{M}(S, (e_1, \dots, e_n), (f_1, \dots, f_m))$ form an orthonormal list in $\mathbb{F}^m$.

PROOF: We prove $(a) \implies (b) \implies (c) \implies (d) \implies (e) \implies (a)$.

STEP 1: (a) $\implies$ (b): If S is an isometry, then for all $v \in V$,

$$\langle (I - S^*S)v, v \rangle = \langle v, v \rangle - \langle S^*Sv, v \rangle = \|v\|^2 - \|Sv\|^2 = 0.$$

Since $I - S^*S$ is self-adjoint, Theorem 7.13 implies $I - S^*S = 0$, so $S^*S = I$.

STEP 2: (b) $\implies$ (c): If $S^*S = I$, then $\langle Su, Sv \rangle = \langle S^*Su, v \rangle = \langle u, v \rangle$ for all $u, v \in V$.

STEP 3: (c) $\implies$ (d): If $\langle Su, Sv \rangle = \langle u, v \rangle$ for all $u, v \in V$, then $\langle Se_j, Se_k \rangle = \langle e_j, e_k \rangle = \delta_{j,k}$, so $Se_1, \dots, Se_n$ is orthonormal in W .

STEP 4: (d) $\implies$ (e): The k -th column of $A = \mathcal{M}(S)$ is the coordinate vector of Se_k with respect to the orthonormal basis $f_1, \dots, f_m$. By Theorem 6.24(c), the Euclidean inner product of column j and column k equals $\langle Se_j, Se_k \rangle = \delta_{j,k}$. Thus, the columns form an orthonormal list in $\mathbb{F}^m$.

STEP 5: (e) $\implies$ (a): Suppose the columns of $\mathcal{M}(S)$ are orthonormal. Then $Se_1, \dots, Se_n$ is an orthonormal list in W . For $v = \sum_{j=1}^n a_j e_j$, by Theorem 6.19,

$$\|Sv\|^2 = \left\| \sum_{j=1}^n a_j Se_j \right\|^2 = \sum_{j=1}^n |a_j|^2 = \|v\|^2.$$

Thus, $\|Sv\| = \|v\|$, so S is an isometry. $\square$

UNITARY OPERATORS (SUB-SECTION 7D2)

DEF 7.31 (unitary operator):

An operator $S \in \mathcal{L}(V)$ is called “unitary” if S is an invertible isometry.

EXAMPLE 7.32 (*rotation of $\mathbb{R}^2$*):

For $\theta \in \mathbb{R}$, the operator $S \in \mathcal{L}(\mathbb{R}^2)$ defined by

$$\mathcal{M}(S) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

has orthonormal columns, so S is an isometry and therefore a unitary operator on $\mathbb{R}^2$.

THM 7.33 (*characterizations of unitary operators*):

Suppose $S \in \mathcal{L}(V)$ and $e_1, \dots, e_n$ is an orthonormal basis of V . Then the following are equivalent:

- (a) S is a unitary operator;
- (b) $S^*S = SS^* = I$;
- (c) S is invertible and $S^{-1} = S^*$;
- (d) $Se_1, \dots, Se_n$ is an orthonormal basis of V ;
- (e) the rows of $\mathcal{M}(S, (e_1, \dots, e_n))$ form an orthonormal basis of $\mathbb{F}^n$;
- (f) S^* is a unitary operator.

THM 7.34 (*eigenvalues of unitary operators*):

Suppose λ is an eigenvalue of a unitary operator. Then $|\lambda| = 1$.

PROOF: Let $S \in \mathcal{L}(V)$ be unitary and let $v \in V$ with $v \neq 0$ satisfy $Sv = \lambda v$. Since S is an isometry,

$$\|v\| = \|Sv\| = \|\lambda v\| = |\lambda| \|v\|. \text{ Dividing by } \|v\| > 0 \text{ gives } |\lambda| = 1. \quad \square$$

THM 7.35 (*description of unitary operators over $\mathbb{C}$*):

Suppose $\mathbb{F} = \mathbb{C}$ and $S \in \mathcal{L}(V)$. Then the following are equivalent:

- (a) S is a unitary operator;
- (b) there is an orthonormal basis of V consisting of eigenvectors of S whose corresponding eigenvalues all have absolute value 1.

PROOF: Since $S^*S = SS^* = I$, every unitary operator is normal. By the Complex Spectral Theorem (Theorem 7.22), S is diagonalizable with respect to an orthonormal basis of eigenvectors. By Theorem 7.34, each eigenvalue has absolute value 1. $\square$

QR FACTORIZATION (SUB-SECTION 7D3)**DEF 7.36** (*unitary matrix*):

A matrix $Q \in \mathbb{F}^{n,n}$ is called “unitary” if its columns form an orthonormal basis of $\mathbb{F}^n$.

THM 7.37 (*QR factorization*):

Suppose $A \in \mathbb{F}^{n,n}$ is an invertible matrix. Then there exist unique matrices $Q, R \in \mathbb{F}^{n,n}$ such that Q is unitary, R is upper-triangular with strictly positive entries on its diagonal, and

$$A = QR.$$

CHOLESKY FACTORIZATION (SUB-SECTION 7D4)**THM 7.38** (*positive invertible operator*):

A self-adjoint operator $T \in \mathcal{L}(V)$ is positive and invertible if and only if $\langle Tv, v \rangle > 0$ for all $v \in V$ with $v \neq 0$.

DEF 7.39 (*positive definite matrix*):

A matrix $B \in \mathbb{F}^{n,n}$ is called “positive definite” if $B^* = B$ and

$$\langle Bx, x \rangle > 0 \quad \text{for all } x \in \mathbb{F}^n \text{ with } x \neq 0.$$

THM 7.40 (*Cholesky factorization*):

Suppose $B \in \mathbb{F}^{n,n}$ is a positive definite matrix. Then there exists a unique upper-triangular matrix $R \in \mathbb{F}^{n,n}$ with strictly positive entries on its diagonal such that

$$B = R^*R.$$

SINGULAR VALUE DECOMPOSITION (SECTION 7E)**SINGULAR VALUES** (SUB-SECTION 7E1)**THM 7.41** (*properties of T^*T*):

Suppose $T \in \mathcal{L}(V, W)$. Then:

- (a) T^*T is a positive operator on V ;
- (b) $\text{null } T^*T = \text{null } T$;
- (c) $\text{range } T^*T = \text{range } T^*$;
- (d) $\dim \text{range } T = \dim \text{range } T^* = \dim \text{range } T^*T$.

PROOF: (a) $(T^*T)^* = T^*(T^*)^* = T^*T$, and $\langle T^*Tv, v \rangle = \langle Tv, Tv \rangle = \|Tv\|^2 \geq 0$ for all $v \in V$. Thus, T^*T is positive.

(b) If $v \in \text{null } T$, then $Tv = 0$, so $T^*Tv = 0$. Conversely, if $T^*Tv = 0$, then $\|Tv\|^2 = \langle T^*Tv, v \rangle = 0$, so $Tv = 0$. Thus, $\text{null } T^*T = \text{null } T$.

(c) $\text{range } T^*T = (\text{null } (T^*T)^*)^\perp = (\text{null } T^*T)^\perp = (\text{null } T)^\perp = \text{range } T^*$ by Theorem 7.6 and part (b).

(d) This follows from the Fundamental Theorem of Linear Maps (Theorem 3.21) and parts (b) and (c). □

DEF 7.42 (*singular values*):

Suppose $T \in \mathcal{L}(V, W)$. The “singular values” of T are the non-negative square roots of the eigenvalues of T^*T , listed in decreasing order:

$$s_1 \geq s_2 \geq \dots \geq s_n \geq 0,$$

where each eigenvalue is repeated according to the dimension of its eigenspace.

THM 7.43 (*role of positive singular values*):

Suppose $T \in \mathcal{L}(V, W)$. Then:

- (a) T is injective if and only if 0 is not a singular value of T ;
- (b) the number of positive singular values of T equals $\dim \text{range } T$;
- (c) T is surjective if and only if the number of positive singular values of T equals $\dim W$.

THM 7.44 (*characterization of isometries via singular values*):

Suppose $S \in \mathcal{L}(V, W)$. Then S is an isometry if and only if all singular values of S equal 1.

SVD FOR LINEAR MAPS AND FOR MATRICES (SUB-SECTION 7E2)**THM 7.45** (*Singular Value Decomposition*):

Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of T are $s_1, \dots, s_m$. Then there exist orthonormal lists $e_1, \dots, e_m$ in V and $f_1, \dots, f_m$ in W such that

$$Tv = \sum_{j=1}^m s_j \langle v, e_j \rangle f_j \quad \text{for all } v \in V.$$

THM 7.46 (*SVD of adjoint and pseudoinverse*):

Suppose $T \in \mathcal{L}(V, W)$ has singular value decomposition $Tv = \sum_{j=1}^m s_j \langle v, e_j \rangle f_j$. Then for all $w \in W$:

- (a) $T^*w = \sum_{j=1}^m s_j \langle w, f_j \rangle e_j$;
- (b) $T^\dagger w = \sum_{j=1}^m \frac{\langle w, f_j \rangle}{s_j} e_j$.

THM 7.47 (*matrix version of SVD*):

Suppose $A \in \mathbb{F}^{p,n}$ has rank $m \geq 1$. Then there exist a matrix $B \in \mathbb{F}^{p,m}$ with orthonormal columns, an $m \times m$ diagonal matrix D with strictly positive entries on the diagonal, and a matrix $C \in \mathbb{F}^{n,m}$ with orthonormal columns such that

$$A = BDC^*.$$

CONSEQUENCES OF SINGULAR VALUE DECOMPOSITION (SECTION 7F)

NORMS OF LINEAR MAPS (SUB-SECTION 7F1)

THM 7.48 (*upper bound for $\|Tv\|$*):

Suppose $T \in \mathcal{L}(V, W)$ and let s_1 be the largest singular value of T . Then

$$\|Tv\| \leq s_1 \|v\| \quad \text{for all } v \in V.$$

DEF 7.49 (*norm of a linear map*):

Suppose $T \in \mathcal{L}(V, W)$. The “operator norm” of T , denoted by $\|T\|$, is defined by

$$\|T\| := \max\{\|Tv\| \mid v \in V \text{ and } \|v\| \leq 1\}.$$

THM 7.50 (*basic properties of norms of linear maps*):

Let $T \in \mathcal{L}(V, W)$. Then:

- (a) $\|T\| \geq 0$;
- (b) $\|T\| = 0 \iff T = 0$;
- (c) $\|\lambda T\| = |\lambda| \|T\|$ for all $\lambda \in \mathbb{F}$;
- (d) $\|S + T\| \leq \|S\| + \|T\|$ for all $S \in \mathcal{L}(V, W)$.

THM 7.51 (*alternative formulas for $\|T\|$*):

Let $T \in \mathcal{L}(V, W)$. Then:

- (a) $\|T\| = s_1$, the largest singular value of T ;
- (b) $\|T\| = \max\{\|Tv\| \mid v \in V \text{ and } \|v\| = 1\}$;
- (c) $\|T\|$ is the smallest real number c such that $\|Tv\| \leq c \|v\|$ for all $v \in V$.

THM 7.52 (*norm of the adjoint*):

Suppose $T \in \mathcal{L}(V, W)$. Then $\|T^*\| = \|T\|$.

APPROXIMATION BY LINEAR MAPS WITH LOWER-DIMENSIONAL RANGE (SUB-SECTION 7F2)

THM 7.53 (*best rank approximation*):

Let $T \in \mathcal{L}(V, W)$ have positive singular values $s_1 \geq \dots \geq s_m > 0$. If $1 \leq k < m$, then

$$\min\{\|T - S\| \mid S \in \mathcal{L}(V, W) \text{ and } \dim \text{range } S \leq k\} = s_{k+1}.$$

Furthermore, if $Tv = \sum_{j=1}^m s_j \langle v, e_j \rangle f_j$ is a singular value decomposition of T , and $T_k \in \mathcal{L}(V, W)$ is defined by

$$T_k v := \sum_{j=1}^k s_j \langle v, e_j \rangle f_j \quad \text{for all } v \in V,$$

then $\dim \text{range } T_k = k$ and $\|T - T_k\| = s_{k+1}$.

THM 7.54 (*polar decomposition*):

Suppose $T \in \mathcal{L}(V)$. Then there exists a unitary operator $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T}.$$

Furthermore, if T is invertible, then S is uniquely determined.

OPERATORS APPLIED TO ELLIPSOIDS AND PARALLELEPIPEDS (SUB-SECTION 7F3)

DEF 7.55 (*open unit ball, B*):

The “open unit ball” in V , denoted by B , is defined by

$$B := \{v \in V \mid \|v\| < 1\}.$$